

NUMERICAL ANALYSIS

(MTS5 B07)

V SEMESTER

B.Sc.

CORE COURSE

2019 Admission onwards



UNIVERSITY OF CALICUT

School of Distance Education

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Study Material

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NUMERICAL ANALYSIS

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PREFACE

The goal of numerical analysis is to provide techniques and algorithms to find approximate numerical solution to problems in several areas of mathematics where it is impossible or hard to find the actual/closed form solution by analytical methods and also to make an error analysis to ascertain the accuracy of the approximate solution. The subject addresses a variety of questions ranging from the approximation of functions and integrals to the approximate solution of algebraic, transcendental, differential and integral equations, with particular emphasis on the stability, accuracy, efficiency and reliability of numerical algorithms. The course enables the students to

- Understand several methods such as bisection method, fixed point iteration method, regula-falsi method etc. to find out the approximate numerical solutions of algebraic and transcendental equations with desired accuracy.
- Understand the concept of interpolation and also learn some well-known interpolation techniques.
- Understand a few techniques for numerical differentiation and integration and also realize their merits and demerits.
- Find out numerical approximations to solutions of initial value problems and also to understand the efficiency of various methods.

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CHAPTER 1

ROUND-OFF ERRORS AND RATES OF CONVERGENCE

Round-Off Errors

The arithmetic performed by a calculator or computer is different from the arithmetic in algebra and calculus courses. You would likely expect that we always have as true statements things such as $2+2 = 4$, $4 \cdot 8 = 32$, and $(\sqrt{3})^2 = 3$. However, with computer arithmetic, we expect exact results for $2+2 = 4$ and $4 \cdot 8 = 32$, but we will not have precisely $(\sqrt{3})^2 = 3$. To understand why this is true, we must explore the world of finite-digit arithmetic.

In our traditional mathematical world, we permit numbers with an infinite number of digits. The arithmetic we use in this world defines $\sqrt{3}$ as that unique positive number that when multiplied by itself produces the integer 3. In the computational world, however, each representable number has only a fixed and finite number of digits. This means, for example, that only rational numbers—and not even all of these—can be represented exactly. Since $\sqrt{3}$ is not rational, it is given an approximate representation, one whose square will not be precisely 3, although it will likely be sufficiently close to 3 to be acceptable in most situations. In most cases, then, this machine arithmetic is satisfactory and passes without notice or concern, but at times problems arise because of this discrepancy.

The error that is produced when a calculator or computer is used to perform real number calculations is called **round-off error**. It occurs because the arithmetic performed in a machine involves numbers with only a finite number of digits, with the result that calculations are performed with only approximate representations of the actual numbers.

Decimal Machine Numbers

We assume that machine numbers are represented in the normalized decimal floating-point form $\pm 0.d_1d_2 \dots d_k \times 10^n$, $1 \leq d_1 \leq 9$, and $0 \leq d_i \leq 9$, for each $i = 2, \dots, k$. Numbers of this form are called k -digit decimal machine numbers.

Any positive real number within the numerical range of the machine can be normalized to the form

$$y = 0.d_1d_2 \dots d_k d_{k+1}d_{k+2} \dots \times 10^n.$$

The floating-point form of y , denoted $fl\{y\}$, is obtained by terminating the mantissa of y at k decimal digits. There are two common ways of performing this termination. One method, called chopping, is to simply chop off the digits $d_{k+1}d_{k+2} \dots$. This produces the floating-point form

$$fl\{y\} = 0.d_1d_2 \dots d_k \times 10^n$$

The other method, called rounding, adds $5 \times 10^{n-(k+1)}$ to y and then chops the result to obtain a number of the form

$$fl\{y\} = 0.\delta_1\delta_2 \dots \delta_k \times 10^n$$

For rounding, when $d_{k+1} \geq 5$, we add 1 to d_k to obtain $fl\{y\}$; that is, we round up. When $d_{k+1} < 5$, we simply chop off all but the first k digits; that is, round down. If we round down, then $\delta_i = d_i$, for each $i = 1, 2, \dots, k$. However, if we round up, the digits (and even the exponent) might change.

Example 1 Determine the five-digit (a) chopping and (b) rounding values of the irrational number π .

Solution The number π has an infinite decimal expansion of the form $\pi = 3.14159265 \dots$

Written in normalized decimal form, we have $\pi = 0.314159265 \dots \times 10^1$

(a) The floating-point form of π using five-digit chopping is $\pi = 0.31415 \times 10^1 = 3.1415$.

(b) The sixth digit of the decimal expansion of π is a 9, so the floating-point form of π

using five-digit rounding is $fl\pi = (0.31415 + 0.00001) \times 10^1 = 3.1416$.

Definition 1.1 Suppose that p^* is an approximation to p . The **actual error** is $p - p^*$, the **absolute error** is $|p - p^*|$ and the **relative error** is $\frac{|p - p^*|}{|p|}$, provided that $p \neq 0$

Consider the actual, absolute, and relative errors in representing p by p^* in the following example.

Example 2 Determine the actual, absolute, and relative errors when approximating p by p^* when

$$(a) p = 0.3000 \times 10^1 \text{ and } p^* = 0.3100 \times 10^1$$

$$(b) p = 0.3000 \times 10^{-3} \text{ and } p^* = 0.3100 \times 10^{-3}$$

$$(c) p = 0.3000 \times 10^4 \text{ and } p^* = 0.3100 \times 10^4$$

Solution

(a) For $p = 0.3000 \times 10^1$ and $p^* = 0.3100 \times 10^1$, the actual error is -0.1 , the absolute error is 0.1 , and the relative error is $0.333\bar{3} \times 10^{-1}$

(b) For $p = 0.3000 \times 10^{-3}$ and $p^* = 0.3100 \times 10^{-3}$, the actual error is -0.1×10^{-4} , the absolute error is 0.1×10^{-4} , and the relative error is $0.333\bar{3} \times 10^{-1}$

(c) For $p = 0.3000 \times 10^4$ and $p^* = 0.3100 \times 10^4$, the actual error is -0.1×10^3 , the absolute error is 0.1×10^3 , and the relative error is again $0.333\bar{3} \times 10^{-1}$.

This example shows that the same relative error, $0.333\bar{3} \times 10^{-1}$, occurs for widely varying absolute errors. As a measure of accuracy, the absolute error can be misleading and the relative error more meaningful because the relative error takes into consideration the size of the value.

An error bound is a nonnegative number larger than the absolute error.

The following definition uses relative error to give a measure of significant digits of accuracy for an approximation.

Definition 1.2 The number p^* is said to approximate p to t **significant digits** (or figures) if t is the largest nonnegative integer for which

$$\frac{|p - p^*|}{|p|} \leq 5 \times 10^{-t}$$

Table 1.1 illustrates the continuous nature of significant digits by listing, for the various values of p , the least upper bound of $|p - p^*|$, denoted $\max |p - p^*|$, when p^* agrees with p to four significant digits.

Table 1.1

P	0.1	0.5	100	1000	5000	9990	10000
$\max p - p^* $	0.00005	0.00025	0.05	0.5	2.5	4.995	5.

Example 3 Suppose that $x = \frac{5}{7}$ and $y = \frac{1}{3}$. Use five-digit chopping for calculating $x + y, x - y, x \times y$ and $x \div y$.

Solution Note that

$$x = \frac{5}{7} = 0.\overline{714285} \text{ and } y = \frac{1}{3} = 0.\overline{3}$$

implies that the five-digit chopping values of x and y are

$$fl(x) = 0.71428 \times 10^0 \text{ and } fl(y) = 0.33333 \times 10^0$$

Thus,

$$\begin{aligned}
 x \oplus y &= fl(fl(x) + fl(y)) \\
 &= fl(0.71428 \times 10^0 + 0.33333 \times 10^0) \\
 &= fl(1.04761 \times 10^0) = 0.10476 \times 10^1.
 \end{aligned}$$

The true value is $x + y = \frac{5}{7} + \frac{1}{3} = \frac{22}{21}$, so we have

$$\text{Absolute Error} = \left| \frac{22}{21} - 0.10476 \times 10^1 \right| = 0.190 \times 10^{-4}.$$

$$\text{Relative Error} = \left| 0.190 \times \frac{10^{-4}}{\frac{22}{21}} \right| = 0.182 \times 10^{-4}.$$

Table 1.2 lists the values of this and the other calculations.

Table 1.2

Operation	Result	Actual value	Absolute error	Relative error
$x \oplus y$	0.10476×10^1	22/21	0.190×10^{-4}	0.182×10^{-4}
$x \ominus y$	0.38095×10^0	8/21	0.238×10^{-5}	0.625×10^{-5}
$x \otimes y$	0.23809×10^0	5/21	0.524×10^{-5}	0.220×10^{-4}
$x \div y$	0.21428×10^1	15/7	0.571×10^{-4}	0.267×10^{-4}

EXAMPLE 4 Suppose that in addition to $x = \frac{5}{7}$ and $u = \frac{1}{3}$ we have

$$u = 0.714251, \quad v = 98765.9, \text{ and } w = 0.111111 \times 10^{-4},$$

so that

$$fl(u) = 0.71425 \times 10^5, fl(v) = 0.98765 \times 10^5, \text{ and } fl(w) = 0.11111 \times 10^{-4}.$$

Determine the five-digit chopping values of $x \ominus u$, $(x \ominus u) \ominus w$, $(x \ominus u) \otimes v$, and $u \oplus v$.

Solution These numbers were chosen to illustrate some problems that can arise with finite-digit arithmetic. Because x and y are nearly the same, their difference is small. The absolute error for $x \ominus u$ is

$$\begin{aligned} |(x - u) - (x \ominus u)| &= |(x - u) - (fl(fl(x) - fl(u)))| \\ &= \left| \left(\frac{5}{7} - 0.714251 \right) \right. \\ &\quad \left. - (fl(0.71428 \times 10^0 - 0.71425 \times 10^0)) \right| \end{aligned}$$

$$|0.347143 \times 10^{-4} - fl(0.00003 \times 10^0)| = 0.47143 \times 10^{-5}$$

This approximation has a small absolute error but a large relative error.

$$\left| \frac{0.47143 \times 10^{-5}}{0.147143 \times 10^{-4}} \right| \leq 0.136$$

The subsequent division by the small number w or multiplication by the large number v magnifies the absolute error without

modifying the relative error. The addition of the large and small numbers u and v produces large absolute error but not large relative error. These calculations are shown in Table 1.3.

Table 1.3

Operation	Result	Actual value	Absolute error	Relative error
$x \ominus u$	0.30000×10^{-4}	0.34714×10^{-4}	0.471×10^{-5}	0.136
$(x \ominus u) \div w$	$0.0.27000 \times 10^1$	0.31242×10^1	0.424	0.136
$(x \ominus u) \otimes y$	0.29629×10^1	0.34285×10^1	0.465	0.136
$u \oplus y$	0.98765×10^5	0.98766×10^5	0.161×10^1	0.163×10^{-4}

Example 5 Let $p = 0.54617$ and $q = 0.54601$. Use four-digit arithmetic to approximate $p - q$ and determine the absolute and relative errors using **(a)** rounding and **(b)** chopping.

Solution The exact value of $r = p - q$ is $r = 0.00016$.

(a) Suppose the subtraction is performed using four-digit rounding arithmetic. Rounding p and q to four digits gives $p^* = 0.5462$ and $q^* = 0.5460$, respectively, and

$r^* = p^* - q^* = 0.0002$ is the four-digit approximation to r . Since

$$\frac{|r - r^*|}{|r|} = \frac{|0.00016 - 0.0002|}{|0.00016|} = 0.25.$$

the result has only one significant digit, whereas p^* and q^* were accurate to four

and five significant digits, respectively.

(b) If chopping is used to obtain the four digits, the four-digit approximations to p, q , and r are $p^* = 0.5461$, $q^* = 0.5460$, and $r^* = p^* - q^* = 0.0001$. This gives

$$\frac{|r - r^*|}{|r|} = \frac{|0.00016 - 0.0001|}{|0.00016|} = 0.375.$$

which also results in only one significant digit of accuracy.

Questions 1.1

Compute the absolute error and relative error in approximations of p by p^* .

a. $p = \pi, p^* = \frac{22}{7}$

b. $p = \pi, p^* = 3.1416$

c. $p = e, p^* = 2.718$

d. $p = \sqrt{2}, p^* = 1.414$

Rates of Convergence

Definition 1.3 Suppose $\{\beta_n\}_{n=1}^{\infty}$ is a sequence known to converge to zero and $\{\alpha_n\}_{n=1}^{\infty}$ converges to a number α . If a positive constant K exists with

$$|\alpha_n - \alpha| \leq K|\beta_n|, \text{ for large } n,$$

Then we say that $\{\alpha_n\}_{n=1}^{\infty}$ converges to α with **rate, or order, of convergence** $O(\beta_n)$. (This expression is read “big oh of β_n ”). It is indicated by writing $\alpha_n = \alpha + O(\beta_n)$.

EXAMPLE 6 Suppose that, for $n \geq 1$, $\alpha_n = \frac{n+1}{n^2}$ and $\hat{\alpha}_n = \frac{n+3}{n^3}$.

Both $\lim_{n \rightarrow \infty} \alpha_n = 0$ and $\lim_{n \rightarrow \infty} \hat{\alpha}_n = 0$, but the sequence $\{\hat{\alpha}_n\}$ converges to this limit much faster than the sequence $\{\alpha_n\}$. Using five-digit rounding arithmetic, we have the values shown in Table 1.4. determine rates of converges for these two sequences.

Table 1.4

η	1	2	3	4	5	6	7
α_n	2.000 00	0.750 00	0.444 44	0.312 50	0.2400 0	0.1944 4	0.1632 7
$\hat{\alpha}_n$	4.000 00	0.625 00	0.222 22	0.109 38	0.0640 00	0.0416 67	0.0291 55

Solution Define the sequences $\beta_n = \frac{1}{n}$ and $\hat{\beta}_n = \frac{1}{n^2}$.

Then $|\alpha_n - 0| = \frac{n+1}{n^2} \leq \frac{n+n}{n^2} = 2 \cdot \frac{1}{n} = 2\beta_n$

and

$$|\hat{\alpha}_n - 0| = \frac{n+3}{n^3} \leq \frac{n+3n}{n^3} = 4 \cdot \frac{1}{n^2} = 4\hat{\beta}_n$$

Hence, the rate of convergence of $\{\alpha_n\}$ to zero is similar to the convergence of $\{\frac{1}{n}\}$ to zero, whereas $\{\hat{\alpha}_n\}$ converges to zero at a rate similar to the more rapidly convergent sequence $\{\frac{1}{n^2}\}$. We express this by writing,

$$\alpha_n = 0 + O\left(\frac{1}{n}\right) \quad \text{and} \quad \hat{\alpha} = 0 + O\left(\frac{1}{n^2}\right).$$

We also use the O (big oh) notation to describe the rate at which functions converge.

Questions 1.2

1. Find the rates of convergence of the following functions as $h \rightarrow 0$.

a. $\lim_{h \rightarrow 0} \frac{\sin h}{h} = 1$

b. $\lim_{h \rightarrow 0} \frac{1 - \cos h}{h} = 0$

c. $\lim_{h \rightarrow 0} \frac{\sin h - h \cos h}{h} = 0$

d. $\lim_{h \rightarrow 0} \frac{1 - e^h}{h} = -1$

CHAPTER 2

SOLUTIONS OF EQUATIONS IN ONE VARIABLE

We deal with one of the most basic problems of numerical approximation, the **root-finding problem**, in this chapter. For a given function f , we need to solve an equation of the form $f(x) = 0$, which is called **root**. A root of this equation is also called a **zero** of the function f .

For example, for an equation $f(x) = x^2 - 2x + 1 = 0$, we can find the root $x = 1$ so that $f(1) = 0$.

For solving such equations, one of the most useful methods are the iterative methods since these methods are suitable for computing using computer programs. The methods we will discuss here are the bisection methods, fixed point iteration method, Newton's method and its extensions.

2.1 The Bisection Method

The bisection method is based on the intermediate value theorem for continuous functions:

Suppose f is a continuous function defined on the interval $[a, b]$ with $f(a)$ and $f(b)$ are of opposite signs. Then a number p exists in (a, b) with $f(p) = 0$. Although the procedure will work when there is more than one root in the interval (a, b) , we assume for simplicity that the root in this interval is unique. The method applies bisecting of subintervals of $[a, b]$ and, at each step, locating the half containing.

To begin, set $a_1 = a$ and $b_1 = b$ and let p_1 be the midpoint of $[a, b]$, that is.

$$p_1 = a_1 + \frac{b_1 - a_1}{2} = \frac{a_1 + b_1}{2}$$

If $f(p_1) = 0$, then $p = p_1$, and we are done.

If $f(p_1) \neq 0$, then $f(p_1)$ has the same sign as either $f(a_1)$ or $f(b_1)$

If $f(p_1)$ and $f(a_1)$ have the same sign, $p \in (p_1, b_1)$. Set $a_2 = p_1$ and $b_2 = b_1$.

If $f(p_1)$ and $f(a_1)$ have opposite signs, $p \in (a_1, p_1)$. Set $a_2 = a_1$ and $b_2 = p_1$

Then re-apply the process to the interval $[a_2, b_2]$.

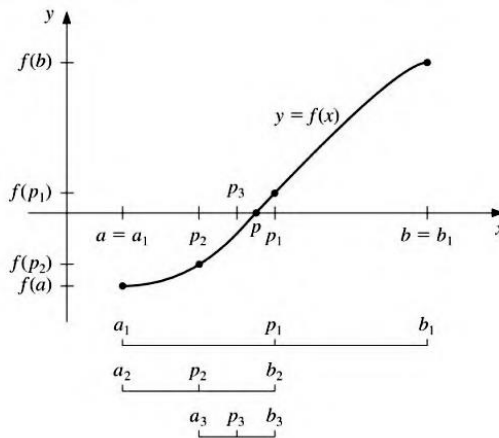


Figure 2.1

The procedure goes on.

Criterion for stopping the process.

We can select a tolerance $\epsilon > 0$ and generate $p_1, p_2, \dots, p_N$ until the following condition is met:

$$\frac{|p_N - p_{N-1}|}{|p_N|} < \epsilon, p_N \neq 0$$

Note that to start the Bisection method, an interval $[a, b]$ must be found with $f(a) \cdot f(b) < 0$. At each step the length of the interval known to contain a zero of f is reduced by a factor of 2; hence, it is advantageous to choose the initial interval $[a, b]$ as small as possible. For example, if $f(x) = 2x^3 - x^2 + x - 1$, we have both

$$f(-4) \cdot f(4) < 0 \text{ and } f(0) \cdot f(1) < 0$$

so, the Bisection Algorithm could be used on $[-4, 4]$ or on $[0, 1]$, Starting the Bisection Algorithm on $[0, 1]$ instead of $[-4, 4]$ will reduce by 3 the number of iterations required to achieve a specified accuracy.

Example 1.

Show that $f(x) = x^3 + 4x^2 - 10 = 0$ has a root in $[1, 2]$ and use the Bisection method to determine an approximation to the root that is accurate to at least within 10^{-4} .

Solution. Since $f(1) = -5$ and $f(2) = 14$, $f(x)$ has a root in $[1, 2]$.

Step 1: We have $p_1 = 1.5$ and $f(1.5) = 2.375$

Step 2: Since $f(p_1) > 0$, we should select the interval $[1, 1.5]$ for the second iteration.

Thus $p_2 = 1.25$ and $f(p_2) = -1.796875$

We can continue in this manner which is given in the Table 2.1

Table 2.1

N	a_n	b_n	p_n	$f(p_n)$
1	1.0	2.0	1.5	2.375
2	1.0	1.5	1.25	-1.79687
3	1.25	1.5	1.375	0.16211
4	1.25	1.375	1.3125	-0.84839
5	1.3125	1.375	1.34375	-0.35098
6	1.34375	1.375	1.359375	-0.09641
7	1.359375	1.375	1.3671875	0.03236
8	1.359375	1.3671875	1.36328125	-0.03215
9	1.36328125	1.3671875	1.365234375	0.000072
10	1.36328125	1.365234375	1.364257813	-0.01605
11	1.364254813	1.365234375	1.364746094	-0.00799
12	1.364746094	1.365234375	1.364990235	-0.00396
13	1.364990235	1.365234375	1.365112305	-0.00194

so, the approximation is correct to at least within 10^{-4} . The correct value of p to nine decimal places is $p = 1.365230013$

Example 2.

Use the bisection method to find p_3 for $f(x) = \sqrt{x} - \cos x = 0$ on $[0,1]$.

Solution. Since $f(0) = -1$ and $f(1) = 0.0001523$, $f(x)$ has a root between 0 and 1.

Step 1: $p_1 = 0.5$ and $f(p_1) = -0.292855$

Step 2: Since $f(p_1) < 0$, we should select the interval $[0.5,1]$ for the second iteration.

Thus $p_2 = 0.75$ and $f(p_2) = 0.134337$

Step 3: Since $f(p_2) > 0$ we select $[0.5,0.75]$ for the third iteration.

Thus $p_3 = 0.625$

Example 3.

Use bisection method to find solutions accurate to within 10^{-2} for $x^3 - 7x^2 + 14x - 6 = 0$ on $[0,1]$.

Solution: Since $f(0) = -6$ and $f(1) = 2$, $f(x)$ has a root between 0 and 1.

Computation is shown in the Table 2.2.

Table 2.2.

n	a_n	b_n	p_n	f(p_n)
1	0	1	0.5	-0.625
2	0.5	1	0.75	0.984375
3	0.5	0.75	0.625	0.259766
4	0.5	0.625	0.5625	-0.16187
5	0.5625	0.625	0.59375	0.054047
6	0.5625	0.59375	0.578125	-0.05262
7	0.578125	0.59375	0.5859375	0.001031

so, the approximation is within 10^{-2} . Thus, the approximate solution is 0.5859.

Example 4.

Find an approximation to $\sqrt{3}$ correct to within 10^{-4} using bisection method.

Solution. Let f be the function defined by $f(x) = x^2 - 3$. Thus, we know that $f(\sqrt{3}) = 0$.

Since $f(1) < 0$ and $f(\sqrt{2}) > 0$, we can consider the interval $[1, 2]$.

Computation is given in the Table 2.3

Table 2.3

n	a_n	b_n
1	1	2
2	1.5	2
3	1.5	1.75
4	1.625	1.75
5	1.6875	1.75
6	1.71875	1.75
7	1.71875	1.734375
8	1.7265625	1.734375
9	1.73046875	1.734375
10	1.73046875	1.732421875
11	1.7314453125	1.732421875
12	1.73193359375	1.732421875
13	1.73193359375	1.732177734375
14	1.73193359375	1.7320556640625
15	1.73199462890625	1.7320556640625

so, that the approximation is correct to within 10^{-4} and the approximation solution is 1.73205

Theorem 2.1 Suppose that $f \in C[a, b]$ and $f(a) \cdot f(b) < 0$. The Bisection method generates a sequence $\{p_n\}_{n=1}^{\infty}$ approximating a zero of f with

$$|p_n - p| \leq \frac{b-a}{2^n}, \text{ when } n \geq 1.$$

Proof. For each $n \geq 1$, we have

$$b_n - a_n = \frac{1}{2^{n-1}}(b - a) \text{ and } p \in (a_n, b_n).$$

Since $p_n = \frac{1}{2}(a_n + b_n)$ for all $n \geq 1$, it follows that

$$|p_n - p| \leq \frac{1}{2}(b_n - a_n) = \frac{b - a}{2^n}.$$

Example 5.

Determine the number of iterations necessary to solve $f(x) = x^3 + 4x^2 - 10 = 0$ with accuracy 10^{-3} using $a_1 = 1$ and $b_1 = 2$

Solution. We will use logarithms to find an integer N that satisfies

$$|p_N - p| \leq 2^{-N}(b - a) = 2^{-N} < 10^{-3}.$$

Logarithms to any base would suffice, but we will use base-10 logarithms because the tolerance is given as a power of 10. Since $2^{-N} < 10^{-3}$ implies that

$$\log_{10} 2^{-N} < \log_{10} 10^{-3} = -3. \text{ We have}$$

$$-N \log_{10} 2 < -3 \text{ and } N > \frac{3}{\log_{10} 2} \approx 9.96$$

Hence, 10 iterations are required for an approximation accurate to within 10^{-3} .

As a final remark, to determine which subinterval of $[a_n, b_n]$ contains a root of f , it is better to make use of the *signum function*, which is defined as

$$\operatorname{sgn}(x) = \begin{cases} -1, & \text{if } x < 0. \\ 0, & \text{if } x = 0. \\ 1, & \text{if } x > 0. \end{cases}$$

The test

$$\operatorname{sgn}(f(a_n)) \operatorname{sgn}(f(p_n)) < 0 \text{ instead of } f(a_n)f(p_n) < 0$$

gives the same result but avoids the possibility of overflow or underflow in the multiplication of $f(a_n)$ and $f(p_n)$.

Questions 2.1

1. Find a positive root of $xe^x = 1$, which lies between 0 and 1 by bisection method.
2. Solve $x^3 - 9x + 1 = 0$ for the root between 2 and 4 by bisection method.
3. Use the Bisection method to find solutions accurate to within 10^{-2} for $x^3 - 7x^2 + 14x - 6 = 0$ on each interval
 - a. $[1, 3.2]$
 - b. $[3.2, 4]$
4. Use the Bisection method to find solutions accurate to within 10^{-5} for the following problems.
 - a. $x - 2^{-x} = 0$ for $0 \leq x \leq 1$

- b. $e^x - x^2 + 3x - 2 = 0$ for $0 \leq x \leq 1$
- c. $2x \cos(2x) - (x + 1)^2 = 0$ for $-3 \leq x \leq -2$
- d. $x \cos x - 2x^2 + 3x - 1 = 0$ for $.2 \leq x \leq 1.3$
5. Find an approximation to $\sqrt[3]{25}$ correct to within 10^{-4} using the Bisection Algorithm. [Hint: Consider $f(x) = x^3 - 25$]
6. Let $f(x) = (x + 2)(x + 1)x(x - 1)^3(x - 2)$. To which zero of does the Bisection method converge when applied on the following intervals?
- a. $[-3, 2.5]$ b. $[-2.5, 3]$
- c. $[-1.75, 1.5]$ d. $[-1.5, 1.75]$

Answers

1. 0.5625
2. 2.9375
3. a. 3.002 b. 3.419
4. a. 0.641182 b. 0.257530
- c. -2.191307 d. 1.256622
5. 2.92401
6. a. 2 b. -2 c. -1 d. 1

2.2 Fixed-Point Iteration

A fixed point for a function is a number at which the value of the function does not change when the function is applied.

Definition 2.1 The number p is a **fixed point** for a given function g if $g(p) = p$.

In this section, we consider the problem of finding solutions to fixed-point problems and the connection between the fixed-point problems and the root-finding problems we wish to solve. Root-finding problems and fixed-point problems are equivalent classes in the following sense:

Given a root-finding problem $f(p) = 0$, we can define functions g with a fixed point at p in a number of ways, for example, as

$g(x) = x - f(x)$ or as $g(x) = x + 3f(x)$.

Conversely, if the function g has a fixed point at p , then the function defined by g has a zero at p .

Example 1 Determine any fixed points of the function $g(x) = x^2 - 2$.

Solution A fixed point p for g has the property that $p = g(p) = p^2 - 2$, which implies that $0 = p^2 - p - 2 = (p + 1)(p - 2)$. A fixed point for g occurs precisely when the graph of $y = g(x)$ intersects the graph of $y = x$, so g has two fixed points, one at $p = -1$ and the other at $p = 2$. These are shown in Figure 2.2.

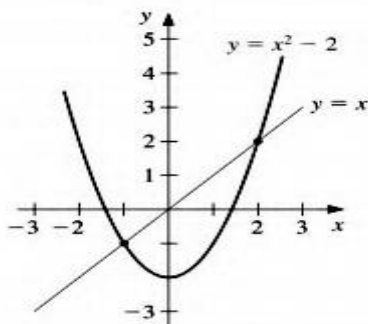


Figure 2.2

Theorem 2.2

(i) If $g \in C[a, b]$ and $g(x) \in [a, b]$ for all $x \in [a, b]$, then g has at least one fixed point in $[a, b]$.

(ii) If, in addition, $g'(x)$ exists on (a, b) and a positive constant $k < 1$ exists with $|g'(x)| \leq k$, for all $x \in (a, b)$,

Then there is exactly one fixed point in $[a, b]$. (See Figure 2.3.)

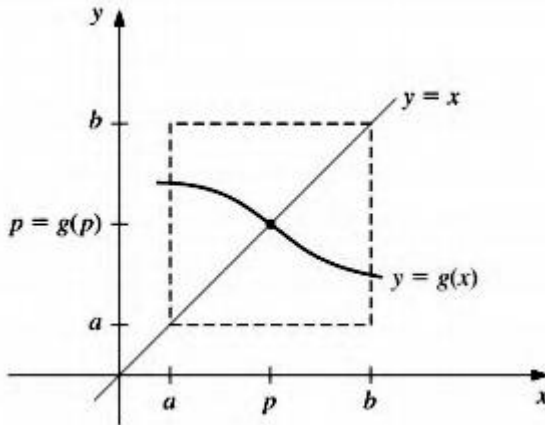


Figure 2.3

Proof

(i) If $g(a) = a$ or $g(b) = b$, then g has a fixed point at endpoint. If not, then $g(a) > a$ and $g(b) < b$. The function $h(x) = g(x) - x$ is continuous on $[a, b]$, with $h(a) = g(a) - a > 0$ and $h(b) = g(b) - b < 0$.

The intermediate value theorem implies that there exists $p \in (a, b)$ for which $h(p) = 0$. This number p is a fixed point for g because

$$0 = h(p) = g(p) - p \text{ implies that } g(p) = p$$

(ii) Suppose, in addition, that $|g'(x)| \leq k < 1$ and that p and q are both fixed points in $[a, b]$. If $p \neq q$, then the Mean Value Theorem implies that a number ξ exists between p and q and hence in $[a, b]$ with

$$\frac{g(p) - g(q)}{p - q} = g'(\xi)$$

Thus,

$$|p - q| = |g(p) - g(q)| = |g'(\xi)||p - q| \leq k|p - q| < |p - q|.$$

which is a contradiction. This contradiction must come from the only supposition, $p \neq q$. Hence, $p = q$, and the fixed point in $[a, b]$ is unique.

Example 2

Show that $g(x) = (x^2 - 1)/3$ has a unique fixed point on the interval $[-1, 1]$.

Solution The maximum and minimum values of $g(x)$ for x in $[-1, 1]$ must occur either when x is an endpoint of the interval or when the derivative is 0. Since $g'(x) = 2x/3$, the function g is continuous, and $g'(x)$ exists on $[-1, 1]$. The maximum and minimum values of $g(x)$ occur at $x = -1$, $x = 0$, or $x = 1$. But $g(-1) = 0$, $g(1) = 0$, and $g(0) = -1/3$, so an absolute

maximum for $g(x)$ on $[-1, 1]$ occurs at $x = -1$ and $x = 1$ and an absolute minimum at $x = 0$.

Moreover,

$$|g'(x)| = \left| \frac{2x}{3} \right| \leq \frac{2}{3}, \text{ for all } x \in (-1, 1).$$

So g satisfies all the hypotheses of Theorem 2.2 and has a unique fixed point in $[-1, 1]$.

For the function in Example 2, the unique fixed point p in the interval $[-1, 1]$ can be determined algebraically. If

$$p = g(p) = \frac{p^2 - 1}{3}, \text{ then } p^2 - 3p - 1 = 0$$

which, by the quadratic formula, implies, as shown on the left graph in Figure 2.4, that

$$p = \frac{1}{2}(3 - \sqrt{13})$$

Note that g also has a unique fixed point $p = \frac{1}{2}(3 + \sqrt{13})$ for the interval $[3, 4]$. However, $g(4) = 5$ and $g'(4) = \frac{8}{3} > 1$, so g does not satisfy the hypotheses of Theorem 2.2 on $[3, 4]$. This demonstrates that the hypotheses of Theorem 2.2 are sufficient to guarantee a unique fixed point but are not necessary. (See the graph on the right in Figure 2.4.)

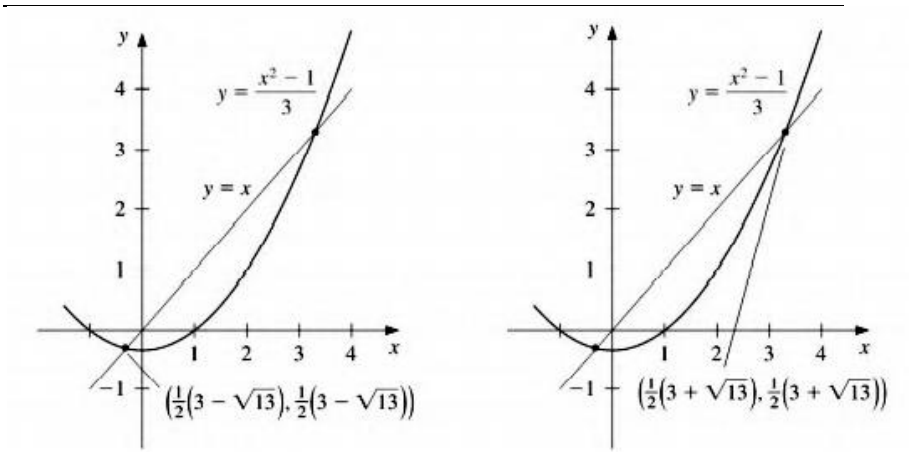


Figure 2.4

Example 3 Show that Theorem 2.2 does not ensure a unique fixed point of $g(x) = 3^{-x}$ on the interval $[0,1]$, even though a unique fixed point on this interval does exist.

Solution $g'(x) = -3^{-x} \ln 3 < 0$ on $[0,1]$, the function g is strictly decreasing on $[0,1]$, So

$$g(1) = \frac{1}{3} \leq g(x) \leq 1 = g(0), \text{ for } 0 \leq x \leq 1$$

Thus, for $x \in [0,1]$, we have $g(x) \in [0,1]$. The first part of Theorem 2.2 ensures that there is at least one fixed point in $[0,1]$,

However,

$$g'(0) = -\ln 3 = -1.098612289,$$

so $|g'(x)| \not\leq 1$ on $(0,1)$, and Theorem 2.2 cannot be used to determine uniqueness. But g is always decreasing, and it is clear from Figure 2.5 that the fixed point must be unique.

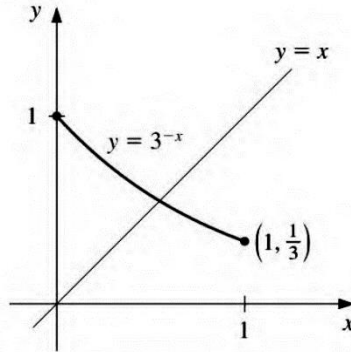


Figure 2.5

We cannot explicitly determine the fixed point in Example 3 because we have no way to solve for p in the equation $p = g(p) = 3^{-p}$. We can, however, determine approximations to this fixed point to any specified degree of accuracy. We will now consider how this can be done.

To approximate the fixed point of a function g , we choose an initial approximation p_0 and generate the sequence $\{p_n\}_{n=0}^{\infty}$ by letting $p_n = g(p_{n-1})$, for each $n \geq 1$. If the sequence converges to p and g is continuous, then

$$p = \lim_{n \rightarrow \infty} p_n = \lim_{n \rightarrow \infty} g(p_{n-1}) = g\left(\lim_{n \rightarrow \infty} p_{n-1}\right) = g(p),$$

and a solution to $x = g(x)$ is obtained. This technique is called fixed-point, or functional iteration. The procedure is illustrated in Figure 2.6

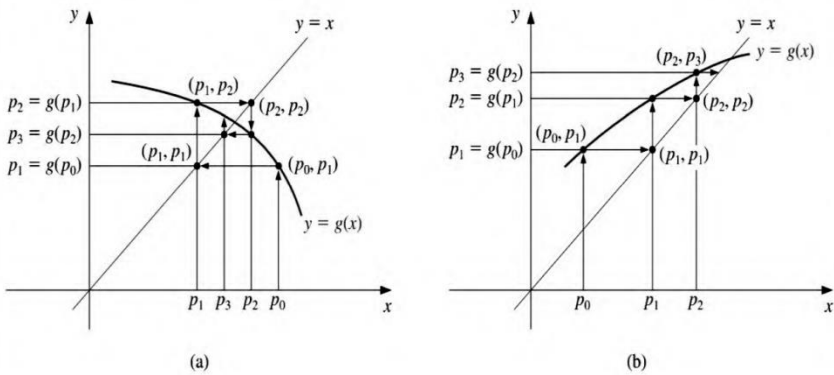


Figure 2.6

Illustration

The equation $x^3 + 4x^2 - 10 = 0$ has a unique root in $[1, 2]$. There are many ways to change the equation to the fixed-point form $x = g(x)$ using simple algebraic manipulation. For example, to obtain the function g described in part (c), we can manipulate the equation $x^3 + 4x^2 - 10 = 0$ as follows:

$$4x^2 = 10 - x^3, \text{ so } x^2 = \frac{1}{4}(10 - x^3), \text{ and } x = \pm \frac{1}{2}(10 - x^3)^{1/2}$$

To obtain a positive solution, $g_3(x)$ is chosen. It is not important for you to derive the functions shown here, but you should verify that the fixed point of each is actually a solution to the original equation, $x^3 + 4x^2 - 10 = 0$.

$$(a) \quad x = g_1(x) = x - x^3 - 4x^2 + 10$$

$$(b) \quad x = g_2(x) = \left(\frac{10}{x} - 4x\right)^{1/2}$$

$$(c) \ x = g_3(x) = \frac{1}{2}(10 - x^3)^{1/2}$$

$$(d) \ x = g_4(x) = \left(\frac{10}{4+x}\right)^{1/2}$$

$$(e) \ x = g_5(x) = x - \frac{x^3+4x^2-10}{3x^2+8x}$$

With $p_0 = 1.5$, Table 2.4 lists the results of the fixed-point iteration for all five choices of g .

Table 2.4

n	(a)	(b)	(c)	(d)	(e)
0	1.5	1.5	1.5	1.5	1.5
1	- 0.875	0.8165	1.286953768	1.348399725	1.373333333
2	6.732	2.9969	1.402540804	1.367376372	1.365262015
3	- 469.7	$(-8.65)^{1/2}$	1.345458374	1.364957015	1.365230014
4	1.03×10^8		1.375170253	1.365264748	1.365230013
5			1.360094193	1.365225594	
6			1.367846968	1.365230576	
7			1.363887004	1.365229942	
8			1.365916734	1.365230022	

9			1.364878217	1.365230012	
10			1.365410062	1.365230014	
15			1.36522368	1.365230013	
20			1.365230236		
25			1.365230006		
30			1.365230013		

The actual root is 1.365230013, as was noted in Example 1 of Section 2.1. Comparing the results to the Bisection Algorithm given in that example, it can be seen that excellent results have been obtained for choices (d) and (e) (the Bisection method requires 27 iterations for this accuracy). It is interesting to note that choice (a) was divergent and that choice (b) became undefined because it involved the square root of a negative number.

The following theorem and its corollary give us some clues concerning the paths we should pursue and, perhaps more important, some we should reject.

Theorem 2.3 (Fixed-Point Theorem)

Let $g \in C[a, b]$ be such that $g(x) \in [a, b]$, for all x in $[a, b]$. Suppose, in addition, that g' exists on (a, b) and that a constant $0 < k < 1$ exists with

$$|g'(x)| \leq k, \text{ for all } x \in (a, b)$$

Then, for any number p_0 in $[a, b]$, the sequence defined by

$$p_n = g(p_{n-1}), n \geq 1,$$

converge to the unique fixed point p in $[a, b]$.

Proof Theorem 2.2 implies that a unique point p exists in $[a, b]$ with $g(p) = p$. Since g maps $[a, b]$ into itself, the sequence $\{p_n\}_{n=0}^{\infty}$ is defined for all $n \geq 0$, and $p_n \in [a, b]$ for all n . Using the fact that $|g'(x)| \leq k$ and the Mean Value Theorem, we have, for each n ,

$$|p_n - p| = |g(p_{n-1}) - g(p)| = |g'(\xi_n)| |p_{n-1} - p| \leq k |p_{n-1} - p|$$

Where $\xi_n \in (a, b)$. Applying this inequality inductively gives

$$|p_n - p| \leq k |p_{n-1} - p| \leq k^2 |p_{n-2} - p| \leq \dots \leq k^n |p_0 - p|,$$

Since $0 < k < 1$, we have $\lim_{n \rightarrow \infty} k^n = 0$ and

$$\lim_{n \rightarrow \infty} |p_n - p| \leq \lim_{n \rightarrow \infty} k^n |p_0 - p| = 0.$$

Hence, $\{p_n\}_{n=0}^{\infty}$ converges to p .

Corollary

If g satisfies the hypotheses of Theorem 2.3, then bounds for the error involved in using p_n to approximate p are given by

$$|p_n - p| \leq k^n \max\{p_0 - a, b - p_0\}$$

and

$$|p_n - p| \leq \frac{k^n}{1 - k} |p_1 - p_0|, \quad \text{for all } n \geq 1$$

Proof Because $p \in [a, b]$, the first bound follows from inequality of the above theorem:

$$|p_n - p| \leq k^n |p_0 - p| \leq k^n \max\{p_0 - a, b - p_0\}$$

For $n \geq 1$, the procedure used in the proof of Theorem 2.3 implies that

$$\begin{aligned} |p_{n+1} - p| &= |g(p_n) - g(p_{n-1})| \leq k |p_n - p_{n-1}| \leq \dots \\ &\leq K^n |p_1 - p_0|. \end{aligned}$$

Thus, for $m > n \geq 1$

$$\begin{aligned} |p_m - p_n| &= |p_m - p_{m-1} - \dots + p_{n+1} - p_n| \\ &\leq |p_m - p_{m-1}| + |p_{m-1} - p_{m-2}| + \dots + |p_{n+1} - p_n| \\ &\leq k^{m-1} |p_1 - p_0| + k^{m-2} |p_1 - p_0| + \dots + k^n |p_1 - p_0| \\ &= k^n |p_1 - p_0| (1 + k + k^2 + \dots + k^{m-n-1}). \end{aligned}$$

By Theorem 2.2, $\lim_{m \rightarrow \infty} p_m = p$, so

$$\begin{aligned} |p - p_n| &= \lim_{m \rightarrow \infty} |p_m - p_n| \leq \lim_{m \rightarrow \infty} k^n |p_1 - p_0| \sum_{i=0}^{m-n-1} k^i \\ &\leq k^n |p_1 - p_0| \sum_{i=0}^{\infty} k^i \end{aligned}$$

But $\sum_{i=0}^{\infty} k^i$ is a geometric series with ratio k and $0 < k < 1$. This sequence converges to $1/(1 - k)$, which gives the second bound:

$$|p - p_n| \leq \frac{k^n}{1 - k} |p_1 - p_0|.$$

Questions 2.2

1. Find a real root of the equation $x^3 + x^2 - 1 = 0$ on the interval $[0, 1]$ with an accuracy of 10^{-4} .
2. Find the root of the equation $2x = \cos x + 3$ correct to three decimal places.
3. Find a positive root of $xe^x = 1$ between 0 and 1 by fixed-point iteration method.
4. Use a fixed-point iteration method to determine a solution accurate to within 10^{-2} for $x^4 - 3x^2 - 3 = 0$ on $[1, 2]$, Use $p_0 = 1$
5. Use a fixed-point iteration method to determine a solution accurate to within 10^{-2} for $x^3 + x - 1 = 0$ on $[1, 2]$, Use $p_0 = 1$
6. Use a fixed-point iteration method to find an approximation to $\sqrt{3}$ that is accurate to within 10^{-4} . Compare your result and the number of iterations required with the answer obtained in example 4 of Section 2.1
7. For each of the following equations, determine an interval $[a, b]$ on which fixed-point iteration will converge. Estimate the number of iterations necessary to obtain approximations accurate to within 10^{-5} and perform the calculations.
 - a. $x = \frac{2 - e^x + x^2}{3}$
 - b. $x = 0.5 (\sin x + \cos x)$

8. Let $f(x) = x^3 - 2x + 1$. To solve $f(x) = 0$, the following four fixed-point problems are proposed. Derive each fixed-point method and compute p_1, p_2, p_3, p_4 . Which methods seem to be appropriate?

a. $x = \frac{1}{2}(x^3 + 1), p_0 = \frac{1}{2}$

b. $x = \frac{2}{x} - \frac{1}{x^2}, p_0 = \frac{1}{2}$

c. $x = \sqrt{2 - \frac{1}{x}}, p_0 = \frac{1}{2}$

d. $x = -\sqrt[3]{1 - 2x}, p_0 = \frac{1}{2}$

Answers

1. 0.7549 2. 1.524

3. 0.5625 4. $g(x) = (3x^2 + 3)^{\frac{1}{4}}, p_0 = 1, p_6 = 1.94332$

5. $g(x) = \frac{1}{x^2 + 1}, p_0 = 1, P_6 = 0.701$

6. for $p_0 = 1$ and $g(x) = 0.5\left(x + \frac{3}{x}\right), \sqrt{3} \approx 1.73205(p_4)$

7. a. With $[0,1]$ and $p_0 = 0, p_9 = 0.257531$

b. With $[0,1]$ and $p_0 = 0, p_6 = 0.704812$

8. a. Solve for $2x$ then divide by 2. $p_1 = 0.5625, p_2 = 0.58898926, p_3 = 0.60216264, p_4 = 0.60917204$

b. Solve for x^3 , divide by x^2 . $p_1 = 0, p_2$ undefined

c. Solve for x^3 , divide by x , then take positive square root. $p_1 = 0$, p_2 undefined

d. Solve for x^3 , then take negative of the cubed root. $p_1 = 0$, $p_2 = -1$, $p_3 = -1.4422496$, $p_4 = -1.57197274$.

2.3 Newton's Method and Its Extension.

Newton-Raphson method

Newton's (or the Newton-Raphson) method is one of the most powerful and well-known numerical methods for solving a root-finding problem. There are many ways of introducing Newton's method.

Suppose that $f \in C^2[a, b]$. Let $p_0 \in [a, b]$ be an approximation to p such that $f'(p_0) \neq 0$ and $|p - p_0|$ is "small." Consider the first Taylor polynomial for $f(x)$ expanded about p_0 and evaluated at $x = p$:

$$f(p) = f(p_0) + (p - p_0)f'(p_0) + \frac{(p - p_0)^2}{2} f''(\xi(p)).$$

Where $\xi(p)$ lies between p and p_0 . Since $f(p) = 0$, this equation gives

$$0 = f(p_0) + (p - p_0)f'(p_0) + \frac{(p - p_0)^2}{2} f''(\xi(p)).$$

Newton's method is derived by assuming that since $|p - p_0|$ is small, the term involving $(p - p_0)^2$ is much smaller, so

$$0 \approx f(p_0) + (p - p_0)f'(p_0)$$

Solving for p gives

$$p \approx p_0 - \frac{f(p_0)}{f'(p_0)} \equiv p_1$$

This sets the stage for Newton's method, which starts with an initial approximation p_0 and generates the sequence $\{p_n\}_{n=0}^{\infty}$, by

$$p_n = p_{n-1} - \frac{f(p_{n-1})}{f'(p_{n-1})}, \text{ for } n \geq 1$$

Figure 2.7 illustrates how the approximations are obtained using successive tangents. Starting with the initial approximation p_0 , the approximation p_1 is the x -intercept of the tangent line to the graph of f at $(p_0, f(p_0))$. The approximation p_2 is the x -intercept of the tangent line to the graph of f at $(p_1, f(p_1))$ and so on.

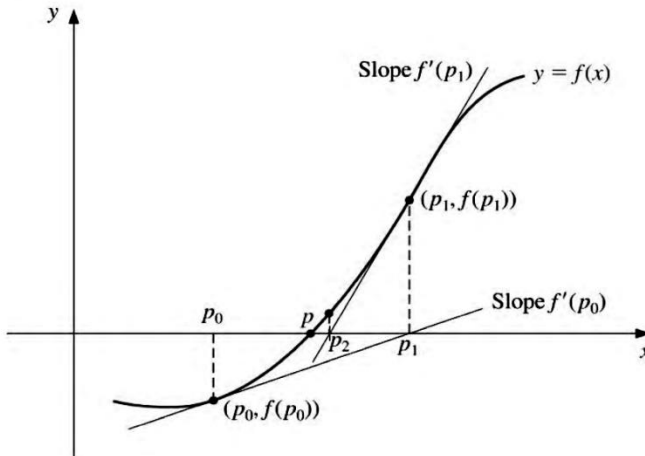


Figure 2.7

It is clear from the approximation formula that Newton's method cannot be continued if $f'(p_{n-1}) = 0$ for some n . In fact, we will

see that the method is most effective when f' is bounded away from zero near p .

Example 1 Consider the function $f(x) = \cos x - x = 0$. Approximate a root of f using (a) a fixed-point method, and (b) Newton's method.

Solution (a) A solution to this root-finding problem is also a solution to the fixed-point problem $x = \cos x$, and the graph in Figure 2.8 implies that a single fixed point p lies in $[0, \pi/2]$.

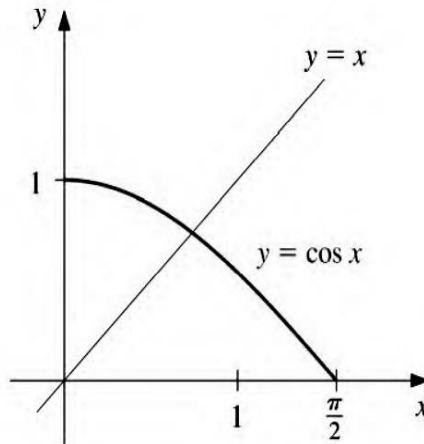


Figure 2.8

Table 2.5 shows the results of fixed-point iteration with $p_0 = \pi/4$. The best we could conclude from these results is that $p \approx 0.74$.

(b) To apply Newton's method to this problem, we need $f'(x) = \sin x - 1$. Starting again with $p_0 = \pi/4$, we have

$$\begin{aligned}
 p_1 &= p_0 - \frac{p_0}{f'(p_0)} \\
 &= \frac{\pi}{4} - \frac{\cos\left(\frac{\pi}{4}\right) - \pi/4}{-\sin\left(\frac{\pi}{4}\right) - 1} \\
 &= \frac{\pi}{4} - \frac{\sqrt{2}/2 - \pi/4}{\sqrt{2}/2 - 1} \\
 &= 0.7395361337 \\
 p_2 &= p_1 - \frac{\cos(p_1) - p_1}{-\sin(p_1) - 1} \\
 &= 0.7390851781
 \end{aligned}$$

We continue generating the sequence by

$$p_n = p_{n-1} - \frac{f(p_{n-1})}{f'(p_{n-1})} = p_{n-1} - \frac{\cos p_{n-1} - p_{n-1}}{-\sin p_{n-1} - 1}$$

This gives the approximations in Table 2.6. An excellent approximation is obtained with $n = 3$.

(Note that the variable in the trigonometric function is in radian measure, not degrees. This will always be the case unless specified otherwise.)

Table 2.5

n	p_n
0	0.7853981635
1	0.7071067810
2	0.7602445972
3	0.7246674808
4	0.7487198858
5	0.7325608446
6	0.7434642113
7	0.7361282565

Table 2.6

Newton's Method	
n	p_n
0	0.7853981635
1	0.7395361337
2	0.7390851781
3	0.7390851332
4	0.7390851332

Convergence Using Newton's Method

The Taylor series derivation of Newton's method at the beginning of the section points out the importance of an accurate initial approximation. The crucial assumption is that the term involving $(p - p_0)^2$ is, by comparison with $|p - p_0|$, so small that it can be deleted. This will clearly be false unless p_0 is a good approximation to p . If p_0 is not sufficiently close to the actual root, there is little reason to suspect that Newton's method will converge to the root. However, in some instances, even poor initial approximations will produce convergence.

The following convergence theorem for Newton's method illustrates the theoretical importance of the choice of p_0 .

Theorem 2.4 Let $f \in C^2[a, b]$. If $p \in (a, b)$ such that $f(p) = 0$ and $f'(p) \neq 0$, then there exists a $\delta > 0$ such that Newton's method generates a sequence $\{p_n\}_{n=1}^{\infty}$ converging to p for any initial approximation $p_0 \in [p - \delta, p + \delta]$.

Proof The proof is based on analyzing Newton's method as the functional iteration scheme $p_n = g(p_{n-1})$, for $n \geq 1$, with

$$g(x) = x - \frac{f(x)}{f'(x)}$$

Let k be in $(0, 1)$. We first find an interval $[p - \delta, p + \delta]$ that g maps into itself and for which $|g'(x)| \leq k$, for all $x \in [p - \delta, p + \delta]$.

Since f' is continuous and $f'(p) \neq 0$, there exists a $\delta_1 > 0$, such that $f'(p) \neq 0$ for $x \in [p - \delta_1, p + \delta_1] \subseteq [a, b]$. Thus, g is defined and continuous on $[p - \delta_1, p + \delta_1]$ Also

$$g'(x) = 1 - \frac{f'(x)f'(x) - f(x)f''(x)}{[f'(x)]^2} = \frac{f(x)f''(x)}{[f'(x)]^2},$$

for $x \in [p - \delta_1, p + \delta_1]$, and, since $f \in C^2[a, b]$, we have $g \in C^1[p - \delta_1, p + \delta_1]$. By assumption, $f(p) = 0$, so

$$g'(p) = \frac{f(p)f''(p)}{[f'(p)]^2} = 0$$

Since g' is continuous and $0 < k < 1$, there exists a δ , with $0 < \delta < \delta_1$, for which

$$|g'(x)| \leq k, \text{ for all } x \in [p - \delta, p + \delta].$$

It remains to show that g maps $[p - \delta, p + \delta]$ into $[p - \delta, p + \delta]$. If $x \in [p - \delta, p + \delta]$, the Mean Value Theorem implies that for some number between x and p , $|g(x) - g(p)| = |g'(\xi)||x - p|$. So,

$$|g(x) - p| = |g(x) - g(p)| = |g'(\xi)||x - p| \leq |x - p| < |x - p|$$

Since $x \in [p - \delta, p + \delta]$ it follows that $|g(x) - p| < \delta$ and that $|g(x) - p| < \delta$. Hence, g maps $[p - \delta, p + \delta]$ into $[p - \delta, p + \delta]$.

All the hypotheses of the Fixed-Point Theorem 2.3 are now satisfied, so the sequence defined by

$$p_n = g(p_{n-1}) = p_{n-1} - \frac{f(p_{n-1})}{f'(p_{n-1})}, \text{ for } n \geq 1,$$

converges to p for any $p_0 \in [p - \delta, p + \delta]$.

Theorem 2.4 states that, under reasonable assumptions, Newton's method converges, provided that a sufficiently accurate initial approximation is chosen. It also implies that the constant k that bounds the derivative of g and, consequently, indicates the speed of convergence of the method decreases to 0 as the procedure continues. This result is important for the theory of Newton's method, but it is seldom applied in practice because it does not tell us how to determine δ .

In a practical application, an initial approximation is selected, and successive approximations are generated by Newton's method. Either these will generally converge quickly to the root or it will be clear that convergence is unlikely.

The Secant Method

Newton's method is an extremely powerful technique, but it has a major weakness; the need to know the value of the derivative of f at each approximation. Frequently, $f'(x)$ is far more difficult and needs more arithmetic operations to calculate than $f(x)$.

To circumvent the problem of the derivative evaluation in Newton's method, we introduce a slight variation. By definition,

$$f'(p_{n-1}) = \lim_{x \rightarrow p_{n-1}} \frac{f(x) - f(p_{n-1})}{x - p_{n-1}}$$

If p_{n-2} is close to p_{n-1} , then

$$f'(p_{n-1}) \approx \frac{f(p_{n-2}) - f(p_{n-1})}{p_{n-2} - p_{n-1}} = \frac{f(p_{n-1}) - f(p_{n-2})}{p_{n-1} - p_{n-2}}$$

Using this approximation for $f'(p_{n-1})$ in Newton's formula gives

$$p_n = p_{n-1} - \frac{f(p_{n-1})(p_{n-1} - p_{n-2})}{f(p_{n-1}) - f(p_{n-2})}$$

This technique is called the Secant method (See Figure 2.9.)

Starting with the two initial approximations p_0 and p_1 , the approximation p_2 is the x -intercept of the line joining $(p_0, f(p_0))$ and $(p_1, f(p_1))$. The approximation p_3 is the

x -intercept of the line joining $(p_1, f(p_1))$ and $(p_2, f(p_2))$ and so on. Note that only one function evaluation is needed per step for the Secant method after p_0 has been determined.

In contrast, each step of Newton's method requires an evaluation of both the function and its derivative.

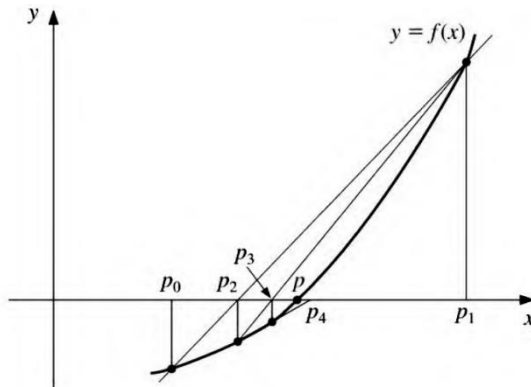


Figure 2.9

Example 2 Use the Secant method to find a solution to $x = \cos x$ and compare the approximations with those given in Example 1, which applied Newton's method.

Solution In Example 1, we compared fixed-point iteration and Newton's method starting with the initial approximation $p_0 = \frac{\pi}{4}$. For the Secant method, we need two initial approximations. Suppose we use $p_0 = 0.5$ and $p_1 = \pi/4$:

$$p_2 = p_1 - \frac{(p_1 - p_0)(\cos p_1 - p_1)}{(\cos p_1 - p_1) - (\cos p_2 - p_2)}$$

$$= \frac{\pi}{4} - \frac{\left(\frac{\pi}{4} - 0.5\right)\left(\cos\left(\frac{\pi}{4}\right) - \frac{\pi}{4}\right)}{\left(\cos\left(\frac{\pi}{4}\right) - \pi/4\right) - (\cos 0.5 - 0.5)}$$

Succeeding approximations are generated by the formula

$$p_n = p_{n-1} - \frac{(p_{n-1} - p_{n-2})(\cos p_{n-1} - p_{n-1})}{(\cos p_{n-1} - p_{n-1}) - (\cos p_{n-2} - p_{n-2})} \text{ for } n \geq 2$$

These give the results in Table 2.7. We note that although the formula for p_2 seems to indicate a repeated computation, once $f(p_0)$ and $f(p_1)$ are computed, they are not computed again.

Table 2.7

Secant Method		Newton's Method	
n	p_n	n	p_n
0	0.5	0	0.7853981635
1	0.7853981635	1	0.7395361337
2	0.7363841388	2	0.7390851781

3	0.7390581392	3	0.7390851332
4	0.7390851493	4	0.7390851332
5	0.7390851332		

Comparing the results in Table 2.7 from the Secant method and Newton's method, we see that the Secant method approximation p_5 is accurate to the 10th decimal place, whereas Newton's method obtained this accuracy by p_3 . For this example, the convergence of the Secant method is much faster than functional iteration but slightly slower than Newton's method. This is generally the case.

Newton's method or the Secant method is often used to refine an answer obtained by another technique, such as the Bisection method, since these methods require good first approximations but generally give rapid convergence.

The Method of False Position

Each successive pair of approximations in the Bisection method brackets a root p of the equation; that is, for each positive integer n , a root lies between a_n and b_n . This implies that, for each n , the Bisection method iterations satisfy

$$|p_n - p| < \frac{1}{2} |a_n - b_n|$$

which provides an easily calculated error bound for the approximations.

Root bracketing is not guaranteed for either Newton's method or the Secant method. In Example 1, Newton's method was applied

to $f(x) = \cos x - x$, and an approximate root was found to be 0.7390851332. Table 2.6 shows that this root is not bracketed by either p_0 and p_1 or p_1 and p_2 . The Secant method approximations for this problem are also given in Table 2.7. In this case, the initial approximations p_0 and p_1 bracket the root, but the pair of approximations p_3 , and p_4 fail to do so.

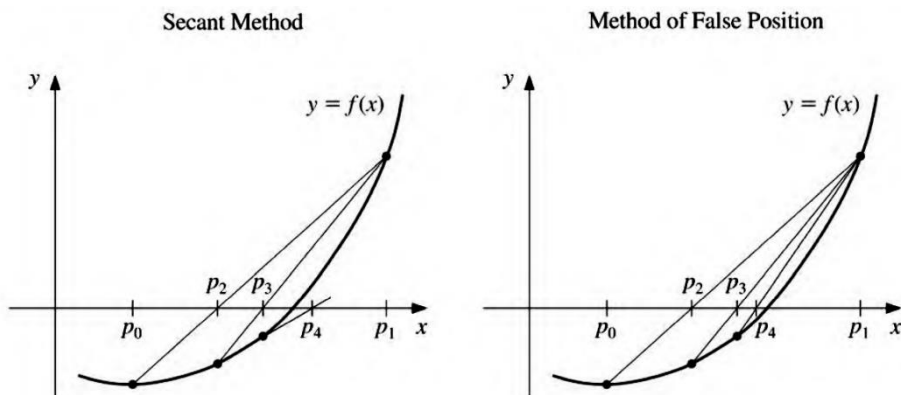
The method of False Position (also called Regula Falsi) generates approximations in the same manner as the Secant method, but it includes a test to ensure that the root is always bracketed between successive iterations.

First, choose initial approximations p_0 and p_1 with $f(p_0) \cdot f(p_1) < 0$. The approximation p_2 is chosen in the same manner as in the Secant method as the x -intercept of the line joining $(p_0, f(p_0))$ and $(p_1, f(p_1))$. To decide which secant line to use to compute p_3 , consider $f(p_2) \cdot f(p_1)$ or, more correctly, $\text{sgn } f(p_2) \cdot \text{sgn } f(p_1)$.

- If $\text{sgn } f(p_2) \cdot \text{sgn } f(p_1) < 0$, then p_1 , and p_2 bracket a root. Choose p_3 as the x -intercept of the line joining $(p_1, f(p_1))$ and $(p_2, f(p_2))$.
- If not, choose p_3 as the x -intercept of the line joining $(p_0, f(p_0))$ and $(p_2, f(p_2))$ and then interchange the indices on p_0 and p_1 .

In a similar manner, once p_3 , is found, the sign of $f(p_3) \cdot f(p_2)$ determines whether we use p_2 and p_3 , or p_3 and p_1 to compute p_4 . In the latter case a relabeling of p_2 and p_1 is performed. The relabeling ensures that the root is bracketed between successive iterations.

Figure 2.10 shows how the iterations can differ from those of the Secant method. In this illustration, the first three approximations are the same, but the fourth approximations differ.



Example 3

Use the method of False Position to find a solution to $x = \cos x$ and compare the approximations with those given in Example 1, which applied fixed-point iteration and Newton's method, and to those found in Example 2, which applied the Secant method.

Solution To make a reasonable comparison we will use the same initial approximations as in the Secant method; that is, $p_0 = 0.5$ and $p_1 = \pi/4$. Table 2.8 shows the results of the method of False Position applied to $f(x) = \cos x - x$ together with those we obtained using the Secant and Newton's methods. Notice that the False Position and Secant approximations agree through p_3 and that the method of False Position requires an additional iteration to obtain the same accuracy as the Secant method.

Table 2.8

	False Position	Secant	Newton
n	p_n	p_n	p_n
0	0.5	0.5	0.785398164
1	0.785398164	0.785398164	0.739536134
2	0.736384139	0.736384139	0.739085178
3	0.739058139	0.739058139	0.739085133
4	0.739084864	0.739085149	0.739085133
5	0.739085131	0.739085133	
6	0.739085133		

Example 4

Using Newton's method solve $f(x) = x^3 + x - 1 = 0$ correct to 6 decimal places. (Let $p_0 = 1$)

Solution

$$f(x) = x^3 + x - 1$$

$$f'(x) = 3x^2 + 1$$

Substituting in $p_n = p_{n-1} - \frac{f(p_{n-1})}{f'(p_{n-1})}$, we get

$$p_n = p_{n-1} - \frac{p_{n-1}^3 + p_{n-1} - 1}{3p_{n-1}^2 + 1} = \frac{2p_{n-1}^3 + 1}{3p_{n-1}^2 + 1}, \quad n = 1, 2, \dots$$

Then $p_1 = \frac{3}{4} = 0.75$; $p_2 = 0.686047$

$p_3 = 0.682340$; $p_4 = 0.682328$ and so on.

An approximate solution of $f(x)$ is 0.682328.

Example 5

Use Newton's method to solve $x = 2 \sin x$ for p_3 .

Solution $f(x) = x - 2 \sin x$, $f'(x) = 1 - 2 \cos x$.

We have $p_n = p_{n-1} - \frac{f(p_{n-1})}{f'(p_{n-1})} = p_{n-1} - \frac{p_{n-1} - 2 \sin p_{n-1}}{1 - 2 \cos p_{n-1}}$

That gives $p_n = \frac{2(\sin p_{n-1} - p_{n-1} \cos p_{n-1})}{1 - 2 \cos p_{n-1}}$, $n = 1, 2, \dots$

Take $p_0 = 2$.

Then $p_1 = \frac{3.483}{1.832} = 1.901$,

$p_2 = \frac{3.125}{1.648} = 1.896$

$p_3 = \frac{3.107}{1.639} = 1.896$

Example 6

Let $f(x) = x^3 - 2x^2 - 5 = 0$; $[1,4]$. Find solutions accurate to within 10^{-4} using,

- a.** Newton's Method **b.** Secant Method **c.** Method of False Position

Solution

a.
$$p_n = p_{n-1} - \frac{p_{n-1}^3 - 2p_{n-1}^2 - 5}{3p_{n-1}^2 - 4p_{n-1}} = \frac{2p_{n-1}^3 - 2p_{n-1}^2 + 5}{3p_{n-1}^2 - 4p_{n-1}}, \quad n = 1, 2, \dots$$

Taking $p_0 = 2$, we obtain

$$p_1 = 3.25, p_2 = 2.811037, p_3 = 2.69799$$

$$p_4 = 2.690677, p_5 = 2.690647, p_6 = 2.690647$$

Thus, solution of $f(x)$ accurate to within 10^{-4} is 2.69065.

b. Using Secant Method.

$$p_n = p_{n-1} - \frac{f(p_{n-1})(p_{n-1} - p_{n-2})}{f(p_{n-1}) - f(p_{n-2})}$$

$$\text{i.e., } p_n = \frac{p_{n-2}f(p_{n-1}) - p_{n-1}f(p_{n-2})}{f(p_{n-1}) - f(p_{n-2})}$$

Take $p_0 = 1$, $p_1 = 4$, we obtain

$$p_2 = 1.545455, p_3 = 1.996935, p_4 = 4.10506,$$

$$p_5 = 2.295136, p_6 = 2.47898, \dots \dots \dots,$$

$p_{11} = 2.69065$ which is the solution of $f(x)$ accurate to within 10^{-4} .

c. Take $p_0 = 1, p_1 = 4$. Then,

$$f(p_0) = -6, f(p_1) = 27.$$

Hence $f(p_0)f(p_1) < 0$, so there is a root between p_0 and p_1 .

Set $f(p_0) = q_0$ and $f(p_1) = q_1$.

$$\text{Take } p = p_1 - \frac{q_1(p_1 - p_0)}{q_1 - q_0}$$

$$\text{i.e., } p = \frac{p_0 q_1 - p_1 q_0}{q_1 - q_0}$$

Hence, $p_2 = 1.545455$

Now $f(p_2) < 0$. So, root lies between p_2 and p_1 . Renaming $p_2 = p_0$ and substituting in the above formula we get

$p_3 = 1.996935$ and $f(p_4) < 0$. Continuing like this, we obtain the required root $p_{16} = 2.69060$

Example 7

Use Secant Method to find solutions accurate to within 10^{-4} for the equation $f(x) = x^3 + 3x^2 - 1$ in $[-3, -2]$.

Solution We know $p_n = \frac{p_{n-2}f(p_{n-1}) - p_{n-1}f(p_{n-2})}{f(p_{n-1}) - f(p_{n-2})}$

Is the iterative technique for secant method.

Take $p_0 = -3, p_1 = -2$, we will get the following iterations.

$$p_2 = -2.75, p_3 = -3.06667, p_4 = -2.86202,$$

$$p_5 = -2.87719, p_6 = -2.87941, p_7 = -2.87939$$

So that the solution accurate to within 10^{-4} is -2.87939

Example 8

Using the method of false position, Find a real root of the equation $x^3 + x - 1$ in $[0,1]$.

Solution Take $p_0 = 0, p_1 = 1$.

Since $f(p_0)f(p_1) < 1$, there is a root in $[p_0, p_1]$

The process of the method of false position is given in the table below:

First, set $q_0 = f(p_0), q_1 = f(p_1)$

$$\text{And } p = \frac{p_0 q_1 - p_1 q_0}{q_1 - q_0}$$

n	p_0	p_1	q_0	q_1	p	$f(p)$
1	0	1	-1	1	0.5	-0.375
2	0.5	1	-0.375	1	0.6364	-0.1058
3	0.6364	1	-0.1058	1	0.6712	-0.0264
4	0.6712	1	-0.0264	1	0.6796	-0.0063

We can accept 0.6796 as an approximate solution of $f(x) = 0$.

Questions 2.3

1. Let $f(x) = x^2 - 6$ and $p_0 = 1$. Use Newton's method to find p_2 .
2. Let $f(x) = x^2 - 6$. With $p_0 = 3$ and $p_1 = 2$, find p_3 .
 - a. Use the Secant method.
 - b. Use the method of False Position.
 - c. Which of part (a) or (b) is closer to $\sqrt{6}$?
3. Use all the three methods in this section to find solution to within 10^{-4} for the following problems.
 - a. $x - \cos x = 0, \left[0, \frac{\pi}{2}\right]$
 - b. $x - 0.8 - 0.2 \sin x = 0, \left[0, \frac{\pi}{2}\right]$
 - c. $3x - e^x = 0, [1, 2]$
 - d. $2x + 3 \cos x - e^x = 0, [1, 2]$
 - e. $-x^3 - \cos x = 0, [-1, 0]$

Answers

1. 2.60714
2. a. 2.45454 b. 2.44444 c. (b) is better
3. a. 0.73909 b. 0.96434
- c. 1.51213455 d. 0.976773017

2.4 Error Analysis for Iterative Methods

Order of Convergence

Definition 2.2 Suppose $\{p_n\}_{n=0}^{\infty}$ is a sequence that converges to p , with $p_n \neq p$ for all n . If positive constants λ and α exist with

$$\lim_{n \rightarrow \infty} \frac{|p_{n+1} - p|}{|p_n - p|^\alpha} = \lambda$$

then $\{p_n\}_{n=0}^{\infty}$ converges to p of order α , with asymptotic error constant λ .

An iterative technique of the form $p_n = g(p_{n-1})$ is said to be of order α if the sequence $\{p_n\}_{n=0}^{\infty}$ converges to the solution $p = g(p)$ of order α .

In general, a sequence with a high order of convergence converges more rapidly than a sequence with a lower order. The asymptotic constant affects the speed of convergence but not to the extent of the order. Two cases of order are given special attention:

- (i) If $\alpha = 1$ (and $\lambda < 1$), the sequence is **linearly convergent**.
- (ii) If $\alpha = 2$, the sequence is **quadratically convergent**.

Illustration

Suppose that $\{p_n\}_{n=0}^{\infty}$ is linearly convergent to 0 with

$$\lim_{n \rightarrow \infty} \frac{|p_{n+1}|}{|p_n|} = 0.5$$

and that $\{\tilde{p}_n\}_{n=0}^{\infty}$ is quadratically convergent to 0 with the same asymptotic error constant,

$$\lim_{n \rightarrow \infty} \frac{|\tilde{p}_{n+1}|}{|\tilde{p}_n|} = 0.5$$

For simplicity, we assume that for each n , we have

$$\frac{|p_{n+1}|}{|p_n|} \approx 0.5 \text{ and } \frac{|\tilde{p}_{n+1}|}{|\tilde{p}_n|} \approx 0.5$$

For the linearly convergent scheme, this means that

$$\begin{aligned} |p_n - 0| = |p_n| &\approx 0.5 |p_{n-1}| \approx (0.5)^2 |p_{n-2}| \approx \dots \\ &\approx (0.5)^n |p_0| \end{aligned}$$

whereas the quadratically convergent procedure has

$$\begin{aligned} |\tilde{p}_n - 0| = |\tilde{p}_n| &\approx 0.5 |\tilde{p}_{n-1}|^2 \approx (0.5)[0.5|\tilde{p}_{n-2}|^2]^2 \\ &= (0.5)^3 |\tilde{p}_{n-2}|^4 \\ &\approx (0.5)^3 [0.5|\tilde{p}_{n-3}|^2]^4 = (0.5)^7 |\tilde{p}_{n-3}|^8 \\ &\approx \dots \approx (0.5)^{2n-1} |\tilde{p}_0|^{2^n} \end{aligned}$$

Table 2.9 illustrates the relative speed of convergence of the sequences to 0 if $|p_0| = |\tilde{p}_0| = 1$

Table 2.9

n	Linear Convergence sequence $\{p_n\}_{n=0}^\alpha$	Quadratic Convergence Sequence $\{p_n\}_{n=0}^\alpha$ $(0.5)2^n - 1$
1	5.0000×10^{-1}	5.0000×10^{-1}
2	2.5000×10^{-1}	1.2500×10^{-1}
3	1.2500×10^{-1}	7.8125×10^{-3}
4	6.2500×10^{-2}	3.0518×10^{-5}
5	3.1250×10^{-2}	4.6566×10^{-10}
6	1.5625×10^{-2}	1.0842×10^{-19}
7	7.8125×10^{-3}	5.8775×10^{-39}

The quadratically convergent sequence is within 10^{-38} of 0 by the seventh term. At least 126 terms are needed to ensure this accuracy for the linearly convergent sequence.

Quadratically convergent sequences are expected to converge much quicker than those that converge only linearly, but the next result implies that an arbitrary fixed-point technique that generates a convergent sequence does so only linearly.

Theorem 2.5 Let $g \in C[a, b]$ be such that $g(x) \in [a, b]$, for all $x \in [a, b]$. Suppose, in addition, that g' is continuous on (a, b) and that a positive constant $k < 1$ exists with $|g'(x)| \leq k$, for all $x \in (a, b)$.

If $g'(p) \neq 0$, then for any number $p_0 \neq p$ in $[a, b]$, the sequence $p_n = g(p_{n-1})$, for $n \geq 1$, converges only linearly to the unique fixed point p in $[a, b]$.

Proof: We know from the Fixed-Point Theorem 2.3 in Section 2.2 that the sequence converges to p . Since g' exists on (a, b) , we can apply the Mean Value Theorem to g to show that for any n ,

$$p_{n+1} - p = g(p_n) - g(p) = g'(\xi_n)(p_n - p),$$

Where ξ_n is between p_n and p . Since $\{p_n\}_{n=0}^{\infty}$ converges to p , we also have $\{\xi_n\}_{n=0}^{\infty}$ converging to p . Since g' is continuous on (a, b) , we have

$$\lim_{n \rightarrow \infty} g'(\xi_n) = g'(p),$$

Thus,

$$\lim_{n \rightarrow \infty} \frac{p_{n+1} - p}{p_n - p} = \lim_{n \rightarrow \infty} g'(\xi_n) = g'(p) \text{ and } \lim_{n \rightarrow \infty} \frac{p_{n+1} - p}{p_n - p} = |g'(p)|.$$

Hence, if $g'(p) \neq 0$, fixed-point iteration exhibits linear convergence with asymptotic error constant $|g'(p)|$

Theorem 2.5 implies that higher-order convergence for fixed-point methods of the form $g(p) = p$ can occur only when $g'(p) = 0$.

Theorem 2.6 Let p be a solution of the equation $x = g(x)$. Suppose that $g'(p) = 0$ and g'' is continuous with $|g''(x)| < M$ on an open interval I containing p . Then there exists a $\delta > 0$ such

that, for $p_0 \in [p - \delta, p + \delta]$, the sequence defined by $p_n = g(p_{n-1})$, when $n \geq 1$, converges at least quadratically to p . Moreover, for sufficiently large values of n ,

$$|p_{n+1} - p| < \frac{M}{2} |p_n - p|^2.$$

Proof Choose k in $(0, 1)$ and $\delta > 0$ such that on the interval $[p - \delta, p + \delta]$, contained in I , we have $|g'(x)| \leq k$ and g'' continuous. Since $|g'(x)| \leq k < 1$, the argument used in the proof of Theorem 2.4 in Section 2.3 shows that the terms of the sequence $\{p_n\}_{n=0}^{\infty}$ are contained in $[p - \delta, p + \delta]$. Expanding $g(x)$ in a linear Taylor polynomial for $x \in [p - \delta, p + \delta]$ gives

$$g(x) = g(p) + g'(p)(x - p) + \frac{g''(\xi)}{2} (x - p)^2,$$

where ξ lies between x and p . The hypotheses $g(p) = p$ and $g'(p) = 0$ imply that

$$g(x) = p + \frac{g''(\xi)}{2} (x - p)^2$$

In particular, when $x = p_n$,

$$p_{n+1} = g(p_n) = p + \frac{g''(\xi_n)}{2} (p_n - p)^2,$$

with ξ_n between p_n and p . Thus,

$$p_{n+1} - p = \frac{g''(\xi_n)}{2} (p_n - p)^2.$$

Since $|g'(x)| \leq k < 1$ on $[p - \delta, p + \delta]$ and g maps $[p - \delta, p + \delta]$ into itself, it follows from the Fixed-Point Theorem that

$\{p_n\}_{n=0}^{\infty}$ converges to p . But ξ_n is between p and p_n for each n , so $\{\xi_n\}_{n=0}^{\infty}$ converges to p , and

$$\lim_{n \rightarrow \infty} \frac{|p_{n+1} - p|}{|p_n - p|^2} = \frac{|g''(p)|}{2}$$

This result implies that the sequence $\{p_n\}_{n=0}^{\infty}$ quadratically convergent if $g''(p) \neq 0$ and of higher-order convergence if $g''(p) = 0$.

Because g'' is continuous and strictly bounded by M on the interval $[p - \delta, p + \delta]$, this also implies that, for sufficiently large values of n ,

$$|p_{n+1} - p| < \frac{M}{2} |p_n - p|^2.$$

Theorems 2.5 and 2.6 tell us that our search for quadratically convergent fixed-point methods should point in the direction of functions whose derivatives are zero at the fixed point. That is,

- For a fixed-point method to converge quadratically, we need to have both $g'(p) = p$, and $g'(p) = 0$.

The easiest way to construct a fixed-point problem associated with a root-finding problem $f(x) = 0$ is to add or subtract a multiple of $f(x)$ from x . Consider the sequence

$$p_n = g(p_{n-1}), \quad \text{for } n \geq 1$$

for g in the form

$$g(x) = x - \phi(x)f(x)$$

where ϕ is a differentiable function that will be chosen later.

For the iterative procedure derived from g to be quadratically convergent, we need to have $g'(p) = 0$ when $f(p) = 0$. Because

$$g'(x) = 1 - \phi(x)f(x) - f'(x)\phi(x)$$

And $f(p) = 0$, we have

$$\begin{aligned} g'(p) &= 1 - \phi'(p)f(p) - f'(p)\phi(p) \\ &= 1 - \phi'(p) \cdot 0 - f'(p)\phi(p) = 1 - f'(p)\phi(p), \end{aligned}$$

And $g'(p) = 0$ if and only if $\phi(p) = 1/f'(p)$

If we let $\phi(x) = 1/f'(x)$, then we will ensure that $\phi(p) = 1/f'(p)$ and produce the quadratically convergent procedure

$$p_n = g(p_{n-1}) = p_{n-1} - \frac{f(p_{n-1})}{f'(p_{n-1})}$$

This, of course, is simply Newton's method. Hence,

- If $f(p) = 0$ and $f'(p) \neq 0$, then for starting values sufficiently close to p , Newton's method will converge at least quadratically.

Multiple Roots

In the preceding discussion, the restriction was made that $f'(p) \neq 0$, where p is the solution to $f(x) = 0$. In particular, Newton's method and the Secant method will generally give problems if $f'(p) = 0$ when $f(p) = 0$.

Definition 2.3

A solution p of $f(x) = 0$ is a zero of multiplicity m of f if for $x \neq p$, we can write $f(x) = (x - p)^m q(x)$, where $\lim_{x \rightarrow p} q(x) \neq 0$.

Theorem 2.7

The function $f \in C^1[a, b]$ has a simple zero at p in (a, b) if and only if $f(p) = 0$ but $f'(p) \neq 0$.

Proof If f has a simple zero at p , then $f(p) = 0$ and $f(x) = (x - p)q(x)$, where $\lim_{x \rightarrow p} q(x) \neq 0$. Since $f \in C^1[a, b]$,

$$\begin{aligned} f'(p) &= \lim_{x \rightarrow p} f'(x) \\ &= \lim_{x \rightarrow p} [q(x) + (x - p)q'(x)] = \lim_{x \rightarrow p} q(x) \neq 0. \end{aligned}$$

Conversely, if $f(p) = 0$ but $f'(p) \neq 0$, expand f in a zeroth Taylor polynomial about p .

Then

$$f(x) = f(p) + f'(\xi(x))(x - p) = (x - p)f'(\xi(x)),$$

where $\xi(x)$ is between x and p . Since $f \in C^1[a, b]$,

$$\lim_{x \rightarrow p} f'(\xi(x)) = f' \left(\lim_{x \rightarrow p} \xi(x) \right) = f'(p) \neq 0.$$

Letting $q = f' \circ \xi$ gives $f(x) = (x - p)q(x)$, where $\lim_{x \rightarrow p} q(x) \neq 0$. Thus, f has a simple zero at p .

Theorem 2.8

The function $f \in C^n[a, b]$ has a zero of multiplicity m at p in (a, b) if and only if

$$0 = f(p) = f'(p) = f''(p) = \dots f^{(m-1)}(p), \\ \text{but } f^{(m)}(p) \neq 0.$$

The result in Theorem 2.8 implies that an interval about p exists where Newton's method converges quadratically to p for any initial approximation $p_0 = p$, provided that p is a simple zero. The following example shows that quadratic convergence might not occur if the zero is not simple.

Example 1

Let $f(x) = e^x - x - 1$. (a) Show that f has a zero of multiplicity 2 at $x = 0$. (b) Show that Newton's method with $p_0 = 1$ converges to this zero but not quadratically.

Solution (a) We have

$$f(x) = e^x - x - 1, \quad f'(x) = e^x - 1, \quad \text{and } f''(x) = e^x$$

so

$$f(0) = e^0 - 0 - 1 = 0, \quad f'(0) = e^0 - 1, \\ \text{and } f''(0) = e^0 = 1.$$

Theorem 2.8 implies that f has a zero of multiplicity 2 at $x = 0$.

(b) The first two terms generated by Newton's method applied to f with $p_0 = 1$ are

$$p_1 = p_0 - \frac{f(p_0)}{f'(p_0)} = 1 - \frac{e - 2}{e - 1} \approx 0.58198$$

And

$$p_2 = p_1 - \frac{f(p_1)}{f'(p_1)} \approx 0.58198 - \frac{0.20760}{0.78957} \approx 0.31906.$$

The first eight terms of the sequence generated by Newton's method are shown in Table 2.10. The sequence is clearly converging to 0 but not quadratically. The graph of f is shown in Figure 2.11.

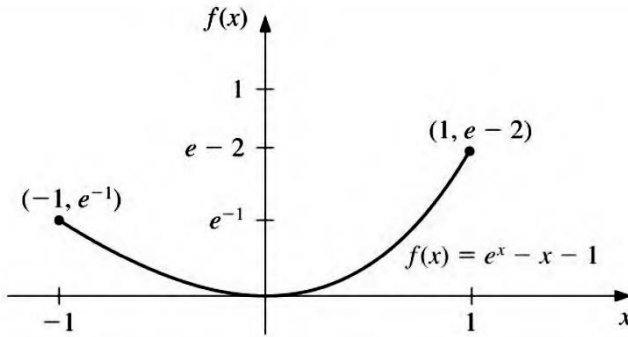


Figure 2.11.

Table 2.10

<i>n</i>	<i>Pu</i>
0	1.0
1	0.58198
2	0.31906
3	0.16800
4	0.08635
5	0.04380
6	0.02206
7	0.01107
8	0.005545
9	2.7750×10^{-3}
10	1.3881×10^{-3}
11	6.9411×10^{-4}
12	3.4703×10^{-4}
13	1.7416×10^{-4}
14	8.8041×10^{-5}
15	4.2610×10^{-5}
16	1.9142×10^{-6}

One method of handling the problem of multiple roots of a function f is to define

$$\mu(x) = \frac{f(x)}{f'(x)}$$

If p is a zero of f of multiplicity m with $f(x) = (x - p)^m q(x)$, then

$$\begin{aligned}\mu(x) &= \frac{(x - p)^m q(x)}{m(x - p)^{m-1} q(x) + (x - p)^m q'(x)} \\ &= (x - p) \frac{q(x)}{mq(x) + (x - p)q'(x)}\end{aligned}$$

also has a zero at p . However, $q(p) \neq 0$, so

$$\frac{q(p)}{mq(p) + (p - p)q'(p)} = \frac{1}{m} \neq 0,$$

and p is a simple zero of $\mu(x)$. Newton's method can then be applied to $\mu(x)$ to give

which simplifies to

$$g(x) = x - \frac{f(x)f'(x)}{[f'(x)]^2 - f(x)f''(x)}$$

If g has the required continuity conditions, functional iteration applied to g will be quadratically convergent regardless of the multiplicity of the zero of f .

Example 2 In Example 1, it was shown that $f(x) = e^x - x - 1$ has a zero of multiplicity 2 at $x = 0$ and that Newton's method with $p_0 = 1$ converges to this zero but not quadratically. Show

that the modification of Newton's method improves the rate of convergence.

Solution Modified Newton's method gives

$$p_1 = p_0 - \frac{f(p_0)f'(p_0)}{f'(p_0)^2 - f(p_0)f''(p_0)} = 1 - \frac{(e-2)(e-1)}{(e-1)^2 - (e-2)e} \\ \approx -2.3421061 \times 10^{-1}.$$

This is considerably closer to 0 than the first term using Newton's method, which was 0.58918. Table 2.11 lists the first five approximations to the double zero at $x = 0$.

Table 2.11

η	p_n
1	$-2.3421061 \times 10^{-1}$
2	$-8.4582788 \times 10^{-3}$
3	$-1.188952.4 \times 10^{-5}$
4	$-6.8638230 \times 10^{-6}$
5	$-2.8085217 \times 10^{-7}$

Illustration

In Section 2.2, we found that a zero of $f(x) = x^3 + 4x^2 - 10 = 0$ is

$$p = 1.36523001.$$

Here we will compare convergence for a simple zero using both Newton's method and the modified Newton method. Let

$$(i) p_n = p_{n-1} - \frac{p_{n-1}^3 + 4p_{n-1}^2 - 10}{3p_{n-1}^2 + 8p_{n-1}}, \text{ from Newton's method,}$$

and, from the modified Newton's method,

$$(ii) p_n = p_{n-1} - \frac{(p_{n-1}^3 + 4p_{n-1}^2 - 10)(3p_{n-1}^2 + 8p_{n-1})}{(3p_{n-1}^2 + 8p_{n-1})^2 - (p_{n-1}^3 + 4p_{n-1}^2 - 10)(6p_{n-1} + 8)}$$

With $p_0 = 1.5$, we have

Newton's method

$$p_1 = 1.37333333, \quad p_2 = 1.36526201, \\ \text{and } p_3 = 1.36523001.$$

Modified Newton's method

$$p_1 = 1.35689898, \quad p_2 = 1.36519585, \\ \text{and } p_3 = 1.36523001.$$

Both methods are rapidly convergent to the actual zero, which is given by both methods as p_3 . Note, however, that in the case of a simple zero, the original Newton's method requires substantially less computation.

Definition 2.4

For a given sequence $\{p_n\}_{n=0}^{\infty}$ the forward difference Δp_n (read "delta p_n ") is defined by

$$\Delta p_n = p_{n+1} - p_n, \text{ for } n \geq 0.$$

Higher powers of the operator Δ are defined recursively by

$$\Delta^k p_n = \Delta(\Delta^{k-1} p_n), \text{ for } k \geq 2.$$

The definition implies that

$$\Delta^2 p_n = \Delta(p_{n+1} - p_n) = \Delta p_{n+1} - \Delta p_n = (p_{n+2} - p_{n+1}) - (p_{n+1} - p_n). \text{ so } \Delta^2 p_n = p_{n+2} - 2p_{n+1} + p_n.$$

Questions 2.4

Use Newton's method to find solutions accurate to within 10^{-5} . Repeat using modified Newton's method also. Tell which method has rapid convergence.

1. $x^2 - 2xe^{-2x} + e^{-2x} = 0, 0 \leq x \leq 1$
2. $x^3 - 3x^2(2^{-x}) + 3x(4^{-x}) - 8^{-x} = 0, 0 \leq x \leq 1$
3. $\cos(x + \sqrt{2}) + x(\frac{x}{2} + \sqrt{2}) = 0, -2 \leq x \leq -1$
4. $e^{6x} + 3(\ln 2)^2 e^{2x} - (\ln 8)e^{4x} - (\ln 2)^3 = 0, -1 \leq x \leq 0.$

Answers

Newton's Method

$$1. \quad p_0 = 0.5, \\ 0.567135$$

$$2. \quad p_0 = 0.5, \\ 0.641166$$

$$3. \quad p_0 = -1.5, \\ -1.414325$$

$$4. \quad p_0 = -0.5, \\ -0.183274$$

Modified Newton's Method

$$p_{13} = \\ p_3 = 0.567143$$

$$p_{22} = \\ p_3 = 0.641274$$

$$p_{23} = \\ p_2 = -1.414158$$

$$p_{23} = \\ p_5 = -0.183319$$

CHAPTER 3

INTERPOLATION AND POLYNOMIAL APPROXIMATION

Introduction

A census of the population of the United States is taken every 10 years. The following table lists the population, in thousands of people, from 1960 to 2010, and the data are also represented in the figure.

Year	1960	1970	1980	1990	2000	2010
Population (in thousands)	179,323	203,302	226,542	249,633	281,422	308,746

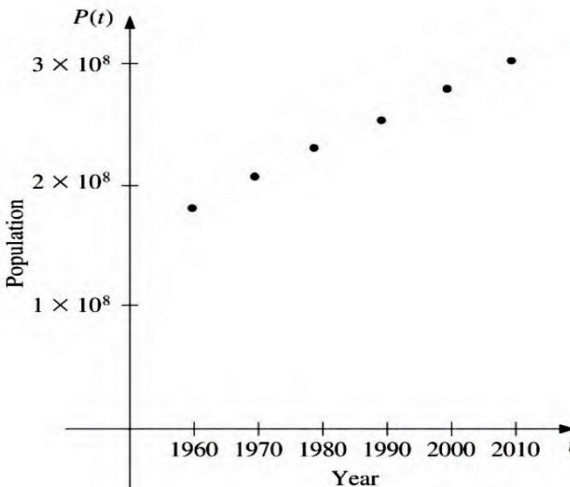


Figure 3.1

In reviewing these data, we might ask whether they could be used to provide a reasonable estimate of the population, say, in 1975 or even in the year 2020. Predictions of this type can be obtained by using a function that fits the given data. This process is called interpolation and is the subject of this chapter.

Now stating generally, the problem is: given a set of tabular values $(x_0, y_0), (x_1, y_1), \dots, (x_n, y_n)$ satisfying $y = f(x)$ where $f(x)$ is not known explicitly and it is asked to find the values of $f(x)$ corresponding to certain given values of x in the range $x_0 < x < x_n$. For this we find a function $g(x)$ in such a way that $f(x)$ and $g(x)$ agree at the set of tabulated values given. Then the required value of $f(x)$ at some point x is taken as the value of $g(x)$ at x . This process is called **interpolation**. If $g(x)$ is a polynomial then the process is called **polynomial interpolation**.

3.1 Interpolation and the Lagrange polynomial

One of the most useful and well-known classes of functions mapping the set of real numbers into itself is the algebraic polynomials, the set of functions of the form

$$P_n(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0.$$

where n is a nonnegative integer and $a_0, \dots, a_n$ real constants. One reason for their importance is that they uniformly approximate continuous functions. By this we mean that given any function, defined and continuous on a closed and bounded interval, there exists a polynomial that is as "close" to the given function as desired.

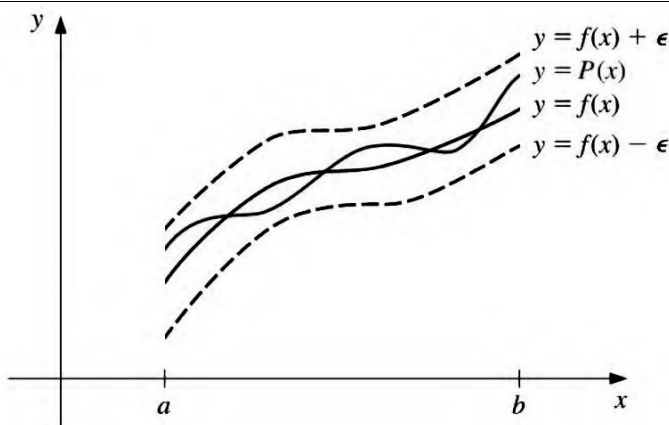


Figure 3.2

Theorem 3.1

(Weierstrass Approximation Theorem)

Suppose f is defined and continuous on $[a, b]$. For each $\epsilon > 0$, there exists a polynomial $P(x)$, with the property that

$$|f(x) - P(x)| < \epsilon, \text{ for all } x \text{ in } [a, b].$$

Lagrange Interpolating Polynomials

The problem of determining a polynomial of degree one that passes through the distinct points (x_0, y_0) and (x_1, y_1) is the same as approximating a function f for which $f(x_0) = y_0$ and $f(x_1) = y_1$ by means of a first-degree polynomial **interpolating**, or agreeing with, the values of f at the given points. Using this polynomial for approximation within the interval given by the endpoints is called **polynomial interpolation**.

Define the functions

$$L_0(x) = \frac{x - x_1}{x_0 - x_1} \text{ and } L_1(x) = \frac{x - x_0}{x_1 - x_0}$$

The linear **Lagrange interpolating polynomial** through (x_0, y_0) and (x_1, y_1) is

$$\begin{aligned} P(x) &= L_0(x)f(x_0) + L_1(x)f(x_1) \\ &= \frac{x - x_1}{x_0 - x_1} f(x_0) + \frac{x - x_0}{x_1 - x_0} f(x_1). \end{aligned}$$

Note that

$$L_0(x_0) = 1, \quad L_0(x_1) = 0, \quad L_1(x_0) = 0, \text{ and } L_1(x_1) = 1,$$

which implies that

$$P(x_0) = 1 \cdot f(x_0) + 0 \cdot f(x_1) = f(x_0) = y_0$$

and

$$P(x_1) = 0 \cdot f(x_0) + 1 \cdot f(x_1) = f(x_1) = y_1$$

So, P is the unique polynomial of degree at most one that passes through (x_0, y_0) and (x_1, y_1) .

Example 1. Determine the linear Lagrange interpolating polynomial that passes through the points $(2, 4)$ and $(5, 1)$

Solution. In this case, we have

$$L_0(x) = \frac{x - 5}{2 - 5} = -\frac{1}{3}(x - 5) \text{ and } L_1(x) = \frac{x - 2}{5 - 2} = \frac{1}{3}(x - 2),$$

So

$$\begin{aligned}
 P(x) &= -\frac{1}{3}(x-5) \cdot 4 + \frac{1}{3}(x-2) \cdot 1 = -\frac{4}{3}x + \frac{20}{3} + \frac{1}{3}x - \frac{2}{3} \\
 &= -x + 6.
 \end{aligned}$$

The graph of $y = P(x)$ is shown in Figure 3.3.

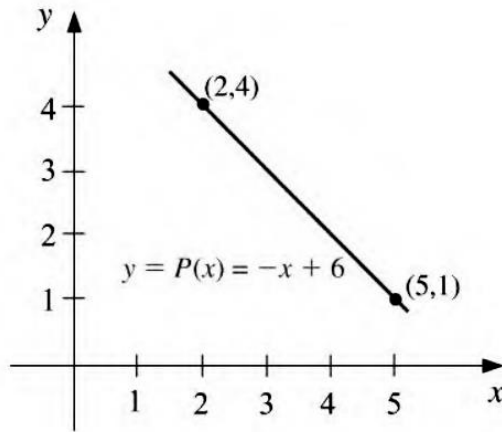


Figure 3.3

To generalize the concept of linear interpolation, consider the construction of a polynomial of degree at most n that passes through the $n + 1$ points

$$(x_0, f(x_0)), (x_1, f(x_1)), \dots, (x_n, f(x_n)). \text{ (See Figure 3.4.)}$$

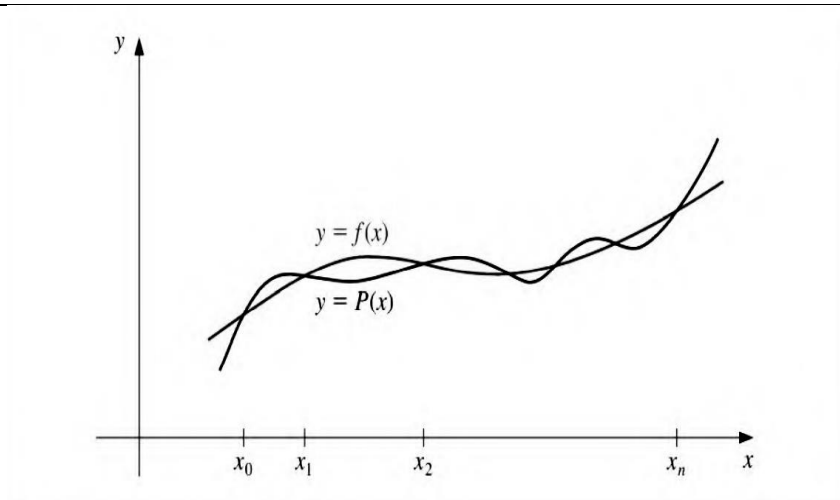


Figure 3.4

In this case, we first construct, for each $k = 0, 1, \dots, n$ a function $L_{n,k}(x)$ with the property that $L_{n,k}(x_i) = 0$ when $i \neq k$ and $L_{n,k}(x_k) = 1$. To satisfy $L_{n,k}(x_i) = 0$ for each $i \neq k$ requires that the numerator of $L_{n,k}(x)$ contain the term

$$(x - x_0)(x - x_1) \dots (x - x_{k-1})(x - x_{k+1}) \dots (x - x_n).$$

To satisfy $L_{n,k}(x_k) = 1$, the denominator of $L_{n,k}(x)$ must be this same term but evaluated at $x = x_k$. Thus,

$$L_{n,k}(x) = \frac{(x - x_0) \dots (x - x_{k-1})(x - x_{k+1}) \dots (x - x_n)}{(x_k - x_0) \dots (x_k - x_{k-1})(x_k - x_{k+1}) \dots (x_k - x_n)}.$$

The interpolating polynomial is easily described once the form of is known. This polynomial, called the **nth Lagrange interpolating polynomial**, is defined in the following theorem.

Theorem 3.2

If $x_0, x_1, \dots, x_n$ are $n + 1$ distinct numbers and f is a function whose values are given at these numbers, then a unique polynomial $P(x)$ of degree at most n exists with

$$f(x_k) = P(x_k), \text{ for each } k = 0, 1, \dots, n.$$

This polynomial is given by.

$$\begin{aligned} P(x) &= f(x_0)L_{n,0}(x) + \dots + f(x_n)L_{n,n}(x) \\ &= \sum_{k=0}^n f(x_k)L_{n,k}(x), \end{aligned} \quad (1)$$

where, for each $k = 0, 1, \dots, n$,

$$\begin{aligned} L_{n,k}(x) &= \frac{(x - x_0)(x - x_1) \dots (x - x_{k-1})(x - x_{k+1}) \dots (x - x_n)}{(x_k - x_0)(x_k - x_1) \dots (x_k - x_{k-1})(x_k - x_{k+1}) \dots (x_k - x_n)} \\ &= \prod_{\substack{i=0 \\ i \neq k}}^n \frac{(x - x_i)}{(x_k - x_i)}. \end{aligned} \quad (2)$$

We will write $L_{n,k}(x)$ simply as $L_k(x)$ when there is no confusion as to its degree.

Example 2.

(a) Use the numbers (called nodes)

$x_0 = 2, x_1 = 2.75$, and $x_2 = 4$ to find the second Lagrange interpolating polynomial for $f(x) = 1/x$.

(b) Use this polynomial to approximate $f(3) = 1/3$.

Solution. (a) We first determine the coefficient polynomials $L_0(x)$, $L_1(x)$, and $L_2(x)$. In nested form, they are

$$L_0(x) = \frac{(x - 2.75)(x - 4)}{(2 - 2.75)(2 - 4)} = \frac{2}{3}(x - 2.75)(x - 4),$$

$$L_1(x) = \frac{(x - 2)(x - 4)}{(2.75 - 2)(2.75 - 4)} = -\frac{16}{15}(x - 2)(x - 4),$$

and

$$L_2(x) = \frac{(x - 2)(x - 2.75)}{(4 - 2)(4 - 2.75)} = \frac{2}{5}(x - 2)(x - 2.75),$$

Also, $f(x_0) = f(2) = \frac{1}{2}$, $f(x_1) = f(2.75) = \frac{4}{11}$, and $f(x_2) = f(4) = 1/4$, so

$$\begin{aligned} P(x) &= \sum_{k=0}^2 f(x_k)L_k(x) \\ &= \frac{1}{3}(x - 2.75)(x - 4) - \frac{64}{165}(x - 2)(x - 4) \\ &\quad + \frac{1}{10}(x - 2)(x - 2.75) \\ &= \frac{1}{22}x^2 - \frac{35}{88}x + \frac{49}{44}. \end{aligned}$$

(b) An approximation to $f(3) = \frac{1}{3}$ (see Figure 3.5) is

$$f(3) \approx P(3) = \frac{9}{22} - \frac{105}{88} + \frac{49}{44} = \frac{29}{88} \approx 0.32955.$$

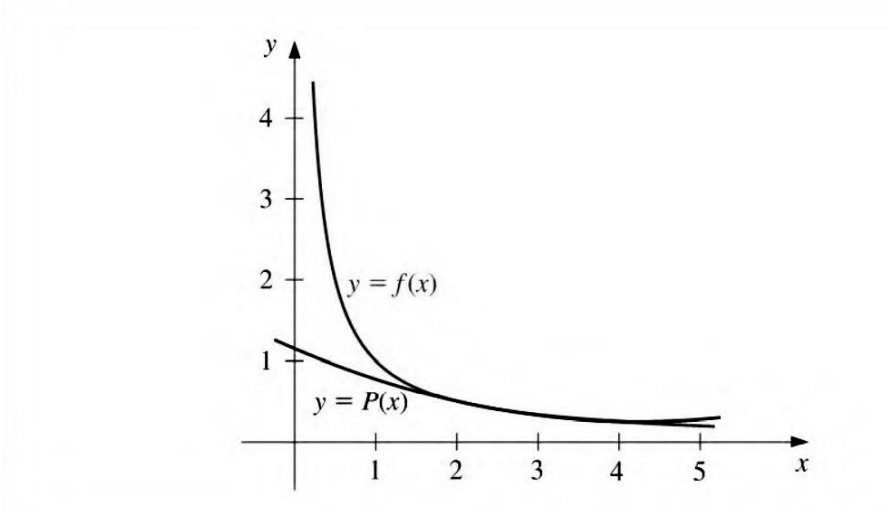


Figure 3.5

The next step is to calculate a remainder term or bound for the error involved in approximating a function by an interpolating polynomial.

Theorem 3.3

Suppose $x_0, x_1, \dots, x_n$ are distinct numbers in the interval $[a, b]$ and $f \in C^{n+1}[a, b]$. Then, for each x in $[a, b]$, a number $\xi(x)$ (generally unknown) between $\min \{x_0, x_1, \dots, x_n\}$, and the $\max \{x_0, x_1, \dots, x_n\}$ and hence in (a, b) , exists with

$$f(x) = P(x) + \frac{f^{(n+1)}(\xi(x))}{(n+1)!} (x - x_0)(x - x_1) \dots (x - x_n), \quad (3)$$

where $P(x)$ is the interpolating polynomial given in Eq. (1).

Proof. Note first that if $x = x_k$, for any $k = 0, 1, \dots, n$, then $f(x_k) = P(x_k)$, and choosing $\xi(x_k)$ arbitrarily in (a, b) yields Eq. (3).

If $x \neq x_k$, for all $k = 0, 1, \dots, n$, define the function g for t in $[a, b]$ by

$$\begin{aligned} g(t) &= f(t) - P(t) \\ &\quad - [f(x) - P(x)] \frac{(t - x_0)(t - x_1) \dots (t - x_n)}{(x - x_0)(x - x_1) \dots (x - x_n)} \\ &= f(t) - P(t) - [f(x) - P(x)] \prod_{i=0}^n \frac{(t - x_i)}{(x - x_i)} \end{aligned}$$

Since $f \in C^{n+1}[a, b]$, and $P \in C^\infty[a, b]$, it follows that $g \in C^{n+1}[a, b]$. For $t = x_k$, we have

$$\begin{aligned} g(x_k) &= f(x_k) - P(x_k) - [f(x) - P(x)] \prod_{i=0}^n \frac{(x_k - x_i)}{(x - x_i)} \\ &= 0 - [f(x) - P(x)] \cdot 0 = 0. \end{aligned}$$

Moreover,

$$\begin{aligned} g(x) &= f(x) - P(x) - [f(x) - P(x)] \prod_{i=0}^n \frac{(x - x_i)}{(x - x_i)} \\ &= f(x) - P(x) - [f(x) - P(x)] = 0. \end{aligned}$$

Thus, $g \in C^{n+1}[a, b]$, and g is zero at the $n + 2$ distinct numbers $x, x_0, x_1, \dots, x_n$. By Generalized Rolle's Theorem, there exists a number ξ in (a, b) for which $g^{(n+1)}(\xi) = 0$. So,

$$\begin{aligned} 0 &= g^{(n+1)}(\xi) \\ &= f^{(n+1)}(\xi) - P^{(n+1)}(\xi) \\ &\quad - [f(x) - P(x)] \frac{d^{n+1}}{dt^{n+1}} \left[\prod_{i=0}^n \frac{(t - x_i)}{(x - x_i)} \right]_{t=\xi} \end{aligned} \quad (4)$$

However, $P(x)$ is a polynomial of degree at most n , so the $(n + 1)$ st derivative, $P^{(n+1)}(x)$, is identically zero. Also, $\prod_{i=0}^n \left[\frac{(t - x_i)}{(x - x_i)} \right]$ is a polynomial of degree $(n + 1)$, so

$$\begin{aligned} \prod_{i=0}^n \frac{(t - x_i)}{(x - x_i)} &= \left[\frac{1}{\prod_{i=0}^n (x - x_i)} \right] t^{n+1} + (\text{lower} \\ &\quad - \text{degree terms in } t) \end{aligned}$$

And

$$\frac{d^{n+1}}{dt^{n+1}} \prod_{i=0}^n \frac{(t - x_i)}{(x - x_i)} = \frac{(n + 1)!}{\prod_{i=0}^n (x - x_i)}$$

Equation (4) now becomes,

$$0 = f^{(n+1)}(\xi) - 0 - [f(x) - P(x)] \frac{(n+1)!}{\prod_{i=0}^n (x - x_i)}$$

and, upon solving for $f(x)$, we have

$$f(x) = P(x) + \frac{f^{(n+1)}(\xi)}{(n+1)!} \prod_{i=0}^n (x - x_i)$$

The Lagrange polynomial of degree n uses information at the distinct numbers $x, x_0, x_1, \dots, x_n$ and, in place of $(x - x_0)^n$, its error formula uses a product of the $n + 1$ terms $(x - x_0), (x - x_1), \dots, (x - x_n)$:

$$\frac{f^{(n+1)}(\xi(x))}{(n+1)!} (x - x_0)(x - x_1) \dots (x - x_n).$$

Example 3.

In Example 2, we found the second Lagrange polynomial for $f(x) = 1/x$ on $[2, 4]$ using the nodes $x_0 = 2, x_1 = 2.75$, and $x_2 = 4$. Determine the error form for this polynomial and the maximum error when the polynomial is used to approximate $f(x)$ for $x \in [2, 4]$.

Solution. Because $f(x) = x^{-1}$ we have

$$f'(x) = -x^{-2}, \quad f''(x) = 2x^{-3}, \quad \text{and} \quad f'''(x) = -6x^{-4}.$$

As a consequence, the second Lagrange polynomial has the error form

$$\begin{aligned} \frac{f'''(\xi(x))}{3!} (x - x_0)(x - x_1)(x - x_2) \\ = -(\xi(x))^{-4} (x - 2)(x - 2.75)(x - 4), \\ \text{for } \xi(x) \text{ in } (2, 4) \end{aligned}$$

The maximum value of $(\xi(x))^{-4}$ on the interval is $2^{-4} = 1/16$. We now need to determine the maximum value on this interval of the absolute value of the polynomial.

$$g(x) = (x - 2)(x - 2.75)(x - 4) = x^3 - \frac{35}{4}x^2 + \frac{49}{2}x - 22.$$

Because

$$\begin{aligned} D_x \left(x^3 - \frac{35}{4}x^2 + \frac{49}{2}x - 22 \right) &= 3x^2 - \frac{35}{2}x + \frac{49}{2} \\ &= \frac{1}{2}(3x - 7)(2x - 7), \end{aligned}$$

the critical points occur at

$$x = \frac{7}{3}, \text{ with } g\left(\frac{7}{3}\right) = \frac{25}{108}, \text{ and } x = \frac{7}{2}, \text{ with } g\left(\frac{7}{2}\right) = -\frac{9}{16}.$$

Hence, the maximum error is

$$\begin{aligned} \frac{f'''(\xi(x))}{3!} |(x - x_0)(x - x_1)(x - x_2)| &\leq \frac{1}{16} \left| -\frac{9}{16} \right| = \frac{9}{256} \\ &\approx 0.03515625. \end{aligned}$$

Example 4.

Suppose a table is to be prepared for the function $(x) = e^x$, for x in $[0, 1]$, Assume the number of decimal places to be given per

entry is $d \geq 8$ and that the difference between adjacent x -values, the step size, is h . What step size h will ensure that linear interpolation gives an absolute error of at most 10^{-6} for all x in $[0, 1]$?

Solution. Let $x_0, x_1, \dots$ be the numbers at which f is evaluated and x be in $[0, 1]$ and suppose j satisfies $x_j \leq x \leq x_{j+1}$. Eq. (3) implies that the error in linear interpolation is

$$\begin{aligned} |f(x) - P(x)| &= \left| \frac{f^{(2)}(\xi)}{2!} (x - x_j)(x - x_{j+1}) \right| \\ &= \frac{|f^{(2)}(\xi)|}{2!} |(x - x_j)|(x - x_{j+1})|. \end{aligned}$$

The step size is h , so $x_j = jh$, $x_{j+1} = (j + 1)h$, and

$$|f(x) - P(x)| \leq \frac{|f^{(2)}(\xi)|}{2!} |(x - jh)(x - (j + 1)h)|.$$

Hence,

$$\begin{aligned} |f(x) - P(x)| &\leq \frac{\max_{\xi \in [0,1]} e^\xi}{2} \max_{x_j \leq x \leq x_{j+1}} |(x - jh)(x - (j + 1)h)| \\ &\leq \frac{e}{2} \max_{x_j \leq x \leq x_{j+1}} |(x - jh)(x - (j + 1)h)| \end{aligned}$$

Consider the function $g(x) = (x - jh)(x - (j + 1)h)$, for $jh \leq x \leq (j + 1)h$. Because

$$g'(x) = (x - (j + 1)h) + (x - jh) = 2 \left(x - jh - \frac{h}{2} \right),$$

the only critical point for g is at $x = jh + h/2$, with $g(jh + h/2) = \left(\frac{h}{2}\right)^2 = h^2/4$.

Since $g(jh) = 0$ and $g((j+1)h) = 0$, the maximum value of $|g'(x)|$ in $[jh, (j+1)h]$ must occur at the critical point, which implies that

$$|f(x) - P(x)| \leq \frac{e}{2} \max_{x_j \leq x \leq x_{j+1}} |g(x)| \leq \frac{e}{2} \cdot \frac{h^2}{4} = \frac{eh^2}{8}.$$

Consequently, to ensure that the error in linear interpolation is bounded by 10^{-6} , it is sufficient for h to be chosen so that

$$\frac{eh^2}{8} \leq 10^{-6}. \text{ This implies that } h < 1.72 \times 10^{-3}.$$

Because $n = (1 - 0)/h$ must be an integer, a reasonable choice for the step size is $h = 0.001$.

Example 5. Use Lagrange interpolating polynomial of degree 3 to approximate $f(9.2)$ if $f(9) = 2.19722$, $f(9.5) = 2.25129$, $f(10) = 2.30259$, $f(11) = 2.39790$

Solution.

$$\begin{aligned} f(9.2) &= \sum_{k=0}^3 f(x_k) L_k(x) \\ &= \frac{(9.2 - 9.5)(9.2 - 10)(9.2 - 11)}{(9 - 9.5)(9 - 10)(9 - 11)} (2.19722) \end{aligned}$$

$$\begin{aligned}
 & + \frac{(9.2 - 9)(9.2 - 10)(9.2 - 11)}{(9.5 - 9)(9.5 - 10)(9.5 - 11)} (2.25129) \\
 & + \frac{(9.2 - 9)(9.2 - 9.5)(9.2 - 11)}{(10 - 9)(10 - 9.5)(10 - 11)} (2.30259) \\
 & + \frac{(9.2 - 9)(9.2 - 9.5)(9.2 - 10)}{(11 - 9)(11 - 9.5)(11 - 10)} (2.39790) \\
 & = 2.21920
 \end{aligned}$$

Example 6. Construct the Lagrange interpolating polynomial of degree two for $f(x) = \ln x$ and let $x_0 = 2, x_1 = 2.5, x_2 = 3$. Also approximate $f(2.7)$ and find absolute error.

Solution.

x	x_0	x_1	x_2
	2	2.5	3
$f(x) = \ln x$	0.69315	0.91629	1.09861

$$\begin{aligned}
 P_2(x) &= f(x_0)L_0(x) + f(x_1)L_1(x) + f(x_2)L_2(x) \\
 &= (0.69315) \frac{(x - 2.5)(x - 3)}{(2 - 2.5)(2 - 3)} \\
 &\quad + (0.91629) \frac{(x - 2)(x - 3)}{(2.5 - 2)(2.5 - 3)} \\
 &\quad + (1.09861) \frac{(x - 2)(x - 2.5)}{(3 - 2)(3 - 2.5)}
 \end{aligned}$$

$$\begin{aligned}
 &= (0.69315)(2x^2 - 11x + 15) \\
 &\quad - (0.91629)(4x^2 - 20x + 24) \\
 &\quad + (1.09861)(2x^2 - 9x + 10) \\
 &= -0.08164x^2 + 0.81366x - 0.60761
 \end{aligned}$$

which is the Lagrange interpolating polynomial of degree two

$$\text{Hence } P_2(2.7) = 0.9941164$$

$$\begin{aligned}
 \text{Error} &= |f(2.7) - P_2(2.7)| = |0.9932518 - 0.9941164| \\
 &= 0.0008646
 \end{aligned}$$

Questions 3.1

1. For the given functions $f(x)$, let $x_0 = 0$, $x_1 = 0.6$, and $x_2 = 0.9$. Construct interpolation polynomials of degree at most one and at most two to approximate $f(0.45)$ and find the absolute error.

- a. $f(x) = \cos x$
- b. $f(x) = \sqrt{1+x}$
- c. $f(x) = \ln(x+1)$
- d. $f(x) = \tan x$

2. Use appropriate Lagrange interpolating polynomials of degrees one, two, and three to approximate each of the following:

- a. $f(8.4)$ if $f(8.1) = 16.94410$, $f(8.3) = 17.56492$, $f(8.6) = 18.50515$, $f(8.7) = 18.82091$

b. $f\left(-\frac{1}{3}\right)$ if $f(-0.75) = -0.07181250$, $f(-0.5) = -0.02475000$, $f(-0.25) = 0.33493750$, $f(0) = 1.10100000$

c. $f(0.25)$ if $f(0.1) = 0.62049958$, $f(0.2) = -0.28398668$, $f(0.3) = 0.00660095$, $f(0.4) = 0.24842440$

d. $f(0.9)$ if $f(0.6) = -0.17694460$, $f(0.7) = 0.01375227$, $f(0.8) = 0.22363362$, $f(1.0) = 0.65809197$

3. The data for Exercise 2 were generated using the following functions. Use the error formula to find a bound for the error and compare the bound to the actual error for the cases $n = 1$ and $n = 2$.

a. $f(x) = x \ln x$

b. $f(x) = x^3 + 4.001x^2 + 4.002x + 1.101$

c. $f(x) = x \cos x - 2x^2 + 3x - 1$

d. $f(x) = \sin(e^x - 2)$

4. Let $P_3(x)$ be the interpolating polynomial for the data $(0, 0)$, $(0.5, y)$, $(1, 3)$, and $(2, 2)$. The coefficient of x^3 in $P_3(x)$ is 6. Find y .

5. Construct the Lagrange interpolating polynomials for the following functions and find a bound for the absolute error on the interval $[x_0, x_n]$.

a. $f(x) = e^{2x} \cos 3x$, $x_0 = 0$, $x_1 = 0.3$, $x_2 = 0.6$, $n = 2$

b. $f(x) = \sin(\ln x)$, $x_0 = 2.0$, $x_1 = 2.4$, $x_2 = 2.6$, $n = 2$

c. $f(x) = \ln x, x_0 = 1, x_1 = 1.1, x_2 = 1.3, x_3 = 1.4, n = 2$

d. $f(x) = \cos x + \sin x, x_0 = 0, x_1 = 0.25, x_2 = 0.5, x_3 = 1.0, n = 3$

6. Let $(x) = e^x$, for $0 \leq x \leq 2$.

a. Approximate $f(0.25)$ using linear interpolation with $x_0 = 0$ and $x_1 = 0.5$.

b. Approximate $f(0.75)$ using linear interpolation with $x_0 = 0.5$ and $x_1 = 1$.

c. Approximate $f(0.25)$ and $f(0.75)$ using the second interpolating polynomial with $x_0 = 0$, and $x_1 = 1$, and $x_2 = 2$.

d. Which approximations are better, and why?

Answers

1. a. $P_1(x) = -0.148878x + 1; P_2(x) = -0.452592x^2 - 0.0131009x + 1;$

$$P_1(0.45) = 0.933005; |f(0.45) - P_1(0.45)| = 0.032558;$$

$$P_2(0.45) = 0.902455; |f(0.45) - P_2(0.45)| = 0.002008$$

b. $P_1(x) = 0.467251x + 1; P_2(x) = -0.0780026x^2 + 0.490652x + 1;$

$$P_1(0.45) = 1.210263; |f(0.45) - P_1(0.45)| = 0.006104;$$

$$P_2(0.45) = 1.204998; |f(0.45) - P_2(0.45)| = 0.000839$$

c. $P_1(x) = 0.874548x; P_2(a) = -0.268961x^2 + 0.955236x;$

$$P_1(0.45) = 0.393546; |f(0.45) - P_1(0.45)| = 0.0212983;$$

$$P_2(0.45) = 0.375392; |f(0.45) - P_2(0.45)| = 0.003828$$

$$d. P_1(x) = 1.031121x; P_2(x) = 0.615092x^2 + 0.846593x;$$

$$P_1(0.45) = 0.464004; |f(0.45) - P_1(0.45)| = 0.019051;$$

$$P_2(0.45) = 0.505523; |f(0.45) - P_2(0.45)| = 0.022468$$

2.

a.

n	$x_0, x_1, \dots, x_n$	$P_n(8.4)$
1	8.3, 8.6	17.87833

2	8.3, 8.6, 8.7	17.87716
---	---------------	----------

3	8.3, 8.6, 8.7, 8.1	17.87714
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b.

n	$x_0, x_1, \dots, x_n$	$P_n(8.4)$
1	-0.5, -0.25	0.21504167

2	-0.5, -0.25, 0.0	0.16988889
---	------------------	------------

3	-0.5, -0.25, 0.0, -0.75	0.17451852
---	-------------------------	------------

c.

n	$x_0, x_1, \dots, x_n$	$P_n(0.25)$
1	0.2, 0.3	-0.13869287

2	0.2, 0.3, 0.4	-0.132559734
---	---------------	--------------

d.

n	$x_0, x_1, \dots, x_n$	$P_n(0.9)$
1	0.8, 1.0	0.44086280

2	0.8, 1.0, 0.7	0.43841352
---	---------------	------------

3	0.2, 0.3, 0.4, 0.1	-0.13277477	3	0.8, 1.0, 0.7, 0.6	0.44198500
a.			b.		
n	Actual Error	Error Bound	n	Actual Error	Error Bound
1	1.180×10^{-3}	1.200×10^{-3}	1	4.052×10^{-2}	4.515×10^{-2}
2	1.367×10^{-5}	1.452×10^{-3}	2	4.630×10^{-3}	4.630×10^{-3}
c.			d.		
n	Actual Error	Error Bound	n	Actual Error	Error Bound
1	5.921×10^{-3}	6.097×10^{-3}	1	2.730×10^{-3}	1.408×10^{-2}
2	1.746×10^{-4}	1.813×10^{-4}	2	5.179×10^{-3}	9.222×10^{-3}

4. $y = 4.25$

5. a. $P_2(x) = -11.22388889x^2 + 3.810500000x + 1$, and an error bound is 0.11371294

b. $P_2(x) = -0.1306344167x^2 + 0.8969979335x - 0.63249693$, and an error bound is 9.45762×10^{-4}

c. $P_3(x) = 0.1970056661x^3 - 1.06259055x^2 + 2.532453189x - 1.666868305$, and an error bound is 10^{-4}

d. $P_3(x) = -0.07932x^3 - 0.545506x^2 + 1.0065992x + 1$, and an error bound is 1.591376×10^{-3}

6. a. 1.32436

b. 2.18350

c. $1.15277, 2.01191$

d. Parts (a) and (b) are better due to the spacing of the nodes.

3.2 Data Approximation and Neville's Method

In the previous section, we found an explicit representation for Lagrange polynomials and their error when approximating a function on an interval. A frequent use of these polynomials involves the interpolation of tabulated data. In this case, an explicit representation of the polynomial might not be needed, only the values of the polynomial at specified points. In this situation, the function underlying the data might not be known, so the explicit form of the error cannot be used.

Illustration

Table 3.2 lists values of a function f at various points. The approximations to $f(1.5)$ obtained by various Lagrange polynomials that use these data will be compared to try and determine the accuracy of the approximation.

Table 3.2

x	$f(x)$
1.0	0.7651977
1.3	0.6200860
1.6	0.4554022
1.9	0.2818186
2.2	0.1103623

The most appropriate linear polynomial uses $x_0 = 1.3$ and $x_1 = 1.6$ because 1.5 is between 1.3 and 1.6. The value of the interpolating polynomial at 1.5 is

$$\begin{aligned}P_1(1.5) &= \frac{(1.5 - 1.6)}{(1.3 - 1.6)} f(1.3) + \frac{(1.5 - 1.3)}{(1.6 - 1.3)} f(1.6) \\&= \frac{(1.5 - 1.6)}{(1.3 - 1.6)} (0.6200860) + \frac{(1.5 - 1.3)}{(1.6 - 1.3)} (0.455022) \\&= 0.5102968\end{aligned}$$

Two polynomials of degree two can reasonably be used, one with $x_0 = 1.3, x_1 = 1.6$, and $x_2 = 1.9$, which gives

$$\begin{aligned}
 P_1(1.5) &= \frac{(1.5 - 1.6)(1.5 - 1.9)}{(1.3 - 1.6)(1.3 - 1.9)} (0.6200860) \\
 &\quad + \frac{(1.5 - 1.3)(1.5 - 1.9)}{(1.6 - 1.3)(1.6 - 1.9)} (0.455022) \\
 &\quad + \frac{(1.5 - 1.3)(1.5 - 1.6)}{(1.9 - 1.3)(1.9 - 1.6)} (0.281886) = 0.5112857,
 \end{aligned}$$

and one with $x_0 = 1.0$, $x_1 = 1.3$, and $x_2 = 1.6$, which gives $\hat{P}_2(1.5) = 0.5124715$.

In the third-degree case, there are also two reasonable choices for the polynomial, one with $x_0 = 1.3$, $x_1 = 1.6$, $x_2 = 1.9$ and $x_3 = 2.2$, which gives $P_3(1.5) = 0.5118302$. The second third-degree approximation is obtained with $x_0 = 1.0$, $x_1 = 1.3$, $x_2 = 1.6$ and $x_3 = 1.9$, which gives $\hat{P}_3 = 0.5118127$.

The fourth-degree Lagrange polynomial uses all the entries in the table. With $x_0 = 1.0$, $x_1 = 1.3$, $x_2 = 1.6$, $x_3 = 1.9$ and $x_4 = 2.2$, the approximation is $P_4(1.5) = 0.5118200$.

Because $P_3(1.5)$, $\hat{P}_3(1.5)$, and $P_4(1.5)$ all agree to within 2×10^{-5} units, we expect this degree of accuracy for these approximations. We also expect $P_4(1.5)$ to be the most accurate approximation since it uses more of the given data.

The function we are approximating is actually the Bessel function of the first kind of order zero, whose value at 1.5 is known to be 0.5118277. Therefore, the true accuracies of the approximations are as follows:

$$|P_1(1.5) - f(1.5)| \approx 1.53 \times 10^{-3},$$

$$|P_2(1.5) - f(1.5)| \approx 5.42 \times 10^{-4},$$

$$|\hat{P}_2(1.5) - f(1.5)| \approx 6.44 \times 10^{-4},$$

$$|P_3(1.5) - f(1.5)| \approx 2.5 \times 10^{-6},$$

$$|\hat{P}_3(1.5) - f(1.5)| \approx 1.50 \times 10^{-5},$$

$$|P_4(1.5) - f(1.5)| \approx 7.7 \times 10^{-6},$$

Although $P_3(1.5)$ is the most accurate approximation, if we had no knowledge of the actual value of $f(1.5)$, we would accept $P_4(1.5)$ as the best approximation since it includes the most data about the function. The Lagrange error term derived in Theorem 3.3 cannot be applied here because we have no knowledge of the fourth derivative of f .

Neville's Method

A practical difficulty with Lagrange interpolation is that the error term is difficult to apply, so the degree of the polynomial needed for the desired accuracy is generally not known until computations have been performed. A common practice is to compute the results given from various polynomials until appropriate agreement is obtained, as was done in the previous illustration. However, the work done in calculating the approximation by the second polynomial does not lessen the work needed to calculate the third approximation, nor is the fourth approximation easier to obtain once the third approximation is known and so on. We will now derive these approximating polynomials in a manner that uses the previous calculations to greater advantage.

Definition

Let f be a function defined at $x_0, x_1, x_2, \dots, x_n$ and suppose that $m_1, m_2, \dots, m_k$ are k distinct integers, with $0 \leq m_i \leq n$ for each i . The Lagrange polynomial that agrees with $f(x)$ at the k points $x_{m_1}, x_{m_2}, \dots, x_{m_k}$ is denoted $P_{m_1, m_2, \dots, m_k}(x)$.

Example 1.

Suppose that $x_0 = 1, x_1 = 2, x_2 = 3, x_3 = 4, x_4 = 6$, and $f(x) = e^x$. Determine the interpolating polynomial denoted $P_{1,2,4}(x)$ and use this polynomial to approximate $f(5)$.

Solution. This is the Lagrange polynomial that agrees with $f(x)$ at $x_1 = 2, x_2 = 3$, and $x_4 = 6$. Hence,

$$P_{1,2,4}(x) = \frac{(x-3)(x-6)}{(2-3)(2-6)}e^2 + \frac{(x-2)(x-6)}{(3-2)(3-6)}e^3 + \frac{(x-2)(x-3)}{(6-2)(6-3)}e^6.$$

So,

$$\begin{aligned} f(5) \approx P(5) &= \frac{(5-3)(5-6)}{(2-3)(2-6)}e^2 + \frac{(5-2)(5-6)}{(3-2)(3-6)}e^3 \\ &\quad + \frac{(5-2)(5-3)}{(6-2)(6-3)}e^6 \\ &= \frac{1}{2}e^2 + e^3 + \frac{1}{2}e^6 \approx 218.105. \end{aligned}$$

The next result describes a method for recursively generating Lagrange polynomial approximations.

Theorem 3.5

Let f be defined at $x_0, x_1, \dots, x_k$ and let x_j and x_i , be two distinct numbers in this set. Then

$$P(x) = \frac{(x - x_j)P_{0,1,\dots,j-1,j+1,k}(x) - (x - x_i)P_{0,1,\dots,i-1,i+1,k}(x)}{(x - x_i)}$$

Is the k th Lagrange polynomial that interpolates f at the $k + 1$ points $x_0, x_1, \dots, x_k$.

Proof.

For ease of notation, let $Q \equiv P_{0,1,\dots,i-1,i+1,k}$ and $\hat{Q} \equiv P_{0,1,\dots,j-1,j+1,k}$. Since $Q(x)$ and $\hat{Q}(x)$ are polynomials of degree $k - 1$ or less, $P(x)$ is of degree at most k .

First, note that $\hat{Q}(x_i) = f(x_i)$ implies that

$$\begin{aligned} P(x_i) &= \frac{(x - x_j)\hat{Q}(x_i) - (x_i - x_i)Q(x_i)}{x_i - x_j} = \frac{(x_i - x_j)}{(x_i - x_j)} f(x_i) \\ &= f(x_i) \end{aligned}$$

Similarly, since $\hat{Q}(x_j) = f(x_j)$, we have $P(x_j) = f(x_j)$

In addition, if $0 \leq r \leq k$ and r is neither i nor j , then $Q(x_r) = \hat{Q}(x_r) = f(x_r)$. So,

$$\begin{aligned} P(x_r) &= \frac{(x_r - x_j)\hat{Q}(x_r) - (x_r - x_i)Q(x_r)}{x_i - x_j} = \frac{(x_i - x_j)}{(x_i - x_j)} f(x_r) \\ &= f(x_r). \end{aligned}$$

But, by definition, $P_{0,1,\dots,k}(x)$ is the unique polynomial of degree at most k that agrees with f at $x_0, x_1, \dots, x_k$. Thus, $P \equiv P_{0,1,\dots,k}$.

Theorem 3.5 implies that the interpolating polynomials can be generated recursively. For example, we have

$$P_{0,1} = \frac{1}{x_1 - x_0} [(x - x_0)]P_1 + (x - x_1)P_0 ,$$

$$P_{1,2} = \frac{1}{x_2 - x_1} [(x - x_1)]P_2 + (x - x_2)P_1 ,$$

$$P_{0,1,2} = \frac{1}{x_2 - x_0} [(x - x_0)]P_{1,2} + (x - x_2)P_{0,1} ,$$

and so on. They are generated in the manner shown in Table 3.3, where each row is completed before the succeeding rows are begun

Table 3.3

x_0	P_0				
x_1	P_1	$P_{0,1}$			
x_2	P_2	$P_{1,2}$	$P_{0,1,2}$		
x_3	P_3	$P_{2,3}$	$P_{1,2,3}$	$P_{0,1,2,3}$	
x_4	P_4	$P_{3,4}$	$P_{2,3,4}$	$P_{1,2,3,4}$	$P_{0,1,2,3,4}$

The procedure that uses the result of Theorem 3.5 to recursively generate interpolating polynomial approximations is called Neville's method. The P notation used in Table 3.3 is cumbersome because of the number of subscripts used to represent the entries. Note, however, that as an array is being constructed, only two subscripts are needed. Proceeding down the table corresponds to using consecutive points x_i with larger i , and proceeding to the right corresponds to increasing the degree of the interpolating polynomial. Since the points appear consecutively in each entry, we need to describe only a starting point and the number of additional points used in constructing the approximation.

To avoid the multiple subscripts, we let $Q_{i,j}(x)$, for $0 \leq j \leq i$, denote the interpolating polynomial of degree j on the $(j + 1)$ numbers $x_{i-j}, x_{i-j+1}, \dots, x_{i-1}, x_i$; that is,

$$Q_{i,j} = P_{i-j, i-j+1, \dots, i-1, i}.$$

Using this notation provides the Q notation array in Table 3.4.

Table 3.4

x_0	$P_0 =$ $Q_{0,0}$			
x_1	$P_1 =$ $Q_{1,0}$	$P_{0,1} =$ $Q_{1,1}$		
x_2	$P_2 =$ $Q_{2,0}$	$P_{1,2} =$ $Q_{2,1}$	$P_{0,1,2} =$ $Q_{2,2}$	

x_3	P_3	$=$	$P_{2,3}$	$=$	$P_{1,2,3}$	$=$	$P_{0,1,2,3}$	$=$	
	$Q_{3,0}$		$Q_{3,1}$		$Q_{3,2}$		$Q_{3,3}$		
x_4	P_4	$=$	$P_{3,4}$	$=$	$P_{2,3,4}$	$=$	$P_{1,2,3,4}$	$=$	$P_{0,1,2,3,4}$
	$Q_{4,0}$		$Q_{4,1}$		$Q_{4,2}$		$Q_{4,3}$		$Q_{4,4}$

Example 2

Values of various interpolating polynomials at $x = 1.5$ were obtained in the illustration at the beginning of this section using the data shown in Table 3.5. Apply Neville's method to the data by constructing a recursive table of the form shown in Table 3.4.

Table 3.5

X	f(x)
1.0	0.7651977
1.3	0.6200860
1.6	0.4554022
1.9	0.2818186
2.2	0.1103623

Solution.

Let $x_0 = 1.0$, $x_1 = 1.3$, $x_2 = 1.6$, $x_3 = 1.9$, and $x_4 = 2.2$, then $Q_{0,0} = f(1.0)$, $Q_{1,0} = f(1.3)$, $Q_{2,0} = f(1.6)$, $Q_{3,0} = f(1.9)$, and $Q_{4,0} = f(2.2)$. These are the five polynomials of degree zero (constants) that approximate $f(1.5)$ and are the same as data given in Table 3.5.

Calculating the first-degree approximation $Q_{1,1}(1.5)$ gives

$$\begin{aligned}
 Q_{1,1}(1.5) &= \frac{(x - x_0)Q_{1,0} - (x - x_1)Q_{0,0}}{x_1 - x_0} \\
 &= \frac{(1.5 - 1.0)Q_{1,0} - (1.5 - 1.3)Q_{0,0}}{1.3 - 1.0} \\
 &= \frac{0.5(0.6200860) - 0.2(0.7651977)}{0.3} = 0.5233449
 \end{aligned}$$

Similarly,

$$\begin{aligned}
 &Q_{2,1}(1.5) \\
 &= \frac{(1.5 - 1.3)(0.4554022) - (1.5 - 1.6)(0.6200860)}{1.6 - 1.3} \\
 &= 0.5102968
 \end{aligned}$$

$$Q_{3,1}(1.5) = 0.5132634, \text{ and } Q_{4,1}(1.5) = 0.5104270.$$

The best linear approximation is expected to be $Q_{2,1}$ because 1.5 is between $x_1 = 1.3$ and $x_2 = 1.6$.

In a similar manner, approximations using higher-degree polynomials are given by

$$\begin{aligned}
 &Q_{2,2}(1.5) \\
 &= \frac{(1.5 - 1.0)(0.5102968) - (1.5 - 1.6)(0.5233449)}{1.6 - 1.0} \\
 &= 0.5124715,
 \end{aligned}$$

$$Q_{3,2}(1.5) = 0.5112857, \text{ and } Q_{4,2}(1.5) = 0.5137361.$$

The higher-degree approximations are generated in a similar manner and are shown in Table 3.6.

Table 3.6

1.0	0.7651977				
1.3	0.6200860	0.5233449			
1.6	0.4554022	0.5102968	0.5124715		
1.9	0.2818186	0.5132634	0.5112857	0.5118127	
2.2	0.1103623	0.5104270	0.5137361	0.5118302	0.5118200

If the latest approximation, $Q_{4,4}$, was not sufficiently accurate, another node, x_5 , could be selected and another row added to the table:

$$x_5 Q_{5,0} Q_{5,1} Q_{5,2} Q_{5,3} Q_{5,4} Q_{5,5}.$$

Then $Q_{4,4}$, $Q_{5,4}$ and $Q_{5,5}$ could be compared to determine further accuracy

Example 3

Table 3.7 lists the values of $f(x) = \ln x$ accurate to the places given. Use Neville's method and four-digit rounding arithmetic to approximate $f(2.1) = \ln 2.1$ by completing the Neville table

Table 3.7

i	x_i	$\ln x_i$
0	2.0	0.6931
1	2.2	0.7885
2	2.3	0.8329

Solution. Because $x - x_0 = 0.1$, $x - x_1 = -0.1$ and $x - x_2 = -0.2$ and we are given $Q_{0,0} = 0.6931$, $Q_{1,0} = 0.7885$ and $Q_{2,0} = 0.8329$, we have

$$Q_{1,1} = \frac{1}{0.2} [(0.1)0.7885 - (-0.1)0.6931] = \frac{0.1482}{0.2} = 0.7410$$

and

$$Q_{2,1} = \frac{1}{0.1} [(-0.1)0.8329 - (-0.2)0.7885] = \frac{0.07441}{0.1} = 0.7441$$

The final approximation we can obtain from this data is

$$Q_{2,1} = \frac{1}{0.3} [(0.1)0.7441 - (-0.2)0.7410] = \frac{0.2276}{0.3} = 0.7420$$

These values are shown in Table 3.8.

Table 3.8

i	x_i	x - x_i	Q_{i0}	Q_{i1}	Q_{i2}
0	2.0	0.1	0.6931		
1	2.2	-0.1	0.7885	0.7410	
2	2.3	-0.2	0.8329	0.7441	0.7420

Questions 3.2

1. Use Neville's method to obtain the approximations for Lagrange interpolating polynomials of degrees one, two, and three to approximate each of the following:

a. $f(8.4)$ if $f(8.1) = 16.94410$, $f(8.3) = 17.56492$, $f(8.6) = 18.50515$, $f(8.7) = 18.82091$

b. $f\left(-\frac{1}{3}\right)$ if $f(-0.75) = -0.07181250$, $f(-0.5) = -0.02475000$, $f(-0.25) = 0.33493750$, $f(0) = 1.10100000$

c. $f(0.25)$ if $f(0.1) = 0.62049958$, $f(0.2) = -0.28398668$, $f(0.3) = 0.00660095$, $f(0.4) = 0.24842440$

d. $f(0.9)$ if $f(0.6) = -0.17694460$, $f(0.7) = 0.01375227$, $f(0.8) = 0.22363362$, $f(1.0) = 0.65809197$

2. Use Neville's method to approximate $\sqrt{3}$ with the following functions and values.

a. $f(x) = 3^x$ and the values $x_0 = -2$, $x_1 = -1$, $x_2 = 0$, $x_3 = 1$ and $x_4 = 2$.

b. $f(x) = \sqrt{x}$ and the values $x_0 = 0$, $x_1 = 1$, $x_2 = 2$, $x_3 = 4$ and $x_4 = 5$.

c. Compare the accuracy of the approximation in parts (a) and (b).

3. Let $P_3(x)$ be the interpolating polynomial for the data $(0, 0)$, $(0.5, y)$, $(1, 3)$, and $(2, 2)$. Find y if the coefficient of x^3 in $P_3(x)$ is 6.

4. Neville's method is used to approximate $f(0.4)$, giving the following table.

$X_0 = 0$	$P_0 = 1$			
$X_1 = 0.25$	$P_1 = 2$	$P_{01} = 2.6$		
$X_2 = 0.5$	P_2	$P_{1.2}$	$P_{0.1,2}$	
$X_3 = 0.75$	$P_3 = 8$	$P_{2.3} = 2.4$	$P_{1.2,3} = 2.96$	$P_{0.1,2,3} = 3.016$

Determine $P_2 = f(0.5)$.

5. Suppose $x_j = j$, for $j = 0, 1, 2, 3$, and it is known that

$P_{0,1}(x) = 2x + 1$, $P_{0,2}(x) = x + 1$, and $P_{1,2,3}(2.5) = 3$.

Find $P_{0,1,2,3}(2.5)$.

Answers

1.

a.

b.

n	$x_0, x_1, \dots, x_n$	$P_n(8.4)$
1	8,3,8.6	17.87833
2	8,3, 8.6, 8.7	17.87716
3	8,3,8.6, 8.7, 8.1	17.87714

n	$x_0, x_1, \dots, x_n$	$P_n(-1/3)$
1	-0.5, -0.25	0.21504167
2	-0.5, -0.25, 0.0	0.16988889
3	-0.5, -0.25, 0.0, -0.75	0.17451852

c.			d.		
n	$x_0, \dots, x_n$	$P_n(0.25)$	n	$x_0, \dots, x_n$	$P_n(0.9)$
1	0.2, 0.3	-0.13869287	1	0.8, 1.0	0.44086280
2	0.2, 0.3, 0.4	-0.132559734	2	0.8, 1.0, 0.7	0.43841352
3	0.2, 0.3, 0.4, 0.1	-0.13277477	3	0.8, 1.0, 0.7, 0.6	0.44198500

2. a. We have $\sqrt{3} \approx P_4(\frac{1}{2}) = 1.7083$.

b. We have $\sqrt{3} \approx P_4(3) = 1.690607$.

c. Absolute error in part (a) is approximately 0.0237, and the absolute error in part (b) is 0.0414, so part (a) is more accurate.

3. $y = 4.25$

4. $P_2 = f(0.5) = 4$

5. $P_{0,1,2,3}(2.5) = 2.875$

3.3 Divided Differences

Iterated interpolation was used in the previous section to generate successively higher-degree polynomial approximations at a specific point. Divided-difference methods introduced in this section are used to successively generate the polynomials themselves.

Definition

For a given sequence $\{p_n\}_{n=0}^{\infty}$ the forward difference Δp_n (read "delta p_n ") is defined by $\Delta p_n = p_{n+1} - p_n$, for $n \geq 0$.

Higher powers of the operator Δ are defined recursively by

$$\Delta^k p_n = \Delta(\Delta^{k-1} p_n), \text{ for } k \geq 2.$$

The definition implies that

$$\begin{aligned} \Delta^2 p_n &= \Delta(p_{n+1} - p_n) = \Delta p_{n+1} - \Delta p_n \\ &= (p_{n+2} - p_{n+1}) - (p_{n+1} - p_n). \end{aligned}$$

$$\text{So } \Delta^2 p_n = p_{n+2} - 2p_{n+1} + p_n.$$

Divided Differences

Suppose that $P_n(x)$ is the n th interpolating polynomial that agrees with the function f at the distinct numbers $x_0, x_1, \dots, x_n$. Although this polynomial is unique, there are alternate algebraic representations that are useful in certain situations. The divided differences of f with respect to $x_0, x_1, \dots, x_n$ are used to express $P_n(x)$ in the form

$$\begin{aligned} P_n(x) &= a_0 + a_1(x - x_0) + a_2(x - x_0)(x - x_1) + \dots \\ &\quad + a_n(x - x_0) \dots (x - x_{n-1}), \end{aligned} \quad (5)$$

for appropriate constants a_0, a_1, a_n . To determine the first of these constants, a_0 , note that if $P_n(x)$ is written in the form of Eq. (5), then evaluating $P_n(x)$ at x_0 leaves only the constant term a_0 ; that is,

$$a_0 = P_n(x_0) = f(x_0);$$

Similarly, when $P(x)$ is evaluated at x_1 , the only nonzero terms in the evaluation of $P_n(x_1)$ are the constant and linear terms

$$f(x_0) + a_1(x_1 - x_0) = P_n(x_1) = f(x_1)$$

So,

$$a_1 = \frac{f(x_1) - f(x_0)}{x_1 - x_0} \quad (6)$$

We now introduce the divided-difference notation.

The zeroth divided difference of the function f with respect to x_i , denoted $f[x_i]$, is simply the value of f at x_i :

$$f[x_i] = f(x_i) \quad (7)$$

The remaining divided differences are defined recursively; the first divided difference of f with respect to x_i and x_{i+1} is denoted $f[x_i, x_{i+1}]$ and defined as

$$f[x_i, x_{i+1}] = \frac{f[x_{i+1}] - f[x_i]}{x_{i+1} - x_i} \quad (8)$$

The second divided difference, $f[x_i, x_{i+1}, x_{i+2}]$, is defined as

$$f[x_i, x_{i+1}, x_{i+2}] = \frac{f[x_{i+1}, x_{i+2}] - f[x_i, x_{i+1}]}{x_{i+2} - x_i}$$

Similarly, after the $(k - 1)$ st divided differences,

$f[x_i, x_{i+1}, \dots, x_{i+k-1}, x_{i+k}]$ and $f[x_{i+1}, x_{i+2}, \dots, x_{i+k-1}, x_{i+k}]$ have been determined, the **k th divided difference** relative to $x_i, x_{i+1}, x_{i+2}, \dots, x_{i+k}$ is

$$= \frac{f[x_{i+1}, x_{i+2}, \dots, x_{i+k}] - f[x_i, x_{i+1}, \dots, x_{i+k-1}]}{x_{i+k} - x_i} \quad (9)$$

The process ends with the single n th divided difference,

$$f[x_0, x_1, \dots, x_n] = \frac{f[x_1, x_2, \dots, x_n] - f[x_0, x_1, \dots, x_{n-1}]}{x_n - x_0}$$

Because of Eq. (6), we can write $a_1 = f[x_0, x_1]$, just as a_0 can be expressed as $a_0 = f(x_0) = f[x_0]$. Hence, the interpolating polynomial in Eq. (5) is

$$P_n(x) = f[x_0] + f[x_0, x_1](x - x_0) + a_2(x - x_0)(x - x_1) + \dots + a_n(x - x_0)(x - x_1) \dots (x - x_{n-1}).$$

As might be expected from the evaluation of a_0 and a_1 , the required constants are

$$a_k = f[x_0, x_1, x_2, \dots, x_k],$$

for each $k = 0, 1, \dots, n$. So, $P_n(x)$ can be rewritten in a form called **Newton's Divided Difference**:

$$P_n(x) = f[x_0] + \sum_{k=1}^n f[x_0, x_1, \dots, x_k](x - x_0) \dots (x - x_{k-1}) \quad (10)$$

The value of $f[x_0, x_1, \dots, x_k]$ is independent of the order of the numbers $x_0, x_1, \dots, x_k$.

The generation of the divided differences is outlined in Table 3.9. Two fourth and one fifth difference can also be determined from these data.

Table 3.9

x	$f(x)$	First divided difference	Second divided difference	Thrid divided difference
x_0	$f[x_0]$			
		$\frac{f[x_0, x_1] - f[x_0]}{x_1 - x_0}$		
x_1	$f[x_1]$		$\frac{f[x_0, x_1, x_2] - f[x_0, x_1]}{x_2 - x_0}$	
		$\frac{f[x_1, x_2] - f[x_1]}{x_2 - x_1}$		$\frac{f[x_0, x_1, x_2, x_3] - f[x_0, x_2]}{x_3 - x_0}$
x_2	$f[x_2]$		$\frac{f[x_1, x_2, x_3] - f[x_1, x_2]}{x_3 - x_1}$	

	$\frac{f[x_2, x_3] - f[x_2]}{x_3 - x_2}$	$\frac{f[x_1, x_2, x_3, x_4] - f[x_1, x_2, x_3]}{x_4 - x_1}$
$x_3 \quad f[x_3]$	$\frac{f[x_2, x_3, x_4] - f[x_2, x_3]}{x_4 - x_2}$	
	$\frac{f[x_3, x_4] - f[x_3]}{x_4 - x_3}$	$\frac{f[x_2, x_3, x_4, x_5] - f[x_2, x_3, x_4]}{x_5 - x_2}$
$x_4 \quad f[x_4]$	$\frac{f[x_3, x_4, x_5] - f[x_3, x_4]}{x_5 - x_3}$	
	$\frac{f[x_4, x_5] - f[x_4]}{x_5 - x_4}$	
$x_5 \quad f[x_5]$		

Example 1. Complete the divided difference table for the data used in Table 3.10 and construct the interpolating polynomial that uses all these data.

Table 3.10

x	$f(x)$
-----	--------

1.0	0.7651977
1.3	0.6200860
1.6	0.4554022
1.9	0.2818186
2.2	0.1103623

Solution. The first divided difference involving x_0 and x_1 is

$$f[x_0, x_1] = \frac{f[x_1] - f[x_0]}{x_1 - x_0} = \frac{0.6200860 - 0.7651977}{1.3 - 1.0} = -0.4837057$$

The remaining first divided differences are found in a similar manner and are shown in the fourth column in Table 3.11.

Table 3.11

i	x_i	$f[x_i]$	$f[x_{i-1}, x_i]$	$f[x_{i-2}, x_{i-1}, x_i]$	$f[x_{i-3}, \dots, x_i]$	$f[x_{i-4}, \dots, x_i]$
0	1.0	0.7651977				
			-0.4837057			

1	1.3	0.6200860	-	0.1087339	
			-0.5489460		0.0658784
2	1.6	0.4554022	-	0.0494433	0.0018251
			-0.5786120		0.0680685
3	1.9	0.2818186		0.0118183	
			-		
			0.57175210		
4	2.2	0.1103623			

The second divided difference involving x_0, x_1 and x_2 is

$$\begin{aligned}
 f[x_0, x_1, x_2] &= \frac{f[x_1, x_2] - f[x_0, x_1]}{x_2 - x_0} \\
 &= \frac{-0.5489460 - (-0.4837057)}{1.6 - 1.0} \\
 &= -0.1087339
 \end{aligned}$$

The remaining second divided differences are shown in the fifth column of Table 3.11. The third divided difference involving x_0, x_1, x_2 and x_3 and the fourth divided difference involving all the data points are, respectively,

$$\begin{aligned}
 f[x_0, x_1, x_2, x_3] &= \frac{f[x_1, x_2, x_3] - f[x_0, x_1, x_2]}{x_3 - x_0} \\
 &= \frac{-0.0494433 - (-0.1087339)}{1.9 - 1.0}
 \end{aligned}$$

$$= 0.0658784,$$

and

$$\begin{aligned} f[x_0, x_1, x_2, x_3, x_4] &= \frac{f[x_1, x_2, x_3, x_4] - f[x_0, x_1, x_2, x_3]}{x_4 - x_0} \\ &= \frac{0.0680685 - 0.0658784}{2.2 - 1.0} \\ &= 0.0018251. \end{aligned}$$

All the entries are given in Table 3.11.

The coefficients of the Newton forward-divided-difference form of the interpolating polynomial are along the diagonal in the table. This polynomial is

$$\begin{aligned} P_4(x) &= 0.7651977 - 0.4837057(x - 1.0) - 0.1087339(x \\ &\quad - 1.0)(x - 1.3) \\ &\quad + 0.0658784(x - 1.0)(x - 1.3)(x - 1.6) \\ &\quad + 0.0018251(x - 1.0)(x - 1.3)(x - 1.6)(x - 1.9). \end{aligned}$$

Theorem 3.6

Suppose that $f \in C^n[a, b]$ and $x_0, x_1, \dots, x_n$ are distinct numbers in $[a, b]$. Then a number ξ exists in (a, b) with

$$f[x_0, x_1, \dots, x_n] = \frac{f^{(n)}(\xi)}{n!}.$$

Proof. Let

$$g(x) = f(x) - P_n(x)$$

Since $f(x_i) = P_n(x_i)$ for each $i = 0, 1, \dots, n$, the function g has $n + 1$ distinct zeros in $[a, b]$. Generalized Rolle's Theorem implies that a number ξ in (a, b) exists with $g^{(n)}(\xi) = 0$, so

$$0 = f^{(n)}(\xi) - P_n^{(n)}(\xi)$$

Since $P_n(x)$ is a polynomial of degree n whose leading coefficient is $f[x_0, x_1, \dots, x_n]$,

$$P_n^{(n)}(x) = n! f[x_0, x_1, \dots, x_n],$$

For all values of x . As a consequence,

$$f[x_0, x_1, \dots, x_n] = \frac{f^{(n)}(\xi)}{n!}.$$

Newton's divided-difference formula can be expressed in a simplified form when the nodes are arranged consecutively with equal spacing. In this case, we introduce the notation $h = x_{i+1} - x_i$, for each $i = 0, 1, \dots, n - 1$ and let $x = x_0 + sh$. Then the difference $x - x_i$ is $x - x_i = (s - i)h$. So, Eq. (10) becomes

$$\begin{aligned} P_n^{(n)}(x) &= P_n(x_0 + sh) \\ &= f[x_0] + shf[x_0, x_1] + s(s-1)h^2f[x_0, x_1, x_2] \\ &\quad + \dots + s(s-1) \dots (s-n+1)h^n f[x_0, x_1, \dots, x_n] \\ &= f[x_0] + \sum_{k=1}^n \frac{s(s-1) \dots (s-k+1)}{k!}, \end{aligned}$$

we can express $P_n(x)$ compactly as

$$\begin{aligned} P_n(x) &= P_n(x_0 + sh) \\ &= f[x_0] \\ &\quad + \sum_{k=1}^n \binom{s}{k} k! h^k f[x_0, x_1, \dots, x_k] \end{aligned} \quad (11)$$

Forward Differences

The Newton forward-difference formula is constructed by making use of the forward difference notation Δ . With this notation.

$$f[x_0, x_1] = \frac{f(x_1) - f(x_0)}{x_1 - x_0} = \frac{1}{h} (f(x_1) - f(x_0)) = \frac{1}{h} \Delta f(x_0)$$

$$f[x_0, x_1, x_2] = \frac{1}{2h} \left[\frac{\Delta f(x_1) - \Delta f(x_0)}{h} \right] = \frac{1}{2h^2} \Delta^2 f(x_0)$$

and, in general,

$$f[x_0, x_1, \dots, x_k] = \frac{1}{k! h^k} \Delta^k f(x_0)$$

Since $f[x_0, x_1, \dots, x_k] = \frac{1}{k! h^k} \Delta^k f(x_0)$.

Since $f[x_0] = f(x_0)$ Eq. (11) has the following form.

Newton Forward-Difference Formula

$$P_n(x) = f(x_0) + \sum_{k=1}^n \binom{s}{k} \Delta^k f(x_0) \quad (12)$$

Backward Differences

If the interpolating nodes are reordered from last to first as $x_n, x_{n-1}, \dots, x_0$, we can write the interpolatory formula as

$$\begin{aligned} P_n(x) = & f[x_n] + f[x_n, x_{n-1}](x - x_n) \\ & + f[x_n, x_{n-1}, x_{n-2}](x - x_{n-1}) \\ & + \dots + f[x_n, \dots, x_0](x - x_n)(x - x_{n-1}) \dots (x - x_1). \end{aligned}$$

If, in addition, the nodes are equally spaced with $x = x_n + sh$ and $x = x_i + (s + n - i)h$, then

$$\begin{aligned} P_n(x) = & P_n(x_n + sh) \\ = & f[x_n] + shf[x_n, x_{n-1}] + s(s+1)h^2f[x_n, x_{n-1}, x_{n-2}] \\ & + \dots \\ & + s(s+1) \dots (s+n-1)h^n f[x_n, \dots, x_0] \end{aligned}$$

This is used to derive a commonly applied formula known as the **Newton backward difference formula**. To discuss this formula, we need the following definition.

Definition 3.7. Given the sequence $\{p_n\}_{n=0}^{\infty}$, define the backward-difference ∇p_n (read nabla p_n) by

$$\nabla p_n = p_n - p_{n-1}, \text{ for } n \geq 1$$

Higher powers are defined recursively by

$$\nabla^k p_n = \nabla(\nabla^{k-1} p_n), \text{ for } k \geq 2.$$

Definition 3.7 implies that

$$f[x_n, x_{n-1}] = \frac{1}{h} \nabla f(x_n), f[x_n, x_{n-1}, x_{n-2}] = \frac{1}{2h^2} \nabla^2 f(x_n),$$

and, in general,

$$f[x_n, x_{n-1}, \dots, x_{n-k}] = \frac{1}{k! h^k} \nabla^k f(x_n)$$

Consequently,

$$\begin{aligned} P_n(x) = f[x_n] + s \nabla f(x_n) + \frac{s(s+1)}{2} \nabla^2 f(x_n) + \dots \\ + \frac{s(s+1) \dots (s+n-1)}{n!} \nabla^n f(x_n). \end{aligned}$$

If we extend the binomial coefficient notation to include all real values of s by letting

$$\begin{aligned} \binom{-s}{k} &= \frac{-s(-s-1) \dots (-s-k+1)}{k!} \\ &= (-1)^k \frac{s(s+1) \dots (s+k-1)}{k!} \end{aligned}$$

Then

$$\begin{aligned} P_n(x) = f[x_n] + (-1)^1 \binom{-s}{1} \nabla f(x_n) + (-1)^2 \binom{-s}{2} \nabla^2 f(x_n) \\ + \dots + (-1)^n \binom{-s}{n} \nabla^n f(x_n) \end{aligned}$$

This gives the following result.

Newton Backward-Difference Formula

$$P_n(x) = f[x_n] + \sum_{k=1}^n (-1)^k \binom{-S}{k} \nabla^k f(x_n) \quad (13)$$

Illustration

The divided-difference Table 3.12 corresponds to the data in Example 1

Table 3.12

	First divided difference	Second divided difference	Third divided difference	Fourth divided differences
1. <u>0.7651977</u> 0				
	<u>0.4837057</u>			
1. 0.6200860 3		<u>0.1087339</u>		
	- 0.5489460		<u>0.0658784</u>	
1. 0.4554022 6		- 0.0494433		<u>0.0018251</u>
	- 0.5786120		<u>0.0680685</u>	
1. 0.2818186 9		<u>0.0118183</u>		

$$\underline{\underline{0.5715210}}$$

$$2. \quad \underline{\underline{0.1103623}}$$

$$2$$

If an approximation to $f(1.1)$ is required, the reasonable choice for the nodes would be $x_0 = 1.0, x_1 = 1.3, x_2 = 1.6, x_3 = 1.9$, and $x_4 = 2.2$, since this choice makes the earliest possible use of the data points closest to $x = 1.1$ and also makes use of the fourth divided difference. This implies that $h = 0.3$ and $s = \frac{1}{3}$, so the Newton forward-divided difference formula is used with the divided differences that have a solid underline () in Table 3.12:

$$\begin{aligned} P_4(1.1) &= P_4\left(1.0 + \frac{1}{3}(0.3)\right) \\ &= 0.7651977 + \frac{1}{3}(0.3)(-0.4837057) \\ &\quad + \frac{1}{3}\left(-\frac{2}{3}\right)(0.3)^2(-0.1087339) \\ &\quad + \frac{1}{3}\left(-\frac{2}{3}\right)\left(-\frac{5}{3}\right)(0.3)^3(0.0658784) \\ &\quad + \frac{1}{3}\left(-\frac{2}{3}\right)\left(-\frac{5}{3}\right)\left(-\frac{8}{3}\right)(0.3)^4(0.0018251) \\ &= 0.7196460. \end{aligned}$$

To approximate a value when x is close to the end of the tabulated values, say, $x = 2.0$, we would again like to make the earliest use

of the data points closest to x . This requires using the Newton backward-divided-difference formula with $s = -\frac{2}{3}$ and the divided differences in Table 3.12 that have a wavy underline (). Notice that the fourth divided difference is used in both formulas:

$$\begin{aligned}
 P_4(2.0) &= P_4\left(2.2 - \frac{2}{3}(0.3)\right) \\
 &= 0.1103623 - \frac{2}{3}(0.3)(-0.5715210) \\
 &\quad - \frac{2}{3}\left(\frac{1}{3}\right)(0.3)^2(0.0118183) \\
 &\quad - \frac{2}{3}\left(\frac{1}{3}\right)\left(\frac{4}{3}\right)(0.3)^3(0.0680685) \\
 &\quad - \frac{2}{3}\left(\frac{1}{3}\right)\left(\frac{4}{3}\right)\left(\frac{7}{3}\right)(0.3)^4(0.0018251) \\
 &= 0.2238754
 \end{aligned}$$

Centered Differences

The Newton forward- and backward-difference formulas are not appropriate for approximating $f(x)$ when x lies near the center of the table because neither will permit the highest-order difference to have x_0 close to x . A number of divided-difference formulas are available for this case, each of which has situations when it can be used to maximum advantage. These methods are known as centered-difference formulas. We will consider only one centered-difference formula, Stirling's method.

For the centered-difference formulas, we choose near the point being approximated and label the nodes directly below x_0 as

$x_1, x_2 \dots$ and those directly above as $x_{-1}, x_{-2}, \dots$, With this convention, Stirling's formula is given by

$$\begin{aligned}
 P_n(x) &= P_{2m+1}(x) \\
 &= f[x_0] + \frac{sh}{2} (f[x_{-1}, x_0] + f[x_0, x_1]) \\
 &\quad + s^2 h^2 f[x_{-1}, x_0, x_1] \quad (3.14) \\
 &\quad + \frac{s(s^2 - 1)h^3}{2} f[x_{-2}, x_{-1}, x_0, x_1] \\
 &\quad + f[x_{-1}, x_0, x_1, x_2]) \\
 &\quad + \dots + s^2 (s^2 - 1)(s^2 \\
 &\quad - 4) \dots (s^2 - (m - 1)^2 h^{2m} f[x_{-m}, \dots, x_m] \\
 &\quad + \frac{s(s^2 - 1) \dots (s^2 - m^2) h^{2m+1}}{2} (f[x_{-m-1}, \dots, x_m] \\
 &\quad + f[x_{-m}, \dots, x_{m+1}])
 \end{aligned}$$

if $n = 2m + 1$ is odd. If $n = 2m$ is even, we use the same formula but delete the last line. The entries used for this formula are underlined in Table 3.13.

Table 3.13

x	$f(x)$	First divided difference	Second divided difference	Third divided difference	Fourth divided difference
x_{-2}	$f[x_{-2}]$				

		$f[x_{-2}, x_{-1}]$		
x_{-1}	$f[x_{-1}]$		$f[x_{-2}, x_{-1}, x_0]$	
		$f[x_{-1}, x_0]$		$f[x_{-2}, x_{-1}, x_0, x_1]$
x_0	$f[x_0]$		$f[x_{-1}, x_0, x_1]$	$f[x_{-2}, x_{-1}, x_0, x_1, x_2]$
		$f[x_0, x_1]$		$f[x_1, x_0, x_1, x_2]$
x_1	$f[x_1]$		$f[x_0, x_1, x_2]$	
		$f[x_1, x_2]$		
x_2	$f[x_2]$			

Example 2

Consider the table of data given in the previous examples. Use Stirling's formula to approximate $f(1.5)$ with $x_0 = 1.6$.

Solution. To apply Stirling's formula, we use the underlined entries in the difference Table 3.14.

Table 3.14

x	$f(x)$	First divided difference	Second divided difference	Third divided difference	Fourth divided differences
1.0	0.7651977				
		-0.4837057			
1.3	0.6200860		-0.1087339		

	<u>-0.5489460</u>	<u>0.0658784</u>	
1.6	<u>0.4554022</u>	<u>-0.0494433</u>	<u>0.0018251</u>
	<u>-0.5786120</u>	<u>0.0680685</u>	
1.9	0.2818186	0.0118183	
	-0.5715210		
2.2	0.1103623		

The formula, with $h = 0.3$, $x_0 = 1.6$, and $s = -\frac{1}{3}$, becomes

$$\begin{aligned}
 f(1.5) &\approx P_4 \left(1.6 + \left(-\frac{1}{3} \right) (0.3) \right) \\
 &= 0.4554022 + \left(-\frac{1}{3} \right) \left(\frac{0.3}{2} \right) ((-0.5489460) \\
 &\quad + (-0.5786120)) \\
 &\quad + \left(-\frac{1}{3} \right)^2 (0.3)^2 (-0.0494433) \\
 &\quad + \frac{1}{2} \left(-\frac{1}{3} \right) \left(\left(-\frac{1}{3} \right)^2 - 1 \right) (0.3)^3 (0.0658784 \\
 &\quad + 0.0680685) \\
 &\quad + \left(-\frac{1}{3} \right)^2 \left(\left(-\frac{1}{3} \right)^2 - 1 \right) (0.3)^4 (0.0018251) \\
 &= 0.5118200
 \end{aligned}$$

Example 3. Using Newton's dividend difference formula, determine the interpolating polynomial for the table given below and find approximate $f(2.1)$

x	0	1	2	4
$f(x)$	1	1	2	5

Solution.

x	$f(x)$	First dividend difference	Second dividend difference	Third dividend difference
0	1	0		
1	1		$+1/2$	
2	2	1		$-1/12$
		$3/2$	$+1/6$	
4	5			

$$\begin{aligned}
 P_3(x) &= 1 + 0(x-0) + \frac{1}{2}(x-0)(x-1) \\
 &\quad - \frac{1}{12}(x-0)(x-1)(x-2) \\
 &= -\frac{1}{12}x^3 + \frac{3}{4}x^2 - \frac{2}{3}x + 1
 \end{aligned}$$

$$\begin{aligned}
 \text{Now } P_3(2.1) &= -\frac{1}{12}(2.1)^3 + \frac{3}{4}(2.1)^2 - \frac{2}{3}(2.1) + 1 \\
 &= 2.135
 \end{aligned}$$

Example 4. Compute the dividend difference table for the data in the table given and construct the interpolating polynomial that uses all these data

x	-1	0	3	6	7
$f(x)$	3	-6	39	822	1611

Solution.

The divided difference table is as follows

x	$f(x)$	First dividend difference	Second dividend difference	Third dividend difference
-1	3	-9		
0	-6	15	6	
3	39	261	41	1
6	822	789	132	
7	1611			

The interpolating polynomial is

$$\begin{aligned}
 P_4(x) &= 3 - 9(x + 1) + 6(x + 1)(x) + 5(x + 1)(x)(x - 3) \\
 &\quad + 1(x + 1)(x)(x - 3)(x - 6) \\
 &= x^4 - 3x^3 + 5x^2 - 6.
 \end{aligned}$$

Example 5. Using Newton forward difference formula find an approximation to $f(2)$ for the data:

x	1	3	5	7
$f(x)$	24	120	336	720

Solution.

We have the newton forward difference formula

$$P_n(x) = f(x_0) + \sum_{k=1}^n \binom{S}{k} \Delta^k f(x_0) \text{ and}$$

$$P_n(x_0 + sh) = f(x_0) + \sum_{k=1}^n \binom{S}{k} k! h^k f[x_0, x_1, \dots, x_k]$$

Construction of divided difference table:

x	$f(x)$	First divided difference	Second divided difference	Third divided difference
1	24	48		
3	120	108	15	
5	336	192	21	1
7	720			

To find an approximation of $f(2)$, we take $h = 2$ and $s = \frac{1}{2}$ so that $sh = 1$ and $P_3(2) = P_3\left(1 + \frac{1}{2}(2)\right)$

$$\text{Thus } P_3(2) = P_3\left(1 + \frac{1}{2}(2)\right) = f(1) + \sum_{k=1}^3 \binom{S}{k} k! h^k f[x_0, x_1, \dots, x_k]$$

$$= 24 + s \times 1 \times h \times 48 + s(s-1) \times 2 \times h^2 \times 15$$

$$\begin{aligned}
 & +s(s-1)(s-2) \times 6 \times h^3 \times 1 \\
 & = 24 + \frac{1}{2} \times 2 \times 48 + \frac{1}{2} \times -\frac{1}{2} \times 2 \times 4 \times 15 + \frac{1}{2} \\
 & \quad \times -\frac{1}{2} \times -\frac{3}{2} \times 6 \times 8 \\
 & = 60.
 \end{aligned}$$

Example 6. Consider the data in Example 5 and find an approximation for $f(6)$

Solution.

By Newton backward differences,

$$\begin{aligned}
 P_n(x) &= P_n(x_n + sh) \\
 &= f[x_n] + sh f[x_n, x_{n-1}] \\
 &\quad + s(s+1)h^2 f[x_n, x_{n-1}, x_{n-2}] + \cdots \\
 &\quad + s(s+1) \dots (s+n-1)h^k f[x_n, \dots, x_0]
 \end{aligned}$$

Hence for approximating $f(6)$, we take $h = 2, s = -\frac{1}{2}$ and thus

$$\begin{aligned}
 P_3(6) &= P_3\left(7 - \frac{1}{2}(2)\right) \\
 &= 720 + \left(-\frac{1}{2}\right) 2 \times 19 + \left(-\frac{1}{2}\right)\left(\frac{1}{2}\right) \times 4 \times 21 \\
 &\quad + \left(-\frac{1}{2}\right)\left(\frac{1}{2}\right)\left(\frac{3}{2}\right) \times 8 \times 1 \\
 &= 504.
 \end{aligned}$$

Questions 3.3

1. Construct an interpolating polynomial which takes on the values -3,3,11,27,57 and 107 when $x = 0,1,2,3,4$ and 5 respectively. Also find an approximation of $f(2.5)$.

2. Use the Newton forward difference formula to give an approximation to $f(2.05)$ for the following data $f(2) = 1.414212, f(2.1) = 1.449138, f(2.2) = 1.483240, f(2.3) = 1.516575, f(2.4) = 1.549193$.

3. Use Newton backward difference formula to given an approximation to $f(7.5)$ for the following data

x	1	2	3	4	5	6	7	8
$f(x)$	1	8	27	64	125	216	343	512

4. Use Eq. (10) to construct interpolating polynomials of degree one, two, and three for the following data. Approximate the specified value using each of the polynomials.

a. $f(8.4)$ if $f(8.1) = 16.94410, f(8.3) = 17.56492, f(8.6) = 18.50515, f(8.7) = 18.82091$

b. $f(0.9)$ if $f(0.6) = -0.17694460, f(0.7) = 0.01375227, f(0.8) = 0.22363362, f(1.0) = 0.65809197$

5. Use the Newton forward-difference formula to construct interpolating polynomials of degree one, two, and three for the following data. Approximate the specified value using each of the polynomials.

a. $f\left(-\frac{1}{3}\right)$ if $f(-0.75) = -0.07181250, f(-0.5) = -0.02475000, f(-0.25) = 0.33493750, f(0) = 1.10100000$

b. $f(0.25)$ if $f(0.1) = -0.62049958, f(0.2) = -0.28398668, f(0.3) = 0.00660095, f(0.4) = 0.24842440$

6. Use the Newton backward-difference formula to construct interpolating polynomials of degree one, two, and three for the following data. Approximate the specified value using each of the polynomials.

a. $f\left(-\frac{1}{3}\right)$ if $f(-0.75) = -0.07181250, f(-0.5) = -0.02475000, f(-0.25) = 0.33493750, f(0) = 1.10100000$

b. $f(0.25)$ if $f(0.1) = -0.62049958, f(0.2) = -0.28398668, f(0.3) = 0.00660095, f(0.4) = 0.24842440$

7. Construct the interpolating polynomial of degree three for the unequally spaced points given in the following table;

x	$f(x)$
-0.1	5.30000
0.0	2.00000
0.2	3.19000
0.3	1.00000

8. a. Approximate $f(0.05)$ using the following data and the Newton forward-difference formula

x	0.0	0.2	0.4	0.6	0.8
$f(x)$	1.00000	1.22140	1.49182	1.82212	2.22554

b. Use the Newton backward-difference formula to approximate $f(0.65)$.

c. Use Stirling's formula to approximate $f(0.43)$.

9. The following data are given for a polynomial $P(x)$ of unknown degree

x	0	1	2
$P(x)$	2	-1	4

Determine the coefficient of x^2 in $P(x)$ if all third-order forward differences are 1.

Answers

1. $P_3(x) = x^3 - 2x^2 + 7x - 3, 17.625$

2. 1.431783

3. 421.875

4. a. $P_1(x) = 16.9441 + 3.1041(x - 8.1); P_1(8.4) = 17.87533; P_2(x) = P_1(x) + 0.06(x - 8.1)(x - 8.3); P_2(8.4) = 17.87713; P_3(x) = P_2(x) - 0.00208333(x - 8.1)(x - 8.3)(x - 8.6); P_3(8.4) = 17.87714$

b. $P_1(x) = -0.1769446 + 1.9069687(x - 0.6); P_1(0.9) = 0.395146;$

$$P_2(x) = P_1(x) + 0.959224(x - 0.6)(x - 0.7); P_2(0.9) = 0.4526995;$$

$$P_3(x) = P_2(x) - 1.785741(x - 0.6)(x - 0.7)(x - 0.8); P_3(0.9) = 0.4419850$$

5. In the following equations, we have $s = \frac{1}{h}(x - x_0)$.

a. $P_1(s) = -0.718125 - 0.0470625s; P_1\left(-\frac{1}{3}\right) = -0.006625$

$$P_2(s) = P_1(s) + 0.312625s(s - 1)/2; P_2\left(-\frac{1}{3}\right) = 0.1803056$$

$$P_3(s) = P_2(s) + 0.09375s(s - 1)(s - 2)/6; P_3\left(-\frac{1}{3}\right) = 0.1745185$$

b. $P_1(s) = 0.62049958 + 0.3365129s; P_1(0.25) = -0.1157302$

$$P_2(s) = P_1(s) - 0.04592527s(s - 1)/2; P_2(0.25) = -0.1329522$$

$$P_3(s) = P_2(s) - 0.00283891s(s - 1)(s - 2)/6; P_3(0.25) = -0.1327748$$

6. In the following equations, we have $s = \frac{1}{h}(x - x_n)$.

a. $P_1(s) = 1.101 + 0.7660625s; f\left(-\frac{1}{3}\right) \approx P_1\left(-\frac{4}{3}\right) = 0.07958333;$

$$P_2(s) = P_1(s) + 0.406375s(s + 1)/2; f\left(-\frac{1}{3}\right) \approx P_2\left(-\frac{4}{3}\right) = 0.1698889;$$

$$P_3(s) = P_2(s) + 0.09375s(s+1)(s+2)/6; f\left(-\frac{1}{3}\right) \\ \approx P_3\left(-\frac{4}{3}\right) = 0.1745185$$

$$\text{b. } P_1(s) = 0.2484244 + 0.2418235s; f(0.25) \approx P_1(-1.5) = -0.1143108$$

$$P_2(s) = P_1(s) - 0.04876419s(s+1)/2; f(0.25) \\ \approx P_2(-1.5) = -0.1325973$$

$$P_3(s) = P_2(s) - 0.0028389s(s+1)(s+2)/6; f(0.25) \\ \approx P_3(-1.5) = -0.1327748$$

$$7. \quad P_3(x) = 5.3 - 33(x+0.1) + 129.8\bar{3}(x+0.1)x - 556.\bar{6}(x+0.1)x(x-0.2)$$

$$8. \text{ a. } f(0.05) \approx 1.05126$$

$$\text{b. } f(0.65) \approx 1.91555$$

$$\text{c. } f(0.43) \approx 1.53725$$

$$9. \text{ The coefficient of } x^2 \text{ is } 3.5.$$

CHAPTER 4**NUMERICAL DIFFERENTIATION AND
INTEGRATION**

Introduction

We know that given any continuous function defined on a closed interval, there exists a polynomial that is arbitrarily close to the function at every point in the interval. Also, the derivatives and integrals of polynomials are easily obtained and evaluated. It should not be surprising, then, that many procedures for approximating derivatives and integrals use the polynomials that approximate the function.

Numerical Differentiation

The derivative of the f at x_0 is

$$f'(x_0) = \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h}.$$

This formula gives an obvious way to generate an approximation to $f'(x_0)$; simply compute

$$\frac{f(x_0 + h) - f(x_0)}{h}$$

for small values of h . Although this may be obvious, it is not very successful due to our old nemesis round-off error. But it is certainly a place to start.

To approximate $f'(x_0)$, suppose first that $x_0 \in (a, b)$, where $f \in C^2[a, b]$, and that $x_1 = x_0 + h$ for some $h \neq 0$ that is sufficiently small to ensure that $x_1 \in [a, b]$. We construct the first Lagrange polynomial $P_{0,1}(x)$ for f determined by x_0 and x_1 , with its error term:

$$\begin{aligned} f(x) &= P_{0,1}(x) + \frac{(x-x_0)(x-x_1)}{2!} f''(\xi(x)), \\ &= \frac{f(x_0)(x-x_0-h)}{-h} + \frac{f(x_0+h)(x-x_0)}{h} + \frac{(x-x_0)(x-x_0-h)}{2} f''(\xi(x)), \end{aligned}$$

For some $\xi(x)$ between x_0 and x_1 . Differentiating gives

$$\begin{aligned} f'(x) &= \frac{f(x_0+h) - f(x_0)}{h} \\ &\quad + D_x \left[\frac{(x-x_0)(x-x_0-h)}{2} f''(\xi(x)) \right] \\ &= \frac{f(x_0+h) - f(x_0)}{h} + \frac{2(x-x_0)-h}{2} f''(\xi(x)) \\ &\quad + \frac{(x-x_0)(x-x_0-h)}{2} D_x f''(\xi(x)) \end{aligned}$$

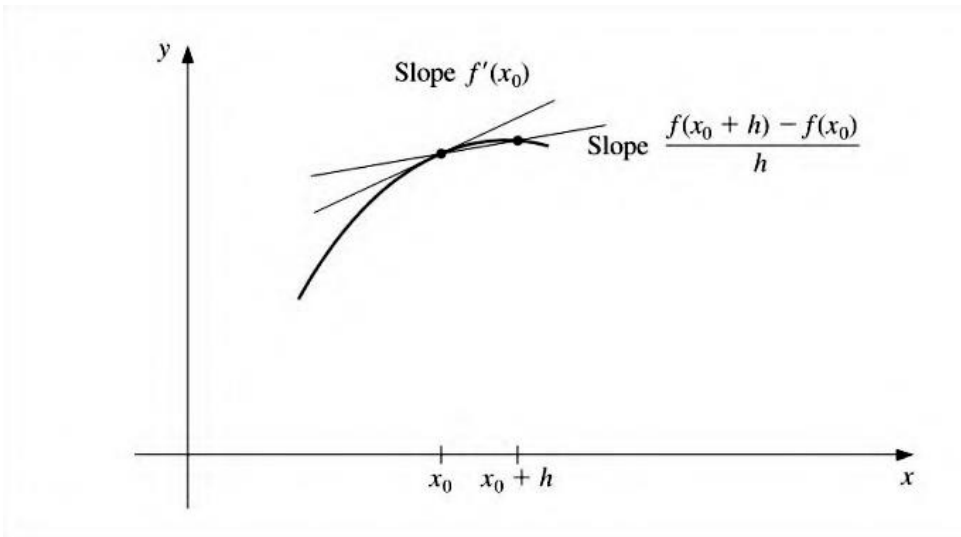
Deleting the terms involving $\xi(x)$ gives

$$f'(x) \approx \frac{f(x_0+h) - f(x_0)}{h}.$$

One difficulty with this formula is that we have no information about $D_x f''(\xi(x))$, so the truncation error cannot be estimated. When x is x_0 , however, the coefficient of $D_x f''(\xi(x))$ is 0, and the formula simplifies to

$$\begin{aligned}
 & f'(x) \\
 &= \frac{f(x_0 + h) - f(x_0)}{h} \\
 &+ \frac{h}{2} f''(\xi(x))
 \end{aligned} \tag{1}$$

For small values of h , the difference quotient $[f(x_0 + h) - f(x_0)]$ can be used to approximate $f'(x_0)$ with an error bounded by $|h|/2$, where M is a bound on $|f''(x)|$ for x between x_0 and $x_0 + h$. This formula is known as the **forward-difference formula** if $h > 0$ (see Figure 4.1) and the **backward-difference formula** if $h < 0$.



Example 1. Use the forward-difference formula to approximate the derivative of $f(x) = \ln x$ at $x_0 = 1.8$ using $h = 0.1$, $h = 0.05$, and $h = 0.01$ and determine bounds for the approximation errors.

Solution. The forward-difference formula

$$\frac{f(1.8 + h) - f(1.8)}{h}$$

With $h = 0.1$ gives

$$\frac{\ln 1.9 - \ln 1.8}{0.1} = \frac{0.64185389 - 0.58778667}{0.1} = 0.5406722.$$

Because $f''(x) = -\frac{1}{x^2}$ and $1.8 < \xi < 1.9$, a bound for this approximation error is

$$\frac{|hf''(\xi)|}{2} = \frac{|h|}{2\xi^2} < \frac{0.1}{2(1.8)^2} = 0.0154321.$$

The approximation and error bounds when $h = 0.05$ and $h = 0.01$ are found in a similar manner, and the results are shown in Table 4.1.

Table 4.1.

h	$f(1.8 + h)$	$\frac{f(1.8 + h) - f(1.8)}{h}$	$\frac{[h]}{2(1.8)^2}$
0.1	0.641853891	0.5406722	0.0154321
0.05	0.61518564	0.5479795	0.0077160
0.01	0.59332685	0.5540180	0.0015432

Since $f'(x) = 1/x$, the exact value of $f'(1.8)$ is $0.55\bar{5}$, and in this case the error bounds are quite close to the true approximation error.

To obtain general derivative approximation formulas, suppose that $\{x_0, x_1, \dots, x_n\}$ are $(n + 1)$ distinct numbers in some interval I and that $f \in C^{n+1}(I)$. From Theorem 3.3 in the chapter 3.3, we have

$$f(x) = \sum_{k=0}^n f(x_k) L_k(x) + \frac{(x - x_0) \dots (x - x_n)}{(n + 1)!} f^{(n+1)}(\xi(x)),$$

for some $\xi(x)$ in I , where $L_k(x)$ denotes the k th Lagrange coefficient polynomial for f at $x_0, x_1, \dots, x_n$. Differentiating this expression gives

$$f'(x) = \sum_{k=0}^n f(x_k) L'_k(x) + D_x \left[\frac{(x-x_0)\dots(x-x_n)}{(n+1)!} \right] f^{(n+1)}(\xi(x)) + \frac{(x-x_0)\dots(x-x_n)}{(n+1)!} D_x [f^{(n+1)}(\xi(x))].$$

We again have a problem estimating the truncation error unless x is one of the numbers x_j . In this case, the term multiplying $D_x[f^{(n+1)}(\xi(x))]$ is 0, and the formula becomes

$$f'(x_j) = \sum_{k=0}^n f(x_k) L'_k(x_j) + \frac{f^{(n+1)}(\xi(x_j))}{(n + 1)!} \prod_{\substack{k=0 \\ k \neq j}}^n (x_j - x_k), \quad (2)$$

which is called an **(n + 1)-point formula** to approximate $f'(x_j)$.

We first derive some useful three-point formulas and consider aspects of their errors. Because

$$L_0(x) = \frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)}, \quad \text{we have } L'_0(x) = \frac{2x-x_1-x_2}{(x_0-x_1)(x_0-x_2)}$$

Similarly,

$$L'_1(x) = \frac{2x-x_0-x_2}{(x_1-x_0)(x_1-x_2)} \quad \text{and} \quad L'_2(x) = \frac{2x-x_0-x_1}{(x_2-x_0)(x_2-x_1)}$$

Hence, from Eq. (2),

$$\begin{aligned} f'(x_j) = & f(x_0) \left[\frac{2x_j - x_1 - x_2}{(x_0 - x_1)(x_0 - x_2)} \right] \\ & + f(x_1) \left[\frac{2x_j - x_0 - x_2}{(x_1 - x_0)(x_1 - x_2)} \right] \\ & + f(x_2) \left[\frac{2x_j - x_0 - x_1}{(x_2 - x_0)(x_2 - x_1)} \right] \\ & + \frac{1}{6} f^3(\xi_j) \prod_{\substack{k=0 \\ k \neq j}}^2 (x_j - x_k), \end{aligned} \quad (3)$$

for each $j = 0, 1, 2$, where the notation ξ_j indicates that this point depends on x_j .

Three-Point Formulas

The formulas from Eq. (3) become especially useful if the nodes are equally spaced, that is, when

$$x_1 = x_0 + h \text{ and } x_2 = x_0 + 2h. \text{ For some } h \neq 0.$$

We will assume equally spaced nodes throughout the remainder of this section. Using Eq. (3) with $x_j = x_0$, $x_1 = x_0 + h$, and $x_2 = x_0 + 2h$ gives

$$f'(x_0) = \frac{1}{h} \left[-\frac{3}{2}f(x_0) + 2f(x_1) - \frac{1}{2}f(x_2) \right] + \frac{h^2}{3} f^{(3)}(\xi_0).$$

Doing the same for $x_j = x_1$ gives

$$f'(x_1) = \frac{1}{h} \left[-\frac{1}{2}f(x_0) + \frac{1}{2}f(x_2) \right] - \frac{h^2}{6} f^{(3)}(\xi_1)$$

And, for $x_j = x_2$,

$$f'(x_2) = \frac{1}{h} \left[\frac{1}{2}f(x_0) - 2f(x_1) + \frac{3}{2}f(x_2) \right] - \frac{h^2}{3} f^{(3)}(\xi_2)$$

Since $x_1 = x_0 + h$ and $x_2 = x_0 + 2h$, these formulas can also be expressed as

$$f'(x_0) = \frac{1}{h} \left[-\frac{3}{2}f(x_0) + 2f(x_0 + h) - \frac{1}{2}f(x_0 + 2h) \right] + \frac{h^2}{3} f^{(3)}(\xi_0).$$

$$f'(x_0 + h) = \frac{1}{h} \left[-\frac{1}{2}f(x_0) + \frac{1}{2}f(x_0 + 2h) \right] + \frac{h^2}{6} f^{(3)}(\xi_1), \text{ and}$$

$$f'(x_0 + 2h) = \frac{1}{h} \left[\frac{1}{2} f(x_0) - 2f(x_0 + h) + \frac{3}{2} f(x_0 + 2h) \right] + \frac{h^2}{3} f^{(3)}(\xi_2).$$

As a matter of convenience, the variable substitution x_0 for $x_0 + h$ is used in the middle equation to change this formula to an approximation for $f'(x_0)$. A similar change, x_0 for $x_0 + 2h$, is used in the last equation. This gives three formulas for approximating $f'(x_0)$:

$$f'(x_0) = \frac{1}{2h} [-3f(x_0) + 4f(x_0 + h) - f(x_0 + 2h)] + \frac{h^2}{3} f^{(3)}(\xi_0),$$

$$f'(x_0) = \frac{1}{2h} [-f(x_0 - h) + f(x_0 + h)] - \frac{h^2}{6} f^{(3)}(\xi_1), \quad \text{and}$$

$$f'(x_0) = \frac{1}{2h} [f(x_0 - 2h) - 4f(x_0 - h) + 3f(x_0)] + \frac{h^2}{3} f^{(3)}(\xi_2).$$

Finally, note that the last of these equations can be obtained from the first by simply replacing h with $-h$, so there are actually only two formulas:

Three-Point Endpoint Formula

$$f'(x_0) = \frac{1}{2h} [-3f(x_0) + 4f(x_0 + h) - f(x_0 + 2h)] + \frac{h^2}{3} f^{(3)}(\xi_0) \quad (4)$$

where ξ_0 lies between x_0 and $x_0 + 2h$.

Three-Point Midpoint Formula

$$f'(x_0) = \frac{1}{2h} [f(x_0 + h) - f(x_0 - h)] - \frac{h^2}{6} f^{(3)}(\xi_1) \quad (5)$$

where ξ_1 lies between $x_0 - h$ and $x_0 + h$.

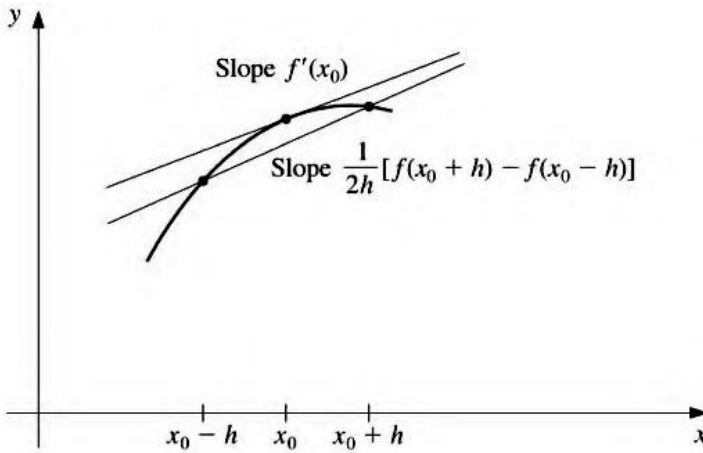


Figure 4.2

Example 2. Values for $f(x) = xe^x$ are given in Table 4.2. Use all the applicable three-point formulas to approximate $f'(2.0)$.

Table 4.2

x	$f(x)$
1.8	10.889365
1.9	12.703199
2.0	14.778112
2.1	17.148957
2.2	19.855030

Solution. The data in the table permit us to find four different three-point approximations. We can use the endpoint formula (4) with $h = 0.1$ or with $h = -0.1$, and we can use the midpoint formula (5) with $h = 0.1$ or with $h = 0.2$. Using the endpoint formula (4) with $h = 0.1$ gives

$$\begin{aligned} \frac{1}{0.2} [-3f(2.0) + 4f(2.1) - f(2.2)] \\ = 5[-3(14.778112) + 4(17.148957) \\ - 19.855030] = 22.032310 \end{aligned}$$

and with $h = -0.1$ gives 22.054525.

Using the midpoint formula (5) with $h = 0.1$ gives

$$\begin{aligned}\frac{1}{0.2}[f(2.1) - f(1.9)] &= 5(17.148957 - 12.7703199) \\ &= 22.228790\end{aligned}$$

and with $h = 0.2$ gives 22.414163.

The true value in this case is $f'(2.0) = (2 + 1)e^2 = 22.167168$, so the approximation error are actually as follows:

Three-point endpoint with $h = 0.1$: 1.35×10^{-1} ;

Three-point endpoint with $h = -0.1$: 1.13×10^{-1} ;

Three-point midpoint with $h = 0.1$: -6.16×10^{-2} ;

Three-point midpoint with $h = 0.2$: -2.47×10^{-1} .

Second Derivative Midpoint Formula

$$\begin{aligned}f''(x_0) &= \frac{1}{h^2}[f(x_0 - h) - 2f(x_0) + f(x_0 + h)] \\ &\quad - \frac{h^2}{12}f^{(4)}(\xi),\end{aligned}\tag{6}$$

for some ξ , where $x_0 - h < \xi < x_0 + h$.

If $f^{(4)}$ is continuous on $[x_0 - h, x_0 + h]$, it is also bounded, and the approximation is $O(h^2)$.

Example 3. In Example 2, we used the data shown in Table 4.3 to approximate the first derivative of $f(x) = xe^x$ at $x = 2.0$. Use the second derivative formula (6) to approximate $f''(2.0)$.

Table 4.2

x	$f(x)$
1.8	10.889365
1.9	12.703199
2.0	14.778112
2.1	17.148957
2.2	19.855030

Solution. The data permit us to determine two approximations for $f''(2.0)$. Using (6) with $h = 0.1$ gives

$$\begin{aligned}
 \frac{1}{0.01} [f(1.9) - 2f(2.0) + f(2.1)] \\
 &= 100[12.703199 - 2(14.778112) \\
 &\quad + 17.148957] \\
 &= 29.593200.
 \end{aligned}$$

and using (6) with $h = 0.2$ gives

$$\begin{aligned}
 \frac{1}{0.04} [f(1.8) - 2f(2.0) + f(2.2)] \\
 &= 25[10.889365 - 2(14.778112) \\
 &\quad + 19.855030] \\
 &= 29.704275.
 \end{aligned}$$

Because $f''(x) = (x + 2)e^x$, the exact value is $f''(x) = 29.556224$. hence the actual errors are -37.0×10^{-2} and -1.48×10^{-1} , respectively.

Round-Off Error Instability

It is particularly important to pay attention to round-off error when approximating derivatives. To illustrate the situation, let us examine the three-point midpoint formula Eq. (5),

$$f'(x_0) = \frac{1}{2h} [f(x_0 + h) - f(x_0 - h)] - \frac{h^2}{6} f^{(3)}(\xi),$$

more closely. Suppose that in evaluating $f(x_0 + h)$ and $f(x_0 - h)$, we encounter round-off errors $e(x_0 + h)$ and $e(x_0 - h)$. Then our computations actually use the values $f(x_0 + h)$ and $f(x_0 - h)$, which are related to the true values $f(x_0 + h)$ and $f(x_0 - h)$ by

$$f(x_0 + h) = f(x_0 + h) + e(x_0 + h) \quad \text{and} \quad f(x_0 - h) = f(x_0 - h) + e(x_0 - h).$$

The total error in the approximation.

$$\begin{aligned} f'(x_0) - \frac{\bar{f}(x_0 + h) - \bar{f}(x_0 - h)}{2h} \\ = \frac{e(x_0 + h) - e(x_0 - h)}{2h} - \frac{h^2}{6} f^{(3)}(\xi), \end{aligned}$$

is due both to round-off error, the first part, and to truncation error. If we assume that the round-off errors $e(x_0 \pm h)$ are bounded by some number $\varepsilon > 0$ and that the third derivative of f is bounded by a number $M > 0$, then

$$\left| f'(x_0) - \frac{\bar{f}(x_0 + h) - \bar{f}(x_0 - h)}{2h} \right| \leq \frac{\varepsilon}{h} + \frac{h^2}{6} M.$$

To reduce the truncation error, $h^2 M/6$, we need to reduce h . But as h is reduced, the round-off error $\frac{\varepsilon}{h}$ grows. In practice, then, it is seldom advantageous to let h be too small, because in that case the round-off error will dominate the calculations.

Example 4. Use the backward-difference formulas to approximate the derivative of $f(x) = 2 \cos(2x) - x$ at $x_0 = -0.1$ using $h = -0.1$ and $h = -0.2$.

Solution. Given $f(x) = 2 \cos(2x) - x$ and so $f(-0.1) = 2.0601$.

$$\text{Now } f'(x) = \frac{f(x_0+h)-f(x_0)}{h}$$

$$\begin{aligned} \frac{f(-0.1-0.1)-f(-0.1)}{-0.1} & \quad \text{if } h = -0.1 \\ &= \frac{f(-0.2) - f(-0.1)}{-0.1} \\ &= \frac{2.0421 - 2.0601}{-0.1} \\ &= 0.18 \end{aligned}$$

$$\begin{aligned} \text{Also } f'(x) &= \frac{f(-0.1-0.2)-f(-0.1)}{-0.2} \quad \text{if } h = -0.2 \\ &= \frac{f(-0.3) - f(-0.1)}{-0.2} \end{aligned}$$

$$= \frac{1.9507 - 2.0601}{-0.2}$$

$$= 0.547$$

Example 5. Use forward-difference formula to approximate $f'(x)$ at $x_0 = 1$ using $h = 0.2$ and $h = 0.4$ where $f(x) = x^2 \ln x + 1$

Solution. For $(x) = x^2 \ln x + 1$

$f(x) = 1$ for $x = x_0 = 1$

h	$x_0 + h$	$f(x_0 + h)$	$f'(x) = \frac{f(x_0 + h) - f(x_0)}{h}$
0.2	1.2	1.2625	1.3125
0.4	1.4	1.6595	1.64875

Example 6. Consider the function $f(x) = x \sin x + x^2 \cos x$. Use the three-point endpoint formula as well as the three-point midpoint formula with $h = 0.1$ to approximate $f'(1.2)$.

Solution. For $f(x) = x \sin x + x^2 \cos x$, we can consider the following values as below.

	x	$f(x)$
$x_0 - h$	1.1	1.52918
x_0	1.2	1.64024
$x_0 + h$	1.3	1.70470

$x_0 + 2h$	1.4	1.71277
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Using the endpoint formula with $h = 0.1$ gives

$$f'(1.2) = \frac{1}{0.2} [-3 \times 1.64024 + 4 \times 1.70470 - 1.71277]$$

$$= 0.92655$$

Using the midpoint formula with $h = 0.1$ gives

$$f'(1.2) = \frac{1}{0.2} [1.70470 - 1.52918]$$

$$= 0.8776$$

Example 7. For the function $f(x) = \ln(x + 2) - (x + 1)^2$. Find the approximation for $f'(7.8)$ with $h = -0.2$ using three-point endpoint formula.

Solution. We use following values of $f(x)$.

x	7.8	7.6	7.4
$f(x)$	-75.1576	-71.6982	-68.3193

$$\text{Then } f'(x) = \frac{1}{2h} [-3f(x_0) + 4f(x_0 + h) - f(x_0 + 2h)]$$

Taking $h = -0.2$ and $x_0 = 7.8$, we get

$$f'(7.8) = \frac{1}{-0.4} [-3(-75.1576) + 4(-71.6982) + (-68.3193)]$$

$$= 324.09825$$

Example 8. Use three-point midpoint formula to approximate the derivate of $f(x) = e^{2x} - \cos 2x$ at $x_0 = -0.1$ with $h = 0.1$

Solution. For this we consider the following values:

x	-0.2	-0.1	0
$f(x)$	-0.25074	-0.16134	0

$$\text{Then } f'(x) = \frac{1}{2h} [f(x_0 + h) - f(x_0 - h)]$$

$$= \frac{1}{0.2} [0. (-0.25074)] \text{ by taking } x_0 = -0.1 \text{ and } h = 0.1$$

$$= 1.2537$$

Example 9. Use the data in example 8 and use the second derivative formula to approximate $f''(-0.1)$

Solution. We have $f''(x_0) \approx \frac{1}{h^2} [(f(x_0 - h) - 2f(x_0) + f(x_0 + h))]$

$$f''(-0.1) = \frac{1}{0.01} [(-0.25074) - 2(-0.16134) + 4]$$

$$= 7.194.$$

Questions 4.1

1. Use the forward-difference formulas and backward-difference formulas to determine each missing entry in the following tables,

a.	x	$f(x)$	$f^1(x)$	b.	x	$f(x)$	$f^1(x)$
	0.5	0.4794			0.0	0.00000	
	0.6	0.5646			0.2	0.74140	
	0.7	0.6442			0.4	1.3718	

2. The data in Exercise 1 were taken from the following functions. Compute the actual errors in Exercise 1, and find error bounds using the error formulas.

a. $f(x) = \sin x$ b. $f(x) = e^x - 2x^2 + 3x - 1$

3. Use the most accurate three-point formula to determine each missing entry in the following tables.

a.	x	$f(x)$	$f^1(x)$	b.	x	$f(x)$	$f^1(x)$
	1.1	9.025013			8.1	16.94410	
	1.2	11.02318			8.3	17.56492	
	1.3	13.46374			8.5	18.19056	
	1.4	16.44465			8.7	18.82091	
c.	x	$f(x)$	$f^1(x)$	d.	x	$f(x)$	$f^1(x)$
	2.9	-4.827866			2.0	3.6887983	
	3.0	-4.240058			2.1	3.6905701	

3.1	-3.496909	2.2	3.6688192
3.2	-2.596792	2.3	3.6245909

4. The data in Exercise 3 were taken from the following functions. Compute the actual errors in Exercise 3 and find error bounds using the error formulas.

a. $f(x) = e^{2x}$

b. $f(x) = x \ln x$

c. $f(x) = x \sin x - x^2 \cos x$
 $3 \sin x$

d. $f(x) = 2(\ln x)^2 +$

5. Use the formulas given in this section to determine, as accurately as possible, approximations for each missing entry in the following tables.

a.

x	$f(x)$	$f'(x)$
2.1	-1.709847	
2.2	-1.373823	
2.3	-1.119214	
2.4	- 0.9160143	
2.5	- 0.7470223	

b.

x	$f(x)$	$f'(x)$
- 3.0	9.367879	
- 2.8	8.233241	
- 2.6	7.180350	
- 2.4	6.209329	
- 2.2	5.320305	

2.6	- 0.6015966	- 2.0	4.513417
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6. The data in Exercise 5 were taken from the following functions. Compute the actual errors in Exercise 5 and find error bounds using the error formulas.

a. $f(x) = \tan x$

b. $f(x) = e^{\frac{x}{3}} + x^2$

7. Use the following data and the knowledge that the first five derivatives of f are bounded on $[1,5]$ by 2, 3, 6, 12, and 23, respectively, to approximate $f'(3)$ as accurately as possible. Find a bound for the error.

x	1	2	3	4	5
$f(x)$	2.4142	2.6734	2.8974	3.0976	3.2804

8. Consider the following table of data:

x	0.2	0.4	0.6	0.8	1.0
$f(x)$	0.979865	0.917771	0.80803	0.638609	0.384373
)	2	0	8	3	5

- a. Use the appropriate formula to approximate $f'(0.4)$ and $f''(0.4)$.

- b. Use the appropriate formula to approximate $f'(0.6)$ and $f''(0.6)$.
9. Let $f(x) = 3xe^x - \cos x$. Use the following data to approximate $f''(1.3)$ with $h = 0.1$ and with $h = 0.01$.

x	1.20	1.29	1.30	1.31	1.40
$f(x)$	11.59006	13.78176	14.04276	14.30741	16.86187

Compare your results to $f''(1.3)$

Answers

1. From the forward-backward difference formula, we have the following approximations:

a. $f'(0.5) \approx 0.8520, f'(0.6) \approx 0.8520, f'(0.7) \approx 0.7960$

b. $f'(0.0) \approx 3.7070, f'(0.2) \approx 3.1520, f'(0.4) \approx 3.1520$

2.

a.

x	Actual Error	Error Bound
0.5	0.0255	0.0282
0.6	0.0267	0.0282

0.7	0.0312	0.0322
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b.

x	Actual Error	Error Bound
0.0	0.2930	0.3000
0.2	0.2694	0.2779
0.4	0.2602	0.2779

3. For the endpoints of the tables, we use Formula (4). The other approximations come from Formula (5).

$$\text{a. } f'(1.1) \approx 17.769705, f'(1.2) \approx 22.193635, f'(1.3) \approx 27.107350, f'(1.4) \approx 32.150850$$

$$\text{b. } f'(8.1) \approx 3.092050, f'(8.3) \approx 3.116150, f'(8.5) \approx 3.139975, f'(8.7) \approx 3.163525$$

$$\text{c. } f'(2.9) \approx 5.101375, f'(3.0) \approx 6.654785, f'(3.1) \approx 8.216330, f'(3.2) \approx 9.786010$$

$$\text{d. } f'(2.0) \approx 0.13533150, f'(2.1) \approx -0.09989550, f'(2.2) \approx -0.3298960, f'(2.3) \approx -0.5546700$$

4.

a.	x	Actual Error	Error Bound	c.	x	Actual Error	Error Bound
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1.1	0.280322	0.359033	2.9	0.011956	0.0180988
1.2	0.147282	0.179517	3.0	0.0049251	0.00904938
1.3	0.179874	0.219262	3.1	0.0004765	0.00493920
1.4	0.378444	0.438524	3.2	0.0013745	0.00987840

b	x	Actual Error	Error Bound	d.	x	Actual Error	Error Bound
	8.1	0.00018594	0.000020322		2.0	0.00252235	0.00410304
	8.3	0.00010551	0.000010161		2.1	0.00142882	0.00205152
	8.5	9.116×10^{-5}	0.000009677		2.2	0.00204851	0.00260034
	8.7	0.00020197	0.000019355		2.3	0.00437954	0.00520068

5. The approximations are:

a. $f'(2.1) \approx 3.899344, f'(2.2) \approx 2.876876, f'(2.3) \approx 2.249704$

$$f'(2.4) \approx 1.837756, f'(2.5) \approx 1.544210, f'(2.6) \approx 1.355496$$

b. $f'(-3.0) \approx -5.877358, f'(-2.8) \approx -5.468933, f'(-2.6) \approx -5.059884$

$$f'(-2.4) \approx -4.650223, f'(-2.2) \approx -4.239911, f'(-2.0) \approx -3.828853$$

6.

a.	x	Actual Error	Error Bound	b.	x	Actual Error	Error Bound
	2.1	0.0242312	0.109271		- 3.0	1.55×10^{-5}	6.33×10^{-7}
	2.2	0.0105138	0.0386885		- 2.8	1.38×10^{-5}	6.76×10^{-7}
	2.3	0.0029352	0.0182120		- 2.6	7.95×10^{-7}	1.05×10^{-7}
	2.4	0.0013262	0.00644808		- 2.4	6.79×10^{-7}	1.13×10^{-7}
	2.5	0.0138323	0.109271		- 2.2	1.28×10^{-5}	6.76×10^{-7}
	2.6	0.0064225	0.0386885		- 2.0	7.76×10^{-5}	6.76×10^{-7}

7. $f'(3) \approx \frac{1}{12} [f(1) - 8f(2) + 8f(4) - f(5)] = 0.21062$ with an error bound given by

$$\max_{1 \leq x \leq 5} \frac{|f^{(5)}(x)|h^4}{30} \leq \frac{23}{30} = 0.7\bar{6}$$

4.2 Elements of Numerical Integration

The need often arises for evaluating the definite integral of a function that has no explicit antiderivative or whose

antiderivative is not easy to obtain. The basic method involved in approximating $\int_a^b f(x)dx$ is called **numerical quadrature**. It uses a sum $\sum_{i=0}^n a_i f(x_i)$ to approximate $\int_a^b f(x)dx$.

The methods of quadrature in this section are based on the interpolation polynomials given in Chapter 3. The basic idea is to select a set of distinct nodes $\{x_0, \dots, x_n\}$ from the interval $[a, b]$. Then integrate the Lagrange interpolating polynomial

$$P_n(x) = \sum_{i=0}^n f(x_i)L_i(x)$$

And its truncation error term over $[a, b]$ to obtain.

$$\begin{aligned} \int_a^b f(x)dx &= \int_a^b \sum_{i=0}^n f(x_i)L_i(x)dx \\ &\quad + \int_a^b \prod_{i=0}^n (x - x_i) \frac{f^{(n+1)}(\xi(x))}{(n+1)!} dx \\ &= \sum_{i=0}^n a_i f(x_i) + \frac{1}{(n+1)!} \int_a^b \prod_{i=0}^n (x - x_i) f^{(n+1)}(\xi(x)) dx, \end{aligned}$$

Where $\xi(x)$ is in $[a, b]$ for each x and

$$a_i = \int_a^b L_i(x)dx, \text{ for each } i = 0, 1, \dots, n.$$

The quadrature formula is, therefore,

$$\int_a^b f(x)dx \approx \sum_{i=0}^n a_i f(x_i),$$

With error given by

$$E(f) = \frac{1}{(n+1)!} \int_a^b \prod_{i=0}^n (x - x_i) f^{(n+1)}(\xi(x)) dx,$$

Let us consider formulas produced by using first and second Lagrange polynomials with equally spaced nodes. This gives the **Trapezoidal rule** and **Simpson's rule**

The Trapezoidal Rule

To derive a Trapezoidal rule for approximating $\int_a^b f(x)dx$, let $x_0 = a$, $x_1 = b$, $h = b - a$ and use the linear Lagrange polynomial:

$$P_1(x) = \frac{(x - x_1)}{(x_0 - x_1)} f(x_0) + \frac{(x - x_0)}{(x_1 - x_0)} f(x_1)$$

Then

$$\begin{aligned} \int_a^b f(x)dx &= \int_{x_0}^{x_1} \left[\frac{(x - x_1)}{(x_0 - x_1)} f(x_0) + \frac{(x - x_0)}{(x_1 - x_0)} f(x_1) \right] dx \\ &\quad + \frac{1}{2} \int_{x_0}^{x_1} f''(\xi(x))(x - x_0)(x - x_1)dx. \end{aligned}$$

(7)

The product $(x - x_0)(x - x_1)$ does not change sign on $[x_0, x_1]$, so the Weighted Mean Value Theorem for Integrals can be applied to the error term to give, for some ξ in $(x - x_0)$

$$\begin{aligned} \int_{x_0}^{x_1} f''(\xi(x))(x - x_0)(x - x_1)dx \\ &= f''(\xi) \int_{x_0}^{x_1} (x - x_0)(x - x_1)dx. \\ &= f''(\xi) \left[\frac{x^3}{3} - \frac{(x_1 + x_0)}{2}x^2 + x_0x_1x \right]_{x_0}^{x_1} \\ &= \frac{h^3}{6} f''(\xi) \end{aligned}$$

Consequently Eq. (7) implies that

$$\begin{aligned} \int_a^b f(x)dx &= \left[\frac{(x - x_1)^2}{2(x_0 - x_1)} f(x_0) + \frac{(x - x_0)^2}{2(x_1 - x_0)} f(x_1) \right]_{x_0}^{x_1} \\ &\quad - \frac{h^3}{12} f''(\xi) \\ &= \frac{(x - x_0)}{2} [f(x_0) + f(x_1)] - \frac{h^3}{12} f''(\xi) \end{aligned}$$

(Weighted Mean Value Theorem for Integrals) Suppose $f \in C[a, b]$, the Riemann integral of g exists on $[a, b]$, and $g(x)$ does not change sign on $[a, b]$. Then there exists a number c in (a, b) with $\int_a^b f(x)g(x)dx = f(c) \int_a^b g(x)dx$. When $g(x) = 1$, this is the usual Mean Value Theorem for Integrals. It gives the

average value of the function f over the interval $[a, b]$, $f(c) = \frac{1}{b-a} \int_a^b f(x) dx$.

Using the notation $h = x_1 - x_0$ gives the following rule:

Trapezoidal Rule:

$$\int_a^b f(x) dx = \frac{h}{2} [f(x_0) + f(x_1)] - \frac{h^3}{12} f''(\xi) \quad (8)$$

This is called the Trapezoidal rule because when f is a function with positive values, $\int_a^b f(x) dx$ is approximated by the area in a trapezoid, as shown in Figure 4.3.

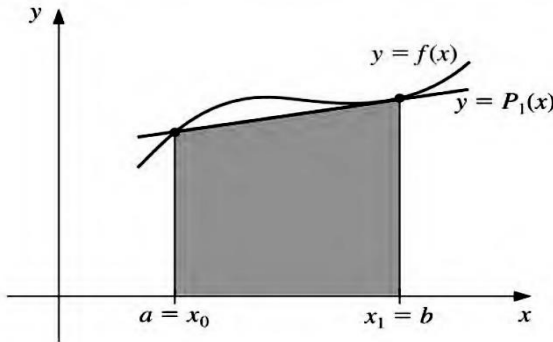


Figure 4.3.

The error term for the Trapezoidal rule involves f'' , so the rule gives the exact result when applied to any function whose second derivative is identically zero, that is, any polynomial of degree one or less.

Simpson's Rule

Simpson's rule results from integrating over $[a, b]$ the second Lagrange polynomial with equally spaced nodes $x_0 = a, x_2 = b$, and $x_1 = a + h$, where $h = \frac{(b-a)}{2}$ (See Figure 4.4.)

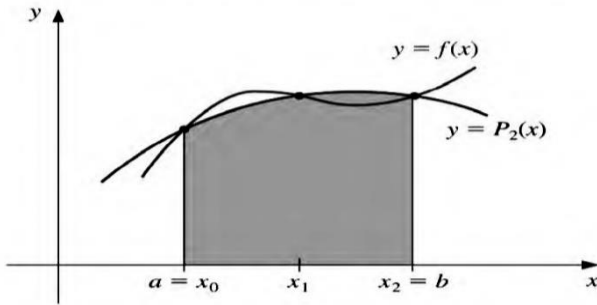


Figure 4.4

Therefore

$$\begin{aligned} \int_a^b f(x) dx &= \int_{x_0}^{x_2} \left[\frac{(x - x_1)(x - x_2)}{(x_0 - x_1)(x_0 - x_2)} f(x_0) \right. \\ &\quad + \frac{(x - x_0)(x - x_2)}{(x_1 - x_0)(x_1 - x_2)} f(x_1) \\ &\quad + \left. \frac{(x - x_0)(x - x_1)}{(x_2 - x_0)(x_2 - x_1)} \right] dx \\ &\quad + \int_{x_0}^{x_2} \frac{(x - x_0)(x - x_1)(x - x_2)}{6} f^{(3)}(\xi(x)) dx \end{aligned}$$

Deriving Simpson's rule in this manner, however, provides only an $O(h^4)$ error term involving $f^{(3)}$. By approaching the problem in another way, a higher-order term involving $f^{(4)}$ can be derived.

To illustrate this alternative method, suppose that f is expanded in the third Taylor polynomial about x_1 . Then, for each x in $[x_0, x_2]$, a number $\xi(x)$ in (x_0, x_2) exists with

$$\begin{aligned} f(x) = & f(x_1) + f'(x_1)(x - x_1) + \frac{f''(x_1)}{2}(x - x_1)^2 \\ & + \frac{f'''(x_1)}{6}(x - x_1)^3 \\ & + \frac{f^{(4)}(\xi(x))}{24}(x - x_1)^4 \end{aligned}$$

and

$$\begin{aligned} \int_{x_0}^{x_2} f(x) dx = & \left[f(x_1)(x - x_1) + \frac{f'(x_1)}{2}(x - x_1)^2 + \frac{f''(x_1)}{6}(x - x_1)^3 \right. \\ & \left. + \frac{f'''(x_1)}{24}(x - x_1)^4 \right]_{x_0}^{x_2} + \frac{1}{24} \int_{x_0}^{x_2} f^{(4)}(\xi(x))(x - x_1)^4 dx. \end{aligned} \quad (4.24)$$

Because $(x - x_1)^4$ is never negative on $[x_0, x_2]$, the Weighted Mean Value Theorem for Integral implies that

$$\frac{1}{24} \int_{x_0}^{x_2} f^{(4)}(\xi(x))(x - x_1)^4 dx = \frac{f^{(4)}(\xi_1)}{24} \int_{x_0}^{x_2} (x - x_1)^4 dx = \frac{f^{(4)}(\xi_1)}{120} (x - x_1)^5 \Big|_{x_0}^{x_2},$$

For some number ξ_1 in (x_0, x_2) .

However, $h = x_2 - x_1 = x_1 - x_0$, so

$$(x_2 - x_1)^2 - (x_0 - x_1)^2 = (x_2 - x_1)^4 - (x_0 - x_1)^4 = 0.$$

whereas

$$(x_2 - x_1)^3 - (x_0 - x_1)^3 = 2h^3 \text{ and } (x_2 - x_1)^5 - (x_0 - x_1)^5 = 2h^5$$

Consequently,

$$\int_{x_0}^{x_2} f(x)dx = 2hf(x_1) + \frac{h^3}{3} f''(x_1) + \frac{f^{(4)}(\xi)}{60} h^5$$

If we replace $f''(x_1)$ by the approximation given Eq. (6) of section 4.1, we have

$$\begin{aligned} \int_{x_0}^{x_2} f(x)dx &= 2hf(x_1) \\ &+ \frac{h^3}{3} \left\{ \frac{1}{h^2} [f(x_0) - 2f(x_1) + f(x_2)] \right. \\ &\quad \left. - \frac{h^2}{12} f^{(4)}(\xi_2) \right\} + \frac{f^{(4)}(\xi_1)}{60} h^5 \\ &= \frac{h}{3} [f(x_0) + 4f(x_1) + f(x_2)] - \frac{h^5}{12} \left[\frac{1}{3} f^{(4)}(\xi_2) - \frac{1}{5} f^{(4)}(\xi_1) \right]. \end{aligned}$$

It can be shown by alternative methods that the values ξ_1 and ξ_2 in this expression can be replaced by a common value ξ in (x_0, x_2) . This gives Simpson's rule.

Simpson's Rule:

$$\begin{aligned} \int_{x_0}^{x_2} f(x)dx &= \frac{h}{3} [f(x_0) + 4f(x_2) + f(x_2)] \\ &\quad - \frac{h^5}{90} f^{(4)}(\xi). \end{aligned} \tag{9}$$

The error term in Simpson's rule involves the fourth derivative of f , so it gives exact results when applied to any polynomial of degree three or less.

Example 1. Compare the Trapezoidal rule and Simpson's rule approximations to $\int_0^2 f(x)dx$ when $f(x)$ is

(a) x^2
(d) $\sqrt{1+x^2}$

(b) x^4
(e) $\sin x$

(c) $(x+1)^2$
(f) e^x

Solution. On $[0, 2]$, the Trapezoidal and Simpson's rules have the forms

Trapezoidal: $\int_0^2 f(x)dx \approx f(0) + f(2)$ and

Simpson's: $\int_0^2 f(x)dx \approx \frac{1}{3} [f(0) + 4f(1) + f(2)]$

When $f(x) = x^2$,

Trapezoidal: $\int_0^2 f(x)dx \approx 0^2 + 2^2 = 4$ and

Simpson's: $\int_0^2 f(x)dx \approx \frac{1}{3} [(0^2) + 4 \cdot 1^2 + 2^2] = \frac{8}{3}$

The approximation from Simpson's rule is exact because its truncation error involves $f^{(4)}$, which is identically 0 when $f(x) = x^2$.

The results to three places for the functions are summarized in Table 4.7. Notice that in each instance, Simpson's rule is significantly superior.

Table 4.7.

	(a)	(b)	(c)	(d)	(c)	(f)
$f(x)$	x^2	$\ln x^2$	$(x+1)^{-1}$	$\sqrt{1+x^2}$	$\sin x$	ℓ^1
Exact value	2.667	6.400	1.099	2.958	1.416	6.389
Trapezoidal	4.000	16.000	1.333	3.326	0.909	8.389
Simpson's	2.667	6.667	1.111	2.964	1.425	6.421

Measuring Precision

The standard derivation of quadrature error formulas is based on determining the class of polynomials for which these formulas produce exact results. The next definition is used to facilitate the discussion of this derivation.

Definition. The **degree of accuracy**, or **precision**, of a quadrature formula is the largest positive integer n such that the formula is exact for x^k , for each $k=0,1, \dots, n$.

This implies that the Trapezoidal and Simpson's rules have degrees of precision one and three respectively.

Integration and summation are linear operations; that is,

$$\int_a^b (\alpha f(x) + \beta g(x)) dx = \alpha \int_a^b f(x) dx + \beta \int_a^b g(x) dx$$

and

$$\sum_{i=0}^n (\alpha f(x_i) + \beta g(x_i)) = \alpha \sum_{i=0}^n f(x_i) + \beta \sum_{i=0}^n g(x_i),$$

for each pair of integrable functions f and g and each pair of real constants α and β . This implies that the degree of precision of a quadrature formula is n if and only if the error is zero for all polynomials of degree $k = 0, 1, \dots, n$, but is not zero for some polynomial of degree $n + 1$.

The Trapezoidal and Simpson's rules are examples of a class of methods known as Newton-Cotes formulas. There are two types of Newton-Cotes formulas: open and closed.

Closed Newton-Cotes Formulas

The $(n + 1)$ point closed Newton-Cotes formula uses nodes $x_1 = x_0 + ih$, for $i = 0, 1, \dots, n$, where $x_0 = a$, $x_n = b$ and $h = \frac{(b-a)}{n}$. (See Figure 4.5.) It is called closed because the endpoints of the closed interval $[a, b]$ are included as nodes.

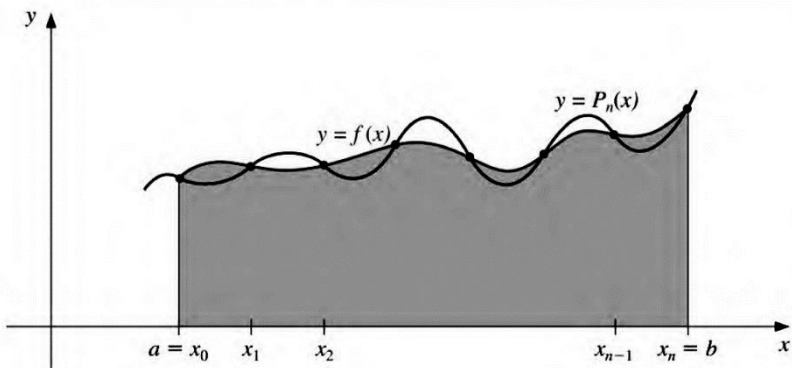


Figure 4.5

The formula assumes the form

$$\int_a^b f(x)dx \approx \sum_{i=0}^n a_i f(x_i),$$

where

$$a_i = \int_{x_0}^{x_n} L_i(x)dx = \int_{x_0}^{x_n} \prod_{\substack{j=0 \\ j \neq i}}^n \frac{(x - x_j)}{(x_i - x_j)} dx.$$

Theorem 4.2 Suppose that $\sum_{i=0}^n a_i f(x_i)$ denotes the $(n+1)$ point closed Newton Cotes formula with $x_0 = a, x_n = b$, and $h = \frac{(b-a)}{n}$. There exists $\xi \in (a, b)$ for which

$$\int_a^b f(x)dx = \sum_{i=0}^n a_i f(x_i) + \frac{h^{n+3} f^{(n+2)}(\xi)}{(n+2)!} \int_0^n t^2(t-1) \dots (t-n)dt,$$

If n is even and $f \in C^{n+2}[a, b]$, and

$$\int_a^b f(x)dx = \sum_{i=0}^n a_i f(x_i) + \frac{h^{n+2} f^{(n+1)}(\xi)}{(n+1)!} \int_0^n t(t-1) \dots (t-n)dt,$$

If n is odd and $f \in C^{n+1}[a, b]$,

Note that when n is an even integer, the degree of precision is $n+1$, although the interpolation polynomial is of degree at most n . When n is odd, the degree of precision is only n .

Some of the common closed **Newton-Cotes formulas** with their error terms are listed. Note that in each case the unknown value ξ lies in (a, b) .

n = 1: Trapezoidal rule

$$\int_{x_0}^{x_1} f(x)dx = \frac{h}{2} [f(x_0) + f(x_1)] - \frac{h^3}{12} f''(\xi),$$

where $x_0 < \xi < x_1$.
(10)

n = 2: Simpson's rule

$$\int_{x_0}^{x_2} f(x)dx = \frac{h}{3} [f(x_0) + 4f(x_1) + f(x_2)] - \frac{h^5}{90} f^{(4)}(\xi),$$

where $x_0 < \xi < x_2$.
(11)

n = 3: Simpson's Three-Eighths rule

$$\int_{x_0}^{x_3} f(x)dx = \frac{3h}{8} [f(x_0) + 3f(x_1) + 3f(x_2) + f(x_3)] - \frac{3h^5}{80} f^{(4)}(\xi),$$

where $x_0 < \xi < x_3$.
(12)

n = 4:

$$\int_{x_0}^{x_4} f(x)dx = \frac{2h}{45} [7f(x_0) + 32f(x_1) + 12f(x_2) + 7f(x_4)] - \frac{8h^7}{945} f^{(6)}(\xi),$$

where $x_0 < \xi < x_4$.
(13)

Open Newton-Cotes Formulas

The open *Newton-Cotes formulas* do not include the endpoints of $[a, b]$ as nodes. They use the nodes $x_i = x_0 + ih$, for each $i = 0, 1, \dots, n$, where $h = (b - a)/(n + 2)$ and $x_0 = a + h$. This implies that $x_n = b - h$, so we label the endpoints by setting $x_{-1} = a$ and $x_{n+1} = b$, as shown in Figure 4.6. Open formulas contain all the nodes used for the approximation within the open interval (a, b) . The formulas become

$$\int_a^b f(x)dx = \int_{x_{-1}}^{x_{n+1}} f(x)dx \approx \sum_{i=0}^n a_i f(x_i).$$

where $a_i = \int_a^b L_i(x)dx$.

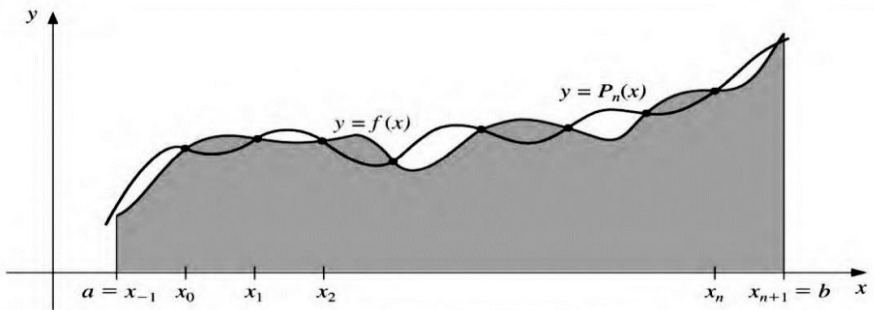


Figure 4.6

Theorem 4.3 Suppose that $\sum_{i=0}^n a_i f(x_i)$ denotes the $(n+1)$ -point open Newton-Cotes formula with $x_{-1} = a$, $x_{n+1} = b$, and $h = (b-a)/(n+2)$. There exists $\xi \in (a, b)$ for which

$$\int_a^b f(x)dx = \sum_{i=0}^n a_i f(x_i) + \frac{h^{n+3}}{(n+2)!} f^{(n+2)}(\xi) \int_{-1}^{n+1} t^2(t-1) \dots (t-n)dt,$$

if n is even and $f \in C^{n+2} [a, b]$, and

$$\int_a^b f(x)dx = \sum_{i=0}^n a_i f(x_i) + \frac{h^{n+2}}{(n+1)!} f^{(n+1)}(\xi) \int_{-1}^{n+1} t^2(t-1) \dots (t-n)dt,$$

if n is odd and $f \in C^{n+1} [a, b]$.

Some of the common **open Newton-Cotes** formulas with their error terms are as follows:

n = 0: Midpoint rule

$$\int_{x_{-1}}^{x_1} f(x)dx = 2hf(x_0) + \frac{h^3}{3} f''(\xi),$$

where $x_{-1} < \xi < x_1$.
(14)

n = 1:

$$\int_{x_{-1}}^{x_2} f(x)dx = \frac{3h}{2} [f(x_0) + f(x_1)] + \frac{3h^3}{4} f'''(\xi),$$

where $x_{-1} < \xi < x_2$.
(15)

n = 2:

$$\int_{x_{-1}}^{x_3} f(x)dx = \frac{4h}{3} [2f(x_0) - f(x_1) + 2f(x_2)] + \frac{14h^5}{45} f^{(4)}(\xi),$$

where $x_{-1} < \xi < x_3$.
(16)

n=3:

$$\begin{aligned} \int_{x_{-1}}^{x_4} f(x)dx &= \frac{5h}{24} [11f(x_0) + f(x_1) + f(x_2) + 11f(x_3)] \\ &+ \frac{95}{144} h^5 f^{(4)}(\xi), \end{aligned}$$

where $x_{-1} < \xi < x_4$.
(17)

Example 2. Compare the results of the closed and open Newton-Cotes formulas listed as Eq. (10) through (13) and Eq. (14) through (17) to approximate

$$\int_0^{\frac{\pi}{4}} \sin x \, dx = 1 - \frac{\sqrt{2}}{2} \approx 0.29289322.$$

Solution. For the closed formulas, we have

$$n = 1: \quad \frac{\left(\frac{\pi}{4}\right)}{2} [\sin 0 + \sin \pi/4] \approx 0.27768018$$

$$n = 2: \quad \frac{\left(\frac{\pi}{8}\right)}{3} [\sin 0 + 4 \sin \frac{\pi}{8} \sin \pi/4] \approx 0.29293264$$

$$n = 3: \frac{3\left(\frac{\pi}{12}\right)}{8} [\sin 0 + 3 \sin \frac{\pi}{12} + 3 \sin \frac{\pi}{6} + \sin \pi/4] \approx 0.29291070$$

$$n = 4: \frac{2\left(\frac{\pi}{16}\right)}{45} [7 \sin 0 + 32 \sin \frac{\pi}{16} + 12 \sin \frac{\pi}{8} + 32 \sin \frac{3\pi}{16} + 7 \sin \frac{\pi}{4}] \approx 0.29289318$$

And for the open formulas, we have

$$n = 0: \quad 2 \left(\frac{\pi}{8}\right) [\sin \frac{\pi}{8}] \approx 0.30055887$$

$$n = 1: \quad \frac{3\left(\frac{\pi}{12}\right)}{2} [\sin \frac{\pi}{12} + \sin \frac{\pi}{6}] \approx 0.29798754$$

$$n = 2: \quad \frac{4\left(\frac{\pi}{16}\right)}{3} [2 \sin \frac{\pi}{16} - \sin \frac{\pi}{8} + 2 \sin \frac{3\pi}{16}] \approx 0.29285866$$

$$n = 4: \quad \frac{5\left(\frac{\pi}{20}\right)}{24} [11 \sin \frac{\pi}{20} + \sin \frac{\pi}{10} + \sin \frac{3\pi}{20} + 11 \sin \frac{\pi}{5}] \approx 0.29286923$$

Table 4.8 summarizes these results and shows the approximation errors

Table 4.8

n	0	1	2	3	4
Closed formulas	0.27768018	0.29293264	0.29291070	0.29289318	
Error	0.01521303	0.00003942	0.00001748	0.00000004	
Open formulas	0.30055887	0.29798754	0.29285866	0.29286923	
Error	0.00766565	0.00509432	0.00003456	0.00002399	1.425

Example 3. Approximate the integral $\int_1^{10} \frac{1}{x} dx$ using closed and open Newton-Cotes formula

Solution. For the closed formula's, we have

$$n = 1: \frac{9}{2} \left[1 + \frac{1}{10} \right] \approx 4.95$$

$$n = 2: \frac{4.5}{3} \left[1 + 4 \times \frac{1}{4.5} + \frac{1}{10} \right] = 2.98333$$

$$n = 3: \frac{9}{8} \left[1 + 3 \times \frac{1}{4} + 3 \times \frac{1}{7} + \frac{1}{10} \right] = 2.56339$$

$$n = 4: \frac{4.5}{4.5} \left[\frac{7}{1} + \frac{32}{3.25} + \frac{12}{5.5} + \frac{32}{7.75} + \frac{7}{10} \right] = 2.38570$$

For the open formulas, we have

$$n = 0: 9 \times \frac{1}{4.5} = 2$$

$$n = 1: \frac{9}{2} \left[\frac{1}{4} + \frac{1}{7} \right] = 1.76786$$

$$n = 2: \frac{9}{3} \left[2 \times \frac{1}{3.25} - \frac{1}{5.5} + 2 \times \frac{1}{7.75} \right] = 2.07489$$

$$n = 3: \frac{9}{24} \left[11 \times \frac{1}{2.8} + \frac{1}{4.6} + \frac{1}{6.4} + 11 \times \frac{1}{8.2} \right] = 2.11638$$

Exercise 4. Approximate $\int_0^1 e^{-x^2} dx$ by means of Trapezoidal rule, Simpson's rule and Mid-point rule.

Solution. Trapezoidal rule: $\int_0^1 e^{-x^2} dx = \frac{1}{2} [e^{-0} + e^{-1}]$

$$= \frac{1}{2} [1 + 0.367879] = 0.683939$$

Simpson's rule: $\int_0^1 e^{-x^2} dx = \frac{0.5}{3} [1 + 4 \times 0.778801 + 0.367879]$

$$= 0.7471805$$

Mid-point rule: $\int_0^1 e^{-x^2} dx = 2 \times 0.5 \times e^{-(0.25)}$

$$= 2 \times 0.5 \times 0.778801$$

$$= 0.778801$$

Questions 4.2

1. Approximate the following integrals using the Trapezoidal rule.

a. $\int_{0.5}^1 x^4 dx$

b. $\int_0^{0.5} \frac{2}{x-4} dx$

c. $\int_1^{1.5} x^2 \ln x dx$

d. $\int_0^1 x^2 e^{-x} dx$

e. $\int_1^{1.6} \frac{2x}{x^2-4} dx$

f. $\int_0^{0.35} \frac{2}{x^2-4} dx$

g. $\int_0^{\pi/4} x \sin x dx$

h. $\int_0^{\pi/4} e^{3x} \sin 2x dx$

2. Find a bound for the error in Exercise 1 using the error formula and compare this to the actual error.

3. Repeat Exercise 1 using Simpson's rule

4. Repeat Exercise 2 using Simpson's rule and the results of Exercise 3.

5. Repeat Exercise 1 using the Midpoint rule.

6. Repeat Exercise 2 using the Midpoint rule and the results of Exercise 5.

7. The Trapezoidal rule applied to $\int_0^2 f(x)dx$ gives the value 4, and Simpson's rule gives the value 2. What is $f(1)$?

8. The Trapezoidal rule applied to $\int_0^2 f(x)dx$ gives the value 5, and the Midpoint rule gives the value 4. What value does Simpson's rule give?

Answers

1. The Trapezoidal rule gives the following approximations

a. 0.265625 b. -0.2678571 c. 0.228074
d. 0.1839397

e. -0.8666667 f. -0.1777643 g. 0.2180895 h.
4.1432597

2. The errors are shown in the table

	Actual Error	Error Bound
a	0.071875	0.125
b	7.943×10^{-4}	9.718×10^{-4}
c	0.0358147	0.0396972
d	0.0233369	0.1666667
e	0.1326975	0.5617284

f	9.443×10^{-4}	1.0707×10^{-3}
g	0.0663431	0.0807455
h	1.554631	2.298827

3. Simpson's rule gives the following approximations

- a. 0.1940104 b. -0.2670635 c. 0.1922453 d. 0.16240168
 e. -0.7391053 f. -0.1768216 g. 0.1513826 h. 2.5836964

4. The errors are shown in the table.

	Actual Error	Error Bound
a	2.604×10^{-4}	2.6042×10^{-4}
b	7.14×10^{-7}	9.92×10^{-7}
c	1.406×10^{-5}	2.170×10^{-5}
d	1.7989×10^{-3}	4.1667×10^{-4}
e	$5.1361.14 \times 10^{-3}$	0.063280
f	1.549×10^{-6}	2.095×10^{-6}
g	3.6381×10^{-4}	4.1507×10^{-4}
h	4.9322×10^{-3}	0.1302826

5. The Midpoint rule gives the following approximations

- a. 0.1582031 b. -0.2666667 c. 0.1743309 d. 0.1516327

e. -0.6753247 f. -0.1768200 g. 0.1180292 h. 1.8039148

6.

	Actual Error	Error Bound
a	0.0355469	0.0625
b	3.961×10^{-4}	4.859×10^{-4}
c	0.0179285	0.0198486
d	89701×10^{-3}	0.0833333
e	0.0564448	0.2808642
f	4.698×10^{-4}	5.353×10^{-4}
g	0.0337172	0.0403728
h	0.7847138	1.1494136

7. $1/2$

8. 6

4.3 Composite Numerical Integration

The Newton-Cotes formulas are generally unsuitable for use over large integration intervals. High-degree formulas would be required, and the values of the coefficients in these formulas are difficult to obtain. Also, the Newton-Cotes formulas are based on interpolatory polynomials that use equally spaced nodes, a procedure that is inaccurate over large

In this section, we discuss a piecewise approach to numerical integration that uses the low-order Newton-Cotes formulas. These are the techniques most often applied.

Example 1. Use Simpson's rule to approximate $\int_0^4 e^x dx$ and compare this to the results obtained by adding the Simpson's rule approximation for $\int_0^2 e^x dx$ and $\int_2^4 e^x dx$ and adding those for $\int_0^1 e^x dx$, $\int_1^2 e^x dx$, $\int_2^3 e^x dx$ and $\int_3^4 e^x dx$.

Solution. Simpson's rule on $[0,4]$ uses $h = 2$ and gives

$$\int_0^4 e^x dx \approx \frac{2}{3}(e^0 + 4e^2 + e^4) = 56.76958.$$

The exact answer in this case is $e^4 - e^0 = 53.59815$, and the error -3.17143 is far larger than we would normally accept.

Applying Simpson's rule on each of the intervals $[0, 2]$ and $[2,4]$ uses $h = 1$ and gives

$$\begin{aligned} \int_0^4 e^x dx &= \int_0^2 e^x dx + \int_2^4 e^x dx \\ &\approx \frac{1}{3}(e^0 + 4e + e^2) + \frac{1}{3}(e^2 + 4e^3 + e^4) \\ &= \frac{1}{3}(e^0 + 4e + 2e^2 + 4e^3 + e^4) \\ &= 53.86385. \end{aligned}$$

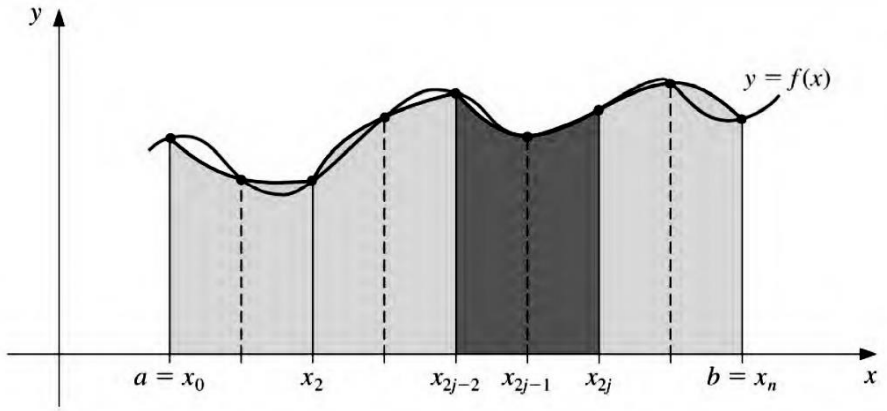
The error has been reduced to -0.26570.

For the integral on $[0,1]$, $[1,2]$, $[2,3]$, and $[3,4]$, we use Simpson's rule four times with $h = \frac{1}{2}$ giving

$$\begin{aligned}
 \int_0^4 e^x dx &= \int_0^1 e^x dx + \int_1^2 e^x dx + \int_2^3 e^x dx + \int_3^4 e^x dx \\
 &\approx \frac{1}{6} \left(e^0 + 4 e^{\frac{1}{2}} + e \right) + \frac{1}{6} \left(e + 4 e^{\frac{3}{2}} + e^2 \right) + \frac{1}{6} \left(e^2 + 4 e^{\frac{5}{2}} + e^3 \right) \\
 &\quad + \frac{1}{6} \left(e^3 + 4 e^{\frac{7}{2}} + e^4 \right) \\
 &= \frac{1}{6} (e^0 + 4 e^{\frac{1}{2}} + 2e + 4 e^{\frac{3}{2}} + 2e^2 + 4 e^{\frac{5}{2}} + 2e^3 + 4 e^{\frac{7}{2}} + e^4) \\
 &= 53.61622.
 \end{aligned}$$

The error has been reduced to -0.01807.

To generalize this procedure for an arbitrary integral $\int_a^b f(x)dx$, choose an even integer n . Subdivide the interval $[a, b]$ into n subintervals and apply Simpson's rule on each consecutive pair of subintervals. (See Figure 4.7.)



With $h = (b - a)/n$ and $x_j = a + jh$, for each $j = 0, 1, \dots, n$, we have

$$\int_a^b f(x) dx = \sum_{j=1}^{\frac{n}{2}} \int_{x_{2j-2}}^{x_{2j}} f(x) dx$$

$$\sum_{j=1}^{\frac{n}{2}} \left\{ \frac{h}{3} \left[f(x_{2j-2}) + 4f(x_{2j-1}) + f(x_{2j}) \right] - \frac{h^5}{90} f^{(4)}(\xi_j) \right\},$$

For some ξ_j with $x_{2j-2} < \xi_j < x_{2j}$, provided that $f \in C^4[a, b]$. Using the fact that for each $j = 1, 2, \dots, \left(\frac{n}{2}\right) - 1$ we have $f(x_{2j})$ appearing in the term corresponding to the interval $[x_{2j-2}, x_{2j}]$ and also in the term corresponding to the interval $[x_{2j}, x_{2j+2}]$, we can reduce this sum to

$$\int_a^b f(x)dx = \frac{h}{3} \left[f(x_0) + 2 \sum_{j=1}^{\left(\frac{n}{2}\right)-1} f(x_{2j}) + 4 \sum_{j=1}^{\frac{n}{2}} f(x_{2j-1}) + f(x_n) \right] - \frac{h^5}{90} \sum_{j=1}^{\frac{n}{2}} f^{(4)}(\xi_j)$$

The error associated with this approximation is

$$E(f) = -\frac{h^5}{90} \sum_{j=1}^{\frac{n}{2}} f^{(4)}(\xi_j),$$

where $x_{2j-2} < \xi_j < x_{2j}$, for each $j = 1, 2, \dots, n/2$

If $f \in C^4[a, b]$, the Extreme Value Theorem implies that $f^{(4)}$ assumes its maximum and minimum in $[a, b]$. Since

$$\min_{x \in [a, b]} f^{(4)}(x) \leq f^{(4)}(\xi_j) \leq \max_{x \in [a, b]} f^{(4)}(x),$$

we have

$$\frac{n}{2} \min_{x \in [a, b]} f^{(4)}(x) \leq \sum_{j=1}^{\frac{n}{2}} f^{(4)}(\xi_j) \leq \frac{n}{2} \max_{x \in [a, b]} f^{(4)}(x),$$

and

$$\min_{x \in [a, b]} f^{(4)}(x) \leq \frac{2}{n} \sum_{j=1}^{\frac{n}{2}} f^{(4)}(\xi_j) \leq \max_{x \in [a, b]} f^{(4)}(x),$$

By the Intermediate Value Theorem, there is a $\mu \in (a, b)$ such that

$$f^{(4)}(\mu) = \frac{2}{n} \sum_{j=1}^{\frac{n}{2}} f^{(4)}(\xi_j)$$

Thus

$$E(f) = \frac{-h^5}{90} \sum_{j=1}^{\frac{n}{2}} f^{(4)}(\xi_j) = -\frac{h^5}{180} n f^{(4)}(\mu).$$

or, since $h = \frac{b-a}{n}$,

$$E(f) = -\frac{(b-a)}{180} h^4 f^{(4)}(\mu).$$

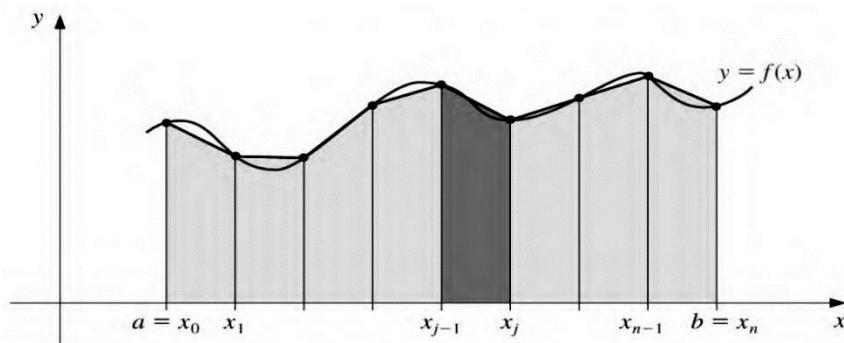
These observations produce the following result.

Theorem 4.4 Let $f \in C^4[a, b]$, n be even, $h = \frac{b-a}{n}$, and $x_j = a + jh$, for each $j = 0, 1, \dots, n$. There exists a $\mu \in (a, b)$ for which the Composite Simpson's rule for n subintervals can be written with its error term as

$$\int_a^b f(x)dx = \frac{h}{3} \left[f(a) + 2 \sum_{j=1}^{\left(\frac{n}{2}\right)-1} f(x_{2j}) + 4 \sum_{j=1}^{\frac{n}{2}} f(x_{2j-1}) + f(b) \right] - \frac{b-a}{180} h^4 f^{(4)}(\mu)$$

For the Composite Simpson's rule, we have $h = \frac{b-a}{n}$, an even integer.

The subdivision approach can be applied to any of the Newton-Cotes formulas. The extensions of the Trapezoidal (see Figure 4.8) and Midpoint rules are given without proof. The Trapezoidal rule requires only one interval for each application, so the integer n can be either odd or even.

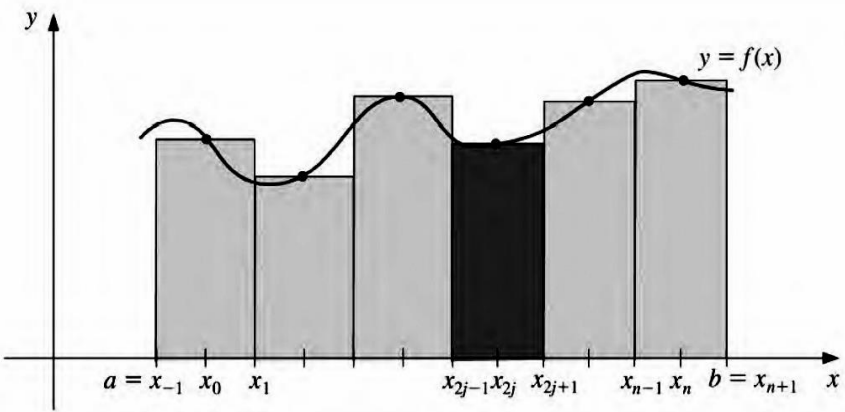


Theorem 4.5 Let $f \in C^2[a, b]$, $h = (b - a)/n$, and $x_j = a + jh$, for each $j = 0, 1, \dots, n$. There exists $\mu \in (a, b)$ for which the

Composite Trapezoidal rule for n subintervals can be written with its error term as

$$\int_a^b f(x)dx = \frac{h}{2} \left[f(a) + 2 \sum_{j=1}^{n-1} f(x_j) + f(b) \right] - \frac{b-a}{12} h^2 f''(\mu)$$

For the Composite Midpoint rule, n must again be even. (See Figure 4.9.)



Theorem 4.6 Let $f \in C^2[a, b]$, n be even, $h = (b - a)/(n + 2)$, and $x_j = a + (j + 1)h$ for each $j = -1, 0, \dots, n + 1$. There exists a $\mu \in (a, b)$ for which the **Composite Midpoint rule** for $n + 2$ subintervals can be written with its error term as

$$\int_a^b f(x)dx = 2h \sum_{j=0}^{\frac{n}{2}} f(x_{2j}) + \frac{(b-a)}{6} h^2 f''(\mu).$$

Example 2. Determine values of h that will ensure an approximation error of less than 0.00002 when approximating $\int_0^\pi \sin x \, dx$ and employing

(a) Composite Trapezoidal rule and **(b)** Composite Simpson's rule.

Solution. **(a)** The error for the Composite Trapezoidal rule for $f(x) = \sin x$ on $[0, \pi]$ is

$$\left| \frac{\pi h^2}{12} f''(\mu) \right| = \left| \frac{\pi h^2}{12} (-\sin \mu) \right| = \frac{\pi h^2}{12} |\sin \mu|$$

To ensure sufficient accuracy with this technique, we need to have

$$\frac{\pi h^2}{12} |\sin \mu| \leq \frac{\pi h^2}{12} < 0.00002.$$

Since $h = \frac{\pi}{n}$, we need

$$\frac{\pi^3}{12n^2} < 0.00002. \quad \text{which implies that } n > \left(\frac{\pi^3}{12(0.00002)} \right)^{\frac{1}{2}} \approx 359.44.$$

and the Composite Trapezoidal rule requires $n \geq 360$.

(b) The error form for the Composite Simpson's rule for $f(x) = \sin x$ on $[0, \pi]$ is

$$\left| \frac{\pi h^4}{180} f^{(4)}(\mu) \right| = \left| \frac{\pi h^4}{180} \sin \mu \right| = \frac{\pi h^4}{180} |\sin \mu|$$

To ensure sufficient accuracy with this technique, we need to have

$$\frac{\pi h^4}{180} |\sin \mu| \leq \frac{\pi h^4}{180} < 0.00002.$$

Using again the fact that $n = \pi/h$ gives

$$\frac{\pi^5}{180n^4} < 0.00002, \text{ which implies that } n > \left(\frac{\pi^5}{180(0.00002)} \right)^{\frac{1}{4}} \approx 17.07.$$

So, the Composite Simpson's rule requires only $n \geq 18$.

The Composite Simpson's rule $n = 18$ gives

$$\begin{aligned} \int_0^\pi \sin x \, dx &\approx \frac{\pi}{54} \left[2 \sum_{j=1}^8 \sin\left(\frac{j\pi}{9}\right) + 4 \sum_{j=1}^9 \sin\left(\frac{(2j-1)\pi}{18}\right) \right] \\ &= 2.0000104. \end{aligned}$$

This is accurate to within about 10^{-5} because the true value is $-\cos(\pi) - (-\cos(0)) = 2$.

Round-Off Error Stability

Suppose we apply the Composite Simpson's rule with n subintervals to a function f on $[a, b]$ and determine the maximum bound for the round-off error. Assume that $f(x_i)$ is approximated by $\tilde{f}(x_i)$ and that

$$f(x_i) = \tilde{f}(x_i) + e_i, \text{ for each } i = 0, 1, \dots, n,$$

where e_i denotes the round-off error associated with using $\bar{f}(x_i)$ to approximate $f(x_i)$. Then the accumulated error, $e(h)$, in the Composite Simpson's rule is

$$\begin{aligned} e(h) &= \left\| \frac{h}{3} \left[e_0 + 2 \sum_{j=1}^{\left(\frac{n}{2}\right)-1} e_{2j} + 4 \sum_{j=1}^{n/2} e_{2j-1} + e_n \right] \right\| \\ &\leq \frac{h}{3} \left[|e_0| + 2 \sum_{j=1}^{\left(\frac{n}{2}\right)-1} |e_{2j}| + 4 \sum_{j=1}^{n/2} |e_{2j-1}| + |e_n| \right] \end{aligned}$$

If the round-off errors are uniformly bounded by ε , then

$$e(h) \leq \frac{h}{3} \left[\varepsilon + 2 \left(\frac{n}{2} - 1 \right) \varepsilon + 4 \left(\frac{n}{2} \right) \varepsilon + \varepsilon \right] = \frac{h}{3} 3n\varepsilon = nh\varepsilon.$$

But $nh = b - a$, so

$$e(h) \leq (b - a)\varepsilon,$$

a bound independent of h (and n). This means that, even though we may need to divide an interval into more parts to ensure accuracy, the increased computation that is required does not increase the round-off error. This result implies that the procedure is stable as h approaches zero. Recall that this was not true of the numerical differentiation procedures considered at the beginning of this chapter.

Summarizing the results, we have:

Composite Simpson's rule

$$\int_{x_0}^{x_n} f(x)dx = \frac{h}{3} [f(x_0) + 4[f(x_1) + f(x_3) + \cdots + f(x_{n-1})] \\ + 2[f(x_2) + f(x_4) + \cdots + f(x_{n-2})] + f(x_n)]$$

$$\text{where } x_i = x_0 + ih, i = 0, 1, 2, \dots, n.$$

and $h = \frac{b-a}{n}$. Note that n should be even.

Composite Trapezoidal rule

$$\int_{x_0}^{x_n} f(x)dx = \frac{h}{2} [f(x_0) \\ + 2[f(x_1) + f(x_2) + \cdots + f(x_{n-1}) + f(x_n)]]$$

$$\text{where } x_i = x_0 + ih, i = 0, 1, 2, \dots, n.$$

and $h = \frac{b-a}{n}$. Here n can be either odd or even.

Composite Midpoint rule

$$\int_{x_{-1}}^{x_{n+1}} f(x)dx = 2h[f(x_0) + f(x_2) + \cdots + f(x_n)]$$

$$\text{where } x_i = x_{-1} + (i + 1)h, i = -1, 0, \dots, n + 1.$$

and $h = \frac{x_{n+1} - x_{-1}}{(n+2)}$. Note that n should be even.

Example 3. Use Composite Trapezoidal rule with $n = 4$ to approximate $\int_1^2 x^2 dx$.

Solution. We have $n = 4$ and $a = x_0 = 1, b = x_n = 2$. So $h = \frac{b-a}{n} = \frac{1}{4}$.

We need the following values:

i	x_i	$f(x_i) = x_i^2$
0	1	1
1	1.25	1.5625
2	1.50	2.25
3	1.75	3.0625
4	2	4

$$\begin{aligned}
 \int_1^2 x^2 dx &= \frac{h}{2} [f(x_0) + f(x_4) + 2(f(x_1) + f(x_2) + f(x_3))] \\
 &= \frac{1}{8} [1 + 4 + 2(6.8750)] \\
 &= 2.3437
 \end{aligned}$$

Example 4. Using Composite Trapezoidal rule approximate $\int_0^1 e^{-x^2} dx$ with $n = 10$.

Solution. Here $h = \frac{1-0}{10} = 0.1$ and need the following data.

i	x_i	$f(x_i) = e^{-x_i^2}$
0	0	1
1	0.1	0.990050
2	0.2	0.960789
3	0.3	0.913931
4	0.4	0.852144
5	0.5	0.778801
6	0.6	0.697676
7	0.7	0.612626
8	0.8	0.527292
9	0.9	0.444858
10	1	0.367879

$$\begin{aligned} \therefore \int_0^1 e^{-x^2} dx &= \frac{0.1}{2} [1 + 0.367879 + 2(6.778167)] \\ &= 0.746211 \end{aligned}$$

Example 5. Using composite Simpson's rule, approximate $\int_0^5 \frac{dx}{4x+5}$ with $n = 10$

Solution. Here $h = \frac{5-0}{10} = 0.5$

Computing the values of $f(x) = \frac{1}{4x+5}$ at the points x_i .

i	x_i	$f(x_i) = \frac{1}{4x_i + 5}$
0	0	0.2
1	0.5	0.1429
2	1	0.1111
3	1.5	0.0909
4	2	0.0769
5	2.5	0.6666
6	3	0.0588
7	3.5	0.0526
8	4	0.0476
9	4.5	0.0434
10	5	0.04

$$\begin{aligned}
 \therefore \int_0^5 \frac{1}{4x+5} dx &= \frac{0.5}{3} [(0.2 + 0.04) \\
 &\quad + 4(0.1429 + 0.0909 + 0.6666 + 0.0526 \\
 &\quad + 0.0434) \\
 &\quad + 2(0.1111 + 0.0769 + 0.0588 + 0.0476)]
 \end{aligned}$$

$$\begin{aligned}
 &= \frac{0.5}{3} [0.24 + 4(0.3963) + 2(0.2944)] \\
 &= 0.4023
 \end{aligned}$$

Example 6. Approximate $\sqrt{\frac{2}{\pi}} \int_0^1 e^{-x^2} dx$ using composite Simpson's rule with $h = 0.125$

Solution. Since $h = 0.125$, we have 8 subintervals. $(h = \frac{b-a}{h})$

i	x_i	$f(x_i) = \sqrt{\frac{2}{\pi}} e^{-x_i^2}$
0	0	0.7979
1	0.125	0.7917
2	0.25	0.7733
3	0.375	0.7437
4	0.5	0.7041
5	0.625	0.6563
6	0.75	0.6023
7	0.875	0.5441
8	1	0.4839

$$\begin{aligned}\sqrt{\frac{2}{\pi}} \int_0^1 e^{-x_i^2} dx &= \frac{0.125}{3} [(f(x_0) + f(x_8)) \\ &\quad + 4(f(x_1) + \cdots + f(x_7)) \\ &\quad + 2(f(x_2) + \cdots + f(x_6))] \\ &= 0.6827\end{aligned}$$

Example 7. Use the Composite Midpoint rule with $x + 2$ subintervals to approximate $\int_{-0.5}^{0.5} \cos^2 x \, dx, x = 4$.

Solution. Here $h = \frac{b-a}{n+2} = \frac{1}{6}$

i	-1	0	1	2	3	4	5
x_i	-0.5	-0.3333	-0.1666	0.0000	0.1666	0.3333	0.5
$f(x_i)$	0.9999	0.9999	0.9999	1	0.9999	0.9999	0.9999
	2	7	9		9	7	2

Hence $\int_{-0.5}^{0.5} \cos^2 dx$ (by composite midpoint rule with 6 subintervals)

$$\begin{aligned}&= 2 \times \frac{1}{6} \times [0.99997 + 1 + 0.99997] \\ &= 0.99998\end{aligned}$$

Questions 4.3

1. Use the Composite Trapezoidal rule with the indicated values of n to approximate the following integrals.

a. $\int_1^2 x \ln x \, dx, \quad n = 4$

b. $\int_{-2}^2 x^3 e^x \, dx, \quad n = 4$

c. $\int_0^2 \frac{2}{x^2+4} \, dx, \quad n = 6$

d. $\int_0^\pi \cos x \, dx, \quad n = 6$

e. $\int_0^2 e^{2x} \sin 3x \, dx, \quad n = 8$

f. $\int_1^3 \frac{x}{x^2+4} \, dx, \quad n = 4$

g. $\int_3^5 \frac{1}{\sqrt{x^2-4}} \, dx, \quad n = 8$

h. $\int_0^{3\pi/8} \tan x \, dx, \quad n = 8$

2. Use the Composite Simpson's rule to approximate the integrals in Exercise 1.

3. Use the Composite Midpoint rule with $n + 2$ subintervals to approximate the integrals in Exercise 1.

4. Approximate $\int_0^2 x^2 \ln(x^2 + 1) \, dx$ using $h = 0.25$. Use

a. Composite Trapezoidal rule.

b. Composite Simpson's rule.

c. Composite Midpoint rule.

5. Approximate $\int_0^2 x^2 e^{-x^2} dx$ using $h = 0.25$. Use

a. Composite Trapezoidal rule.

b. Composite Simpson's rule.

c. Composite Midpoint rule

6. Determine the values of n and h required to approximate

$$\int_0^2 e^{2x} \sin 3x dx$$

to within 10^{-4} . Use

a. Composite Trapezoidal rule.

b. Composite Simpson's rule.

c. Composite Midpoint rule

7. Determine the values of n and h required to approximate

$$\int_0^2 \frac{1}{x+4} dx$$

to within 10^{-5} and compute the approximation. Use

a. Composite Trapezoidal rule.

b. Composite Simpson's rule.

c. Composite Midpoint rule

ANSWERS

1. The Composite Trapezoidal rule approximations are;

- a. 0.639900 b. 31.3653 c. 0.784241
 d. -6.42872
- e. -13.5760 f. 0.476977 g. 0.605498
 h. 0.970926

2. The Composite Trapezoidal rule approximations are;

- a. 0.6363098 b. 22.477713 c. 0.7853980 d. -6.274868
- e. -14.18334 f. 0.4777547 g. 0.6043941 h. 0.9610554

3. The Composite Midpoint rule approximations are

- a. 0.633096 b. 11.1568 c. 0.786700
 d. -6.11274
- e. -14.9985 f. 0.478751 g. 0.602961
 h. 0.947868

4. a. 3.15947567 b. 3.10933713 c. 3.00906003

5.

a. The Composite Trapezoidal rule requires $h < 0.000922295$ and $n \geq 2168$.

b. The Composite Simpson's rule requires $h < 0.037658$ and $n \geq 54$.

c. The Composite Midpoint rule requires $h < 0.00065216$ and $n \geq 3066$.

6.

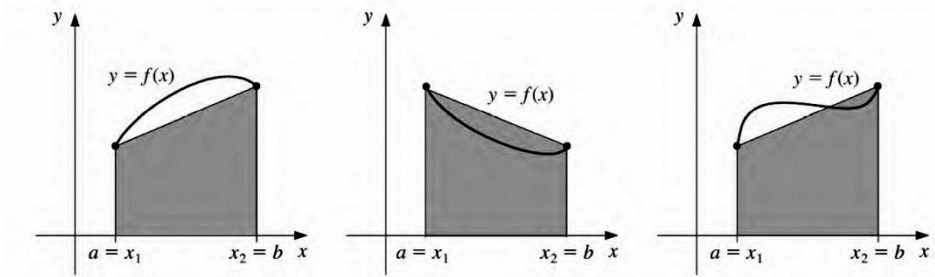
- a. The Composite Trapezoidal rule requires $h < 0.04382$ and $n > 46$. The approximation is 0.405471.
- b. The Composite Simpson's rule requires $h < 0.44267$ and $n > 6$. The approximation is 0.405466.
- c. The Composite Midpoint rule requires $h < 0.03098$ and $n > 64$. The approximation is 0.405460

4.4 Gaussian Quadrature

The Newton-Cotes formulas in Section 4.3 were derived by integrating interpolating polynomials. The error term in the interpolating polynomial of degree n involves the $(n + 1)$ st derivative of the function being approximated, so a Newton-Cotes formula is exact when approximating the integral of any polynomial of degree less than or equal to n .

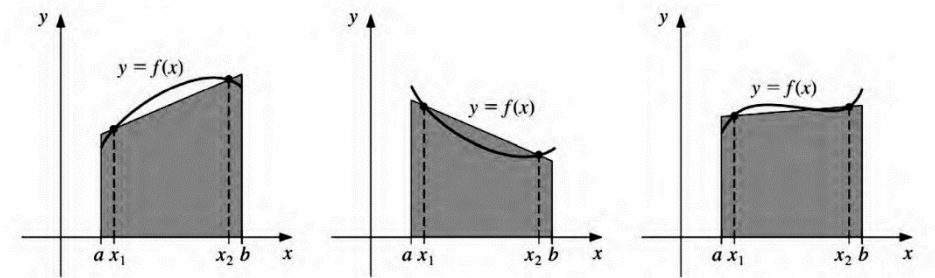
All the Newton-Cotes formulas use values of the function at equally spaced points. This restriction is convenient when the formulas are combined to form the composite rules, but it can significantly decrease the accuracy of the approximation. Consider, for example, the Trapezoidal rule applied to determine the integrals of the functions whose graphs are shown in Figure 4.10.

Figure 4.10



The Trapezoidal rule approximates the integral of the function by integrating the linear function that joins the endpoints of the graph of the function. But this is not likely the best line for approximating the integral. Lines such as those shown in Figure 4.11 would likely give much better approximations in most cases.

Figure 4.11



In Gaussian quadrature, the points for evaluation are chosen in an optimal rather than an equally spaced way. The nodes $x_1, x_2, \dots, x_n$ in the interval $[a, b]$ and coefficients $c_1, c_2, \dots, c_n$ are chosen to minimize the expected error obtained in the approximation

$$\int_a^b f(x)dx \approx \sum_{i=1}^n c_i f(x_i)$$

To measure this accuracy, we assume that the best choice of these values produces the exact result for the largest class of polynomials, that is, the choice that gives the greatest degree of precision.

The coefficients $c_1, c_2, \dots, c_n$ in the approximation formula are arbitrary, and the nodes $x_1, x_2, \dots, x_n$ are restricted only by the fact that they must lie in $[a, b]$, the interval of integration. This gives us $2n - 1$ parameters to choose. If the coefficients of a polynomial are considered parameters, the class of polynomials of degree at most $2n - 1$ also contains $2n$ parameters. This, then, is the largest class of polynomials for which it is reasonable to expect a formula to be exact. With the proper choice of the values and constants, exactness on this set can be obtained.

To illustrate the procedure for choosing the appropriate parameters, we will show how to select the coefficients and nodes when $n = 2$ and the interval of integration is $[-1, 1]$. We will then discuss the more general situation for an arbitrary choice of nodes and coefficients and show how the technique is modified when integrating over an arbitrary interval.

Suppose we want to determine c_1, c_2, x_1 and x_2 so that the integration formula

$$\int_{-1}^1 f(x)dx \approx c_1 f(x_1) + c_2 f(x_2)$$

gives the exact result whenever $f(x)$ is a polynomial of degree $2(2) - 1 = 3$ or less, that is, when

$$f(x) = a_0 + a_1x + a_2x^2 + a_3x^3.$$

for some collection of constants, a_0, a_1, a_2 , and a_3 . Because

$$\begin{aligned} \int (a_0 + a_1x + a_2x^2 + a_3x^3)dx \\ = a_0 \int 1dx \\ + a_1 \int xdx + a_2 \int x^2dx + a_3 \int x^3dx. \end{aligned}$$

This is equivalent to showing that the formula gives exact results when $f(x) = 1, x, x^2$, and x^3 . Hence, we need c_1, c_2, x_1 and x_2 , so that

$$c_1 \cdot 1 + c_2 \cdot 1 = \int_{-1}^1 1 dx = 2 \quad c_1 \cdot x_1 + c_2 \cdot x_2 = \int_{-1}^1 x dx = 0.$$

$$c_1 \cdot x_1^2 + c_2 \cdot x_2^2 = \int_{-1}^1 x^2 dx = \frac{2}{3}. \text{ And } \quad c_1 \cdot x_1^3 + c_2 \cdot x_2^3 = \int_{-1}^1 x^3 dx = 0.$$

A little algebra shows that this system of equations has the unique solution

$$c_1 = 1. \quad c_2 = 1. \quad x_1 = -\frac{\sqrt{3}}{3}, \quad \text{and } x_2 = \frac{\sqrt{3}}{3}$$

which gives the approximation formula

$$\int_{-1}^1 f(x)dx \approx f\left(-\frac{\sqrt{3}}{3}\right) + f\left(\frac{\sqrt{3}}{3}\right)$$

This formula has degree of precision three; that is, it produces the exact result for every polynomial of degree three or less.

Legendre Polynomials

Consider Legendre polynomials, a collection $\{P_0(x), P_1(x), \dots, P_n(x), \dots\}$ with the following properties:

- (1) For each n , $P_n(x)$ is a monic polynomial of degree n
- (2) $\int_{-1}^1 P(x)P_n(x) = 0$ whenever $P(x)$ is a polynomial of degree less than n .

The first few Legendre polynomials are

$$P_0(x) = 1, \quad P_1(x) = x, \quad P_2(x) = x^2 - \frac{1}{3},$$

$$P_3(x) = x^3 - \frac{3}{5}x, \text{ and } P_4(x) = x^4 - \frac{6}{7}x^2 + \frac{3}{35}$$

The roots of these polynomials are distinct, lie in the interval $(-1, 1)$, have a symmetry with respect to the origin, and, most important, are the correct choice for determining the parameters that give us the nodes and coefficients for our quadrature method.

The nodes $x_1, x_2, \dots, x_n$ needed to produce an integral approximation formula that gives exact results for any polynomial of degree less than $2n$ are the roots of the n th-degree Legendre polynomial.

Theorem 4.7 Suppose that $x_1, x_2, \dots, x_n$ are the roots of the n th Legendre polynomial $P_n(x)$ and that for each $i = 1, 2, \dots, n$, the numbers c_i , are defined by

$$c_i = \int_{-1}^1 \prod_{\substack{j=1 \\ j \neq i}}^n \frac{x - x_j}{x_i - x_j} dx$$

If $P(x)$ is any polynomial of degree less than $2n$, then

$$\int_{-1}^1 P(x) dx = \sum_{i=1}^n c_i P(x_i).$$

Proof Let us first consider the situation for a polynomial $P(x)$ of degree less than n . Rewrite $P(x)$ in terms of $(n-1)$ st Lagrange coefficient polynomials with nodes at the roots of the n th Legendre polynomial $P_n(x)$. The error term for this representation involves the n th derivative of $P(x)$. Since $P(x)$ is of degree less than n , the n th derivative of $P(x)$ is 0, and this representation of is exact. So,

$$P(x) = \sum_{i=1}^n P(x_i) L_i(x) = \sum_{i=1}^n \prod_{\substack{j=1 \\ j \neq i}}^n \frac{x - x_j}{x_i - x_j} P(x_i).$$

and

$$\int_{-1}^1 P(x) dx = \int_{-1}^1 \left[\sum_{i=1}^n \prod_{\substack{j=1 \\ j \neq i}}^n \frac{x - x_j}{x_i - x_j} P(x_i) \right] dx$$

$$= \sum_{i=1}^n \left[\int_{-1}^1 \prod_{\substack{j=1 \\ j \neq i}}^n \frac{x - x_j}{x_i - x_j} dx \right] P(x_i) = \sum_{i=1}^n c_i P(x_i).$$

Hence, the result is true for polynomials of degree less than n .

Now consider a polynomial $P(x)$ of degree at least n but less than $2n$. Divide $P(x)$ by n th Legendre polynomial $P_n(x)$. This gives two polynomials $Q(x)$ and $R(x)$, each of degree less than n , with

$$P(x) = Q(x)P_n(x) + R(x).$$

Note that x_i is a root of $P_n(x)$ for each $i = 1, 2, \dots, n$. So we have

$$P(x_i) = Q(x_i)P_n(x_i) + R(x_i) = R(x_i)$$

We now invoke the unique power of the Legendre polynomials. First, the degree of the polynomial $Q(x)$ is less than n , so (by Legendre property (2)),

$$\int_{-1}^1 Q(x)P_n(x)dx = 0.$$

Then, since $R(x)$ is a polynomial of degree less than n , the opening argument implies that

$$\int_{-1}^1 R(x)dx = \sum_{i=1}^n c_i R(x_i).$$

Putting these facts together verifies that the formula is exact for the polynomial $P(x)$:

$$\begin{aligned}\int_{-1}^1 P(x)dx &= \int_{-1}^1 [Q(x)P_n(x) + R(x)]dx \\ &= \int_{-1}^1 R(x)dx = \sum_{i=1}^n c_i R(x_i) = \sum_{i=1}^n c_i P(x_i).\end{aligned}$$

The constants c_i , needed for the quadrature rule can be generated from the Equation in Theorem 4.7, but both these constants and the roots of the Legendre polynomials are extensively tabulated. Table 4.12 lists these values for $n = 2, 3, 4$, and 5.

Table 4.12

n	Roots $r_{n,i}$	Coefficients $c_{n,i}$
2	0.5773502692	1.0000000000
	-0.5773502692	1.0000000000
3	0.7745966692	0.5555555556
	0.0000000000	0.8888888889
	-0.7745966692	0.5555555556
4	0.8611363116	0.3478548451
	0.3399810436	0.6521451549
	-0.3399810436	0.6521451549
	-0.8611363116	0.3478548451

5	0.90617984559	0.2369268850
	0.5384693101	0.4786286705
	0.0000000000	0.5688888889
	-0.5384693101	0.4786286705
	-0.9061798459	0.2369268850

Example 1. Approximate $\int_{-1}^1 e^x \cos x \, dx$ using Gaussian quadrature with $n = 3$.

Solution. The entries in Table 4.12 gives us

$$\begin{aligned}
 \int_{-1}^1 e^x \cos x \, dx &\approx 0.5 e^{0.774596692} \cos 0.774596692 + 0.8 \cos 0 \\
 &\quad + 0.5 e^{-0.774596692} \cos(-0.774596692) \\
 &= 1.9333904.
 \end{aligned}$$

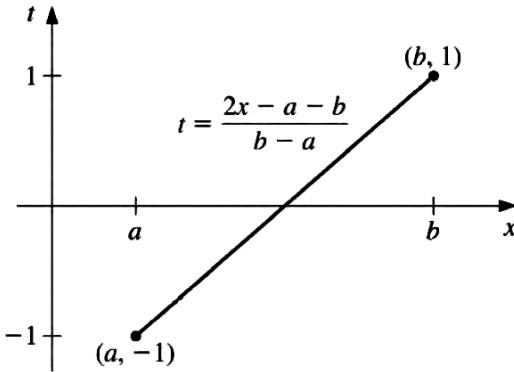
Integration by parts can be used to show that the true value of the integral is 1.9334214, so the absolute error is less than 3.2×10^{-5} .

Gaussian Quadrature on Arbitrary Intervals

An integral $\int_a^b f(x) dx$ over an arbitrary $[a, b]$ can be transformed into an integral over $[-1, 1]$ by using the change of variable (see Figure 4.12):

$$t = \frac{2x - a - b}{b - a} \Leftrightarrow x = \frac{1}{2}[(b - a)t + a + b].$$

Figure 4.12



The permits Gaussian quadrature to be applied to any interval $[a, b]$ because

$$\int_a^b f(x)dx = \int_{-1}^1 f\left(\frac{(b-a)t+(b+a)}{2}\right)\frac{(b-a)}{2}dt \quad (18)$$

Example 2. Consider the integral $\int_1^3 x^6 - x^2 \sin(2x)dx = 317.3442466$.

(a) Compare the results for the closed Newton-Cotes formula with $n = 1$, the open Newton-Cotes formula with $n = 1$, and Gaussian quadrature when $n = 2$.

(b) Compare the results for the closed Newton-Cotes formula with $n = 2$, the open Newton-Cotes formula with $n = 2$, and Gaussian quadrature when $n = 3$.

Solution.

(a) Each of the formulas in this part requires two evaluations of the function $f(x) = x^6 - x^2 \sin(2x)$. The Newton-Cotes approximations are

$$\text{Closed } n = 1: \frac{2}{2} [f(1) + f(3)] = 732.6054420;$$

$$\text{Open } n = 1: \frac{3(\frac{2}{3})}{2} \left[f\left(\frac{5}{3}\right) + f\left(\frac{7}{3}\right) \right] = 188.7856682.$$

Gaussian quadrature applied to this problem requires that the integral first be transformed into a problem whose interval of integration is $[-1, 1]$. Using Eq. (18) gives

$$\begin{aligned} \int_1^3 x^6 - x^2 \sin(2x) dx \\ = \int_{-1}^1 (t+2)^6 - (t+2)^2 \sin(2(t+2)) dt. \end{aligned}$$

Gaussian quadrature with $n = 2$ then gives

$$\begin{aligned} \int_1^3 x^6 - x^2 \sin(2x) dx \\ \approx f(-0.5773502692 + 2) + f(0.5773502692 + 2) \\ = 306.8199344. \end{aligned}$$

(b) Each of the formulas in this part requires three function evaluations. The Newton-Cotes approximations are

$$\text{Closed } n = 2: \frac{1}{3} [f(1) + 4f(2) + f(3)] = 333.2380940;$$

Open $n = 2$: $\frac{4(\frac{1}{2})}{3} [2f(1.5) + f(2) + 2f(2.5)] = 303.5912023.$

Gaussian quadrature with $n = 3$, once the transformation has been done, gives

$$\begin{aligned} \int_1^3 x^6 - x^2 \sin(2x) dx &\approx 0.5 \bar{f}(-0.7745966692 + 2) \\ &\quad + 0.8 \bar{f}(2) + 0.5 \bar{f}(-0.7745966692 + 2) \\ &= 317.2641516. \end{aligned}$$

The Gaussian quadrature results are clearly superior in each instance.

Questions 4.4

Approximate the following integrals using Gaussian Quadrature with $n = 2$ and $n = 3$ and compare your results with exact values of integrals.

1. $\int_1^{1.5} x^2 \ln x \, dx$

2. $\int_0^1 x^2 e^{-x} \, dx$

3. $\int_0^{0.35} \frac{2}{x^2 - 4} \, dx$

4. $\int_0^{\frac{\pi}{4}} x^2 \sin x \, dx$

Answers

With $n = 2$

-
1. 0.1922687
 2. 0.1594104
 3. -0.1768190
 4. 0.08926302

With $n = 3$

1. 0.1922594
2. 0.1605954
3. -0.1768200
4. 0.08875385

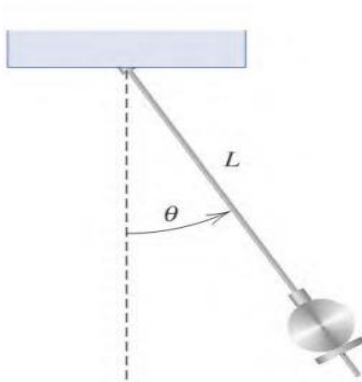
CHAPTER 5

INITIAL VALUE PROBLEMS FOR ORDINARY DIFFERENTIAL EQUATIONS

Introduction

The motion of a swinging pendulum under certain simplifying assumptions is described by the second-order differential equation

$$\frac{d^2\theta}{dt^2} + \frac{g}{L} \sin \theta = 0$$



where L is the length of the pendulum, $g \approx 32.17 \text{ ft/s}^2$ is the gravitational constant of the earth, and θ is the angle the pendulum makes with the vertical. If, in addition, we specify the position of the pendulum when the motion begins, $\theta(t_0) = \theta_0$, and its

velocity at that point, $\theta'(t_0) = \theta'_0$, we have what is called an **initial-value problem**.

For small values of θ , the approximation $\theta \approx \sin \theta$ can be used to simplify this problem to the linear initial-value problem

$$\frac{d^2\theta}{dt^2} + \frac{g}{L}\theta = 0, \quad \theta(t_0) = \theta_0, \quad \theta'(t_0) = \theta'_0$$

This problem can be solved by a standard differential-equation technique. For larger values of θ , the assumption that $\theta = \sin \theta$ is not reasonable, so approximation methods must be used.

The first part of this chapter is concerned with approximating the solution $y(t)$ to a problem of the form

$$\frac{dy}{dt} = f(t, y), \quad \text{for } a \leq t \leq b,$$

subject to an initial condition $y(a) = \alpha$. Later in the chapter, we deal with the extension of these methods to a system of first-order differential equations in the form

$$\frac{dy_1}{dt} = f_1(t, y_1, y_2, \dots, y_n)$$

$$\frac{dy_2}{dt} = f_2(t, y_1, y_2, \dots, y_n)$$

$$\vdots$$

$$\frac{dy_n}{dt} = f_n(t, y_1, y_2, \dots, y_n)$$

for $a \leq t \leq b$, subject to the initial conditions

$$y_1(a) = \alpha_1, \quad y_2(a) = \alpha_2, \dots, y_n(a) = \alpha_n.$$

We also examine the relationship of a system of this type to the general n th-order initial-value problem of the form

$$y^{(n)} = f(t, y, y', y'', \dots, y^{(n-1)}),$$

For $a \leq t \leq b$, subject to the initial conditions

$$y(a) = \alpha_1, \quad y'(a) = \alpha_2, \quad \dots, y^{n-1}(a) = \alpha_n.$$

5.1 The Elementary Theory of Initial-Value Problems

Differential equations are used to model problems in science and engineering that involve the change of some variable with respect to another. Most of these problems require the solution of an initial-value problem, that is, the solution to a differential equation that satisfies a given initial condition

The methods that we consider in this chapter do not produce a continuous approximation to the solution of the initial-value problem. Rather, approximations are found at certain specified and often equally spaced points. Some method of interpolation, commonly Hermite, is used if intermediate values are needed.

Definition

A function $f(t, y)$ is said to satisfy a **Lipschitz condition** in the variable y on a set $D \subset \mathbb{R}^2$ if a constant $L > 0$ exists with

$$|f(t, y_1) - f(t, y_2)| \leq L|y_1 - y_2|$$

whenever (t, y_1) and (t, y_2) are in D . The constant L is called a **Lipschitz constant** for f .

Example 1.

Show that $f(t, y) = t|y|$ satisfies a Lipschitz condition on the interval

$$D = \{(t, y) | 1 \leq t \leq 2 \text{ and } -3 \leq y \leq 4\}$$

Solution. For each pair of points (t, y_1) and (t, y_2) in D , we have

$$\begin{aligned} |f(t, y_1) - f(t, y_2)| &= |t|y_1| - t|y_2|| = |t|||y_1| - |y_2|| \\ &\leq 2|y_1 - y_2|. \end{aligned}$$

Thus, f satisfies a Lipschitz condition on D in the variable y with Lipschitz constant 2. The smallest value possible for the Lipschitz constant for this problem is $L = 2$ because, for example,

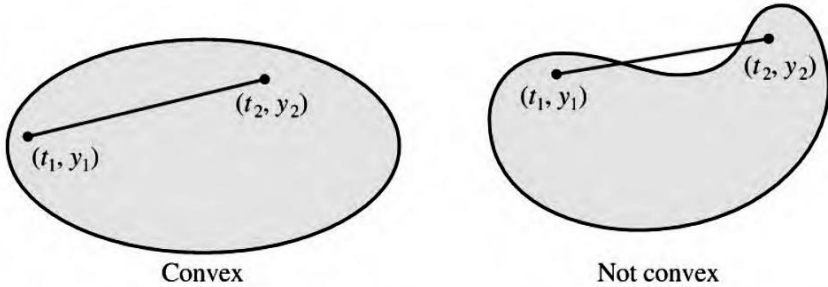
$$|f(2, 1) - f(2, 0)| = |2 - 0| = 2|1 - 0|.$$

Definition

A set $D \subset \mathbb{R}^2$ is said to be **convex** if whenever (t_1, y_1) and (t_2, y_2) belong to D , then $((1 - \lambda)t_1 + \lambda t_2, (1 - \lambda)y_1 + \lambda y_2)$ also belongs to D for every λ in $[0, 1]$.

In geometric terms, the above Definition states that a set is convex provided that whenever two points belong to the set, the entire straight-line segment between the points also belongs to the set. (See Figure 5.1) The sets we consider in this chapter are generally of the form $D = \{(t, y) | a \leq t \leq b \text{ and } -\infty < y < \infty\}$ for some constants a and b . It is easy to verify that these sets are convex.

Figure 5.1



Theorem 5.1 Suppose $f(t, y)$ is defined on a convex set $D \subset \mathbb{R}^2$. If a constant $L > 0$ exists with

$$\left| \frac{\partial f}{\partial y}(t, y) \right| \leq L, \text{ for all } (t, y) \in D, \quad (1)$$

then f satisfies a Lipschitz condition on D in the variable y with Lipschitz constant L .

We should note that Theorem 5.1 gives only sufficient conditions for a Lipschitz condition to hold

The following theorem is a version of the fundamental existence and uniqueness theorem for first-order ordinary differential equations

Theorem 5.2

Suppose that $D = \{(t, y) | a \leq t \leq b \text{ and } -\infty < y < \infty\}$ and that $f(t, y)$ is continuous on D . If f satisfies a Lipschitz condition on D in the variable y , then the initial-value problem

$$y'(t) = f(t, y), \quad a \leq t \leq b, \quad y(a) = \alpha,$$

has a unique solution $y(t)$ for $a \leq t \leq b$.

Example 2

Use Theorem 5.2 to show that there is a unique solution to the initial-value problem

$$y' = 1 + t \sin(ty), \quad 0 \leq t \leq 2, \quad y(0) = 0.$$

Solution. Holding t constant and applying the Mean Value Theorem to the function

$$f(t, y) = 1 + t \sin(ty)$$

we find that when $y_1 < y_2$, a number ξ in (y_1, y_2) exists with

$$\frac{f(t, y_2) - f(t, y_1)}{y_2 - y_1} = \frac{\partial}{\partial y} f(t, \xi) = t^2 \cos(\xi t).$$

Thus,

$$|f(t, y_2) - f(t, y_1)| = |y_2 - y_1| |t^2 \cos(\xi t)| \leq 4 |y_2 - y_1|,$$

and f satisfies a Lipschitz condition in the variable y with Lipschitz constant $L = 4$. Additionally, $f(t, y)$ is continuous when $0 \leq t \leq 2$ and $-\infty < y < \infty$, so Theorem 5.2 implies that a unique solution exists to this initial-value problem.

Well-Posed Problems

Now that we have, to some extent, taken care of the question of when initial-value problems have unique solutions, we can move to the second important consideration when approximating the

solution to an initial-value problem. Initial-value problems obtained by observing physical phenomena generally only approximate the true situation, so we need to know whether small changes in the statement of the problem introduce correspondingly small changes in the solution. This is also important because of the introduction of round-off error when numerical methods are used.

Definition

The initial-value problem

$$\begin{aligned} \frac{dy}{dt} &= f(t, y), \quad a \leq t \leq b, \\ y(a) &= \alpha, \end{aligned} \quad (2)$$

is said to be a well-posed problem if:

- A unique solution, $y(t)$, to the problem exists, and
- There exist constants $\varepsilon_0 > 0$ and $k > 0$ such that for any ε , in $(0, \varepsilon_0)$, whenever $\delta(t)$ is continuous with $|\delta(t)| < \varepsilon$ for all t in $[a, b]$ and when $|\delta_0| < \varepsilon$, the initial-value problem

$$\begin{aligned} \frac{dz}{dt} &= f(t, z) + \delta(t), \quad a \leq t \leq b, \quad z(a) = \alpha + \delta_0, \\ (3) \end{aligned}$$

has a unique solution $z(t)$ that satisfies

$$|z(t) - y(t)| < k\varepsilon \text{ for all } t \text{ in } [a, b].$$

The problem specified by Eq. (3) is called a **perturbed problem** associated with the original problem in Eq. (2). It assumes the possibility of an error being introduced in the statement of the

differential equation as well as an error (δ_0 being present in the initial condition.

Theorem 5.3

Suppose $D = \{(t, y) | a \leq t \leq b \text{ and } -\infty < y < \infty\}$. If f is continuous and satisfies a Lipschitz condition in the variable y on the set D , then the initial-value problem

$$\frac{dy}{dt} = f(t, y), \quad a \leq t \leq b, \quad y(a) = \alpha$$

is well posed

Example 3. Show that the initial-value problem

$$\frac{dy}{dt} = y - t^2 + 1, \quad 0 \leq t \leq 2, \quad y(0) = 0.5, \quad (4)$$

is well posed on $D = \{(t, y) | 0 \leq t \leq 2 \text{ and } -\infty < y < \infty\}$.

Solution. Because

$$\left| \frac{\partial(y - t^2 + 1)}{\partial y} \right| = |1| = 1,$$

Theorem 5.1 implies that $f(t, y) = y - t^2 + 1$ satisfies a Lipschitz condition in y on D with Lipschitz constant 1. Since f is continuous on D , Theorem 5.3 implies that the problem is well posed.

As an illustration, consider the solution to the perturbed problem

$$\begin{aligned}\frac{dz}{dt} &= z - t^2 + 1 + \delta, & 0 \leq t \leq 2, \\ z(0) &= 0.5 + \delta_0,\end{aligned}\tag{5}$$

where δ and δ_0 are constants. The solutions to Eqs. (4) and (5) are

$$y(t) = (t - 1)^2 - 0.5e^t \quad \text{and} \quad z(t) = (t + 1)^2 + (\delta + \delta_0 - 0.5)e^t - \delta.$$

respectively.

Suppose that ε is a positive number. If $|\delta| < \varepsilon$ and $|\delta_0| < \varepsilon$, then

$$\begin{aligned}|y(t) - z(t)| &= |(\delta + \delta_0)e^t - \delta| \leq |\delta + \delta_0| e^2 - |\delta| \\ &\leq (2e^2 + 1)\varepsilon\end{aligned}$$

for all t . This implies that problem (5.4) is well posed with $k(\varepsilon) = 2e^2 + 1$ for all $\varepsilon > 0$

Example 4. Show that $f(t, y) = t^{-2}(\sin 2t - 2ty)$, $1 \leq t \leq 2$, $y(1) = 2$ has a unique solution.

Solution. $f(t, y) = t^{-2}(\sin 2t - 2ty) = \frac{\sin 2t}{t^2} - \frac{2y}{t}$

$$\frac{\partial f}{\partial y} = -\frac{2}{t}$$

Thus, $|f(t, y_2) - f(t, y_1)| = |y_2 - y_1| \left| -\frac{2}{t} \right|$ by MVT

$$\leq 2|y_2 - y_1|,$$

and f satisfies a Lipschitz condition in the variable y with Lipschitz constant

$$L = 2 \text{ on } D = \{(t, y) / 1 \leq t \leq 2, 0 - \infty < y < \infty\}$$

Also, f is continuous on D .

So there exists a unique solution.

Example 5. Does $f(t, y) = \cos(yt)$ satisfy a Lipschitz condition on

$$D = \{(t, y) \mid 0 \leq t \leq 1, -\infty < y < \infty\}?$$

Also, is $y' = f(t, y), 0 \leq t \leq 1, y(0) = 1$ well posed?

Solution. $f(t, y) = \cos(yt)$

$$\frac{\partial f}{\partial y} = t \cos(yt)$$

Thus $|f(t, y_2) - f(t, y_1)| = |y_2 - y_1| |t \cos(yt)|$ by MVT

$$\leq |y_2 - y_1|,$$

And f satisfies the Lipschitz condition with Lipschitz constant $L=1$ in y on D .

Since f is continuous, the given IVP is well posed.

Questions 5.1

1. Use Theorem 5.2 to show that each of the following initial-value problems has a unique solution and find the solution.

a. $y' = y \cos t, \quad 0 \leq t \leq 1, \quad y(0) = 1$

b. $y' = \frac{2}{t} y + t^2 e^t, \quad 1 \leq t \leq 2, \quad y(1) = 0$

c. $y' = -\frac{2}{t}y + t^2e^t, \quad 1 \leq t \leq 2, \quad y(1) = \sqrt{2}e$

d. $y' = \frac{4t^3y}{1+t^4}, \quad 0 \leq t \leq 1, \quad y(0) = 1$

2. For each choice of $f(t, y)$ given in parts (a)-(d):

(i) Does f satisfy a Lipschitz condition on $D = \{(t, y) | 0 \leq t \leq 1, -\infty < y < \infty\}$?

(i) Can Theorem 5.3 be used to show that the initial-value problem

$$y' = f(t, y), \quad 0 \leq t \leq 1, \quad y(0) = 1,$$

is well posed?

a. $f(t, y) = t^2y + 1$

b. $f(t, y) = ty$

c. $f(t, y) = 1 - y$

d. $f(t, y) = -ty + \frac{4t}{y}$

Answers

1. a. Since $f(t, y) = y \cos t$, we have $\frac{\partial f}{\partial y}(t, y) = \cos t$, and f satisfies a Lipschitz condition in y with $L = 1$ on

$$D = \{(t, y) | 0 \leq t \leq 1, -\infty < y < \infty\}.$$

Also, f is continuous on D , so there exists a unique solution, which is $y(t) = e^{\sin t}$.

b. Since $f(t, y) = \frac{2}{t}y + t^2e^t$, we have $\frac{\partial f}{\partial y} = \frac{2}{t}$, and f satisfies a Lipschitz condition in y with $L = 2$ on

$$D = \{(t, y) | 1 \leq t \leq 2, -\infty < y < \infty\}$$

Also, f is continuous on D , so there exists a unique solution, which is $y(t) = t^2(e^t - e)$.

c. Since $f(x, y) = -\frac{2}{t}y + t^2e^t$, we have $\frac{\partial f}{\partial y} = -\frac{2}{t}$, and f satisfies a Lipschitz condition in y with $L = 2$ on

$$D = \{(t, y) | 1 \leq t \leq 2, -\infty < y < \infty\}$$

Since, f is continuous on D so there exists a unique solution, which is

$$y(t) = \frac{t^4e^t - 4t^3e^t + 12t^2e^t - 24te^t + 24e^t + (\sqrt{2} - 9)e}{t^2}.$$

d. Since $f(t, y) = \frac{4t^3y}{1+t^4}$, we have $\frac{\partial f}{\partial y} = \frac{4t^3}{1+t^4}$, and f satisfies a Lipschitz condition in y with $L = 2$ on

$$D = \{(t, y) | 0 \leq t \leq 1, -\infty < y < \infty\}.$$

Also, f is continuous on D , so there exists a unique solution, which is $y(t) = 1 + t^4$.

2. a. Lipschitz constant $L = 1$; it is a well-posed problem.

b. Lipschitz constant $L = 1$; it is a well-posed problem.

c. Lipschitz constant $L = 1$; it is a well-posed problem.

d. The function f does not satisfy a Lipschitz condition, so Theorem 5.3 cannot be used

5.2 Euler's Method

Euler's method is the most elementary approximation technique for solving initial-value problems. Although it is seldom used in practice, the simplicity of its derivation can be used to illustrate the techniques involved in the construction of some of the more advanced

techniques, without the cumbersome algebra that accompanies these constructions. The object of Euler's method is to obtain approximations to the well-posed initial-value problem

$$\frac{dy}{dt} = f(t, y), \quad a \leq t \leq b, \quad y(a) = \alpha \quad (6)$$

A continuous approximation to the solution $y(t)$ will not be obtained; instead, approximations to y will be generated at various values, called **mesh points**, in the interval $[a, b]$. Once the approximate solution is obtained at the points, the approximate solution at other points in the interval can be found by interpolation.

We first make the stipulation that the mesh points are equally distributed throughout the interval $[a, b]$. This condition is ensured by choosing a positive integer N , setting $h = (b - a)/N$, and selecting the mesh points

$$t_i = a + ih, \quad \text{for each } i = 0, 1, 2, \dots, N.$$

The common distance between the points $h = t_{i+1} - t_i$, is called the step size.

We will use Taylor's Theorem to derive Euler's method. Suppose that $y(t)$, the unique solution to (6), has two continuous derivatives on $[a, b]$, so that for each $i = 0, 1, 2, \dots, N - 1$

$$y(t_{i+1}) = y(t_i) + (t_{i+1} - t_i)y'(t_i) + \frac{(t_{i+1} - t_i)^2}{2}y''(\xi_i),$$

for some number ξ_i in (t_i, t_{i+1}) . Because $h = t_{i+1} - t_i$, we have

$$y(t_{i+1}) = y(t_i) + hy'(t_i) + \frac{h^2}{2}y''(\xi_i),$$

and, because $y(t)$ satisfies the differential equation (6),

$$\begin{aligned} & y(t_{i+1}) \\ &= y(t_i) + hf(t_i, y(t_i)) \\ & \quad + \frac{h^2}{2}y''(\xi_i) \end{aligned} \quad (7)$$

Euler's method constructs $w_i \approx y(t_i)$, for each $i = 1, 2, \dots, N$, by deleting the remainder term. Thus, Euler's method is

$$w_0 = \alpha$$

$$w_{i+1} = w_i + hf(t_i, w_i), \text{ for each } i = 0, 1, \dots, N - 1. \quad (8)$$

Illustration

Use Euler's method to approximate the solution to

$$y' = y - t^2 + 1, \quad 0 \leq t \leq 2, \quad y(0) = 0.5,$$

at $t = 2$. Here we will simply illustrate the steps in the technique when we have $h = 0.5$. For this problem, $f(t, y) = y - t^2 + 1$; so,

$$w_0 - y(0) = 0.5;$$

$$w_1 = w_0 + 0.5 (w_0 - (0.0)^2 + 1) = 0.5 + 0.5(1.5) = 1.25;$$

$$w_2 = w_1 + 0.5 (w_1 - (0.5)^2 + 1) = 1.25 + 0.5(2.0) \\ = 2.25;$$

$$w_3 = w_2 + 0.5 (w_2 - (1.0)^2 + 1) = 2.25 + 0.5(2.25) \\ = 3.375;$$

and

$$y(2) \approx w_4 = w_3 + 0.5 (w_3 - (1.5)^2 + 1) \\ = 3.375 + 0.5(2.125) = 4.4375.$$

Equation (8) is called the **difference equation** associated with Euler's method.

To interpret Euler's method geometrically, note that when w_i is a close approximation to $y(t_i)$, the assumption that the problem is well posed implies that $f(t_i, w_i) \approx y'(t_i) = f(t_i, y(t_i))$

The graph of the function highlighting $y(t_i)$ is shown in Figure 5.2. One step in Euler's method appears in Figure 5.3, and a series of steps appears in Figure 5.4.

Figure 5.2

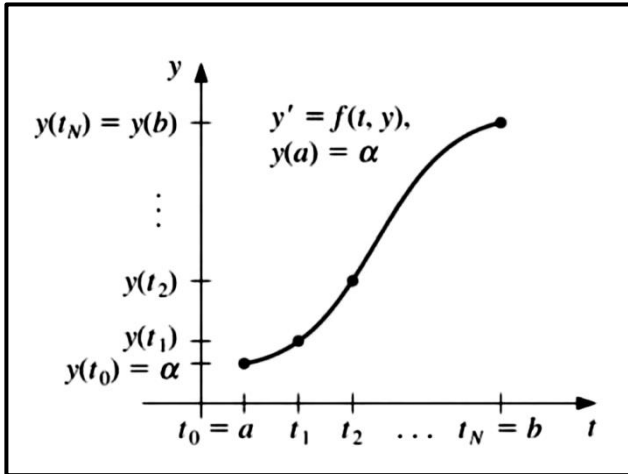


Figure 5.3

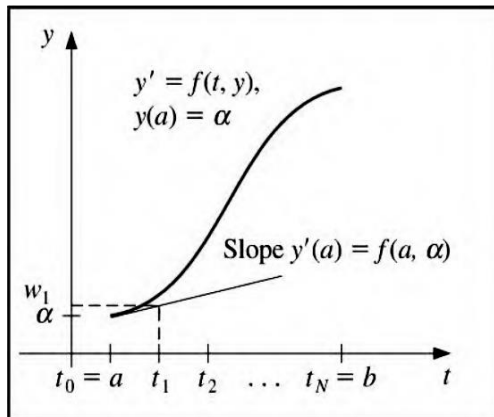
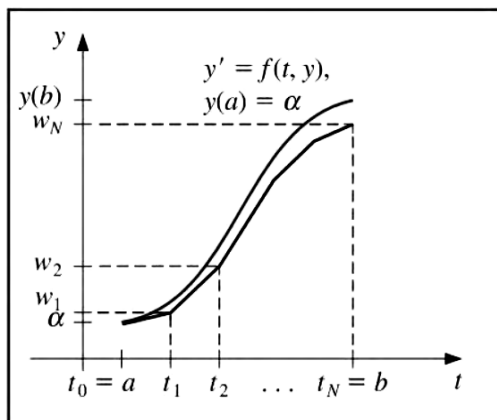


Figure 5.4



Example 1.

Euler's method was used in the first illustration with $h = 0.5$ to approximate the solution to the initial-value problem

$$y' = y - t^2 + 1, \quad 0 \leq t \leq 2, \quad y(0) = 0.5.$$

Use $N = 10$ to determine approximations and compare these with the exact values given by $y(t) = (t + 1)^2 - 0.5e^t$.

Solution. With $N = 10$, we have $h = 0.2, t_i = 0.2i, w_0 = 0.5$, and

$$\begin{aligned} w_{i+1} &= w_i + h(w_i - t_i^2 + 1) = w_i + 0.2[w_i - 0.04i^2 + 1] \\ &= 1.2w_i - 0.008i^2 + 0.2, \end{aligned}$$

for $i = 0, 1, \dots, 9$. So,

$$\begin{aligned} w_1 &= 1.2(0.5) - 0.008(0)^2 + 0.2 = 0.8, w_2 \\ &= 1.2(0.8) - 0.008(1)^2 + 0.2 = 1.152, \end{aligned}$$

and so on. Table 5.1 shows the comparison between the approximate values at t , and the actual values.

Table 5.1

η	w_1	$y_1 = y(h)$	$(y_1 - w_1)$
0.0	0.5000000	0.5000000	0.0000000
0.2	0.8000000	0.8292986	0.0292986
0.4	1.1520000	1.2140877	0.0620877
0.6	1.5504000	1.6489406	0.0985406
0.8	1.9884800	2.1272295	0.1387495
1.0	2.4581760	2.6408591	0.1826831
1.2	2.9498112	3.1799415	0.2301303
1.4	3.4517734	3.7324000	0.2806266
1.6	3.9501281	4.2834838	0.3333557
1.8	4.4281538	4.8151763	0.3870225
2.0	4.8657845	5.3054720	0.4396874

Error Bounds for Euler's Method

Although Euler's method is not accurate enough to warrant its use in practice, it is sufficiently elementary to analyze the error that is produced from its application.

Lemma 5.4

For all $x \geq -1$ and any positive m , we have $0 \leq (1+x)^m \leq e^{mx}$

Proof. Applying Taylor's Theorem with $f(x) = e^x$, $x_0 = 0$, and $n = 1$ gives

$$e^x = 1 + x + \frac{1}{2}x^2e^\xi$$

where ξ is between x and zero. Thus,

$$= 0 \leq 1 + x \leq 1 + x + \frac{1}{2}x^2e^\xi = e^x$$

and, because $1 + x \geq 0$, we have

$$0 \leq (1+x)^m \leq (e^x)^m = e^{mx}.$$

Lemma 5.5 If s and t are positive real numbers, $\{a\}_{i=0}^k$ is a sequence satisfying $a_0 \geq -\frac{t}{s}$ and $a_{i+1} \leq (1+s)a_i + t$, for each $i = 0, 1, 2, \dots, k-1$ (9)

Then

$$a_{i+1} \leq e^{(i+1)s} \left(a_0 + \frac{t}{s} \right) - \frac{t}{s}. \quad (10)$$

Proof. For a fixed integer i . Inequality (9) implies that

$$\begin{aligned} & a_{i+1} \leq (1+s)a_i + t \\ & \leq (1+s)[(1+s)a_{i+1} + t] + t \\ & = (1+s)^2 a_{i+1} + [1 + (1+s)]t \end{aligned}$$

$$\leq (1+s)^2 a_{i-2} + [1 + (1+s) + (1+s)^2]t$$

$\vdots$

$$\leq (1+s)^{i+1} a_0 + [1 + (1+s) + (1+s)^2 + \cdots + (1+s)^i]t$$

But

$$1 + (1+s) + (1+s)^2 + \cdots + (1+s)^i = \sum_{j=0}^i (1+s)^j$$

is a geometric series with ratio $(1+s)$ that sums to

$$\frac{1 - (1+s)^{i+1}}{1 - (1+s)} = \frac{1}{s} [(1+s)^{i+1} - 1]$$

Thus,

$$a_{i+1} \leq (1+s)^{i+1} a_0 + \frac{(1+s)^{i+1}}{s} t = (1+s)^{i+1} \left(a_0 + \frac{t}{s} \right) - \frac{t}{s},$$

and using Lemma 5.4 with $x = 1+s$ gives

$$a_{i+1} \leq e^{(i+1)s} \left(a_0 + \frac{t}{s} \right) - \frac{t}{s}$$

Theorem 5.6 Suppose f is continuous and satisfies a Lipschitz condition with constant L on

$$D = \{(t, y) \mid a \leq t \leq b \text{ and } -\infty < y < \infty\}$$

and that a constant M exists with

$$|y''(t)| \leq M, \text{ for all } t \in [a, b],$$

where $y(t)$ denotes the unique solution to the initial-value problem

$$y' = f(t, y), \quad a \leq t \leq b, \quad y(a) = \alpha.$$

Let $w_0, w_1, \dots, w_N$ be the approximations generated by Euler's method for some positive integer N . Then, for each $i = 0, 1, 2, \dots, N$,

$$|y(t_i) - w_i| \leq \frac{hM}{2L} [e^{L(t_i-a)} - 1] \quad (11)$$

Proof. When $i = 0$, the result is clearly true since $y(t_0) = w_0 = \alpha$. From Eq. (7), we have

$$y(t_{i+1}) = y(t_i) + hf(t_i, y(t_i)) + \frac{h^2}{2} y''(\xi_i),$$

For $i = 0, 1, \dots, N - 1$, and from the equations in (8)

$$w_{i+1} = w_i + hf(t_i, w_i)$$

Using the notation $y_i = y(t_i)$ and $y_{i+1} = y(t_{i+1})$, we subtract these two equations to obtain

$$y_{i+1} - w_{i+1} = y_i - w_i + h[f(t_i, y_i) - f(t_i, w_i)] + \frac{h^2}{2} y''(\xi_i)$$

Hence,

$$\begin{aligned} |y_{i+1} - w_{i+1}| &\leq |y_i - w_i| + h|f(t_i, y_i) - f(t_i, w_i)| \\ &\quad + \frac{h^2}{2} |y''(\xi_i)| \end{aligned}$$

Now f satisfies a Lipschitz condition in the second variable with constant L , and $\|y''(t)\| \leq M$, so

$$|y_{i+1} - w_{i+1}| \leq (1 + hL)|y_i - w_i| + \frac{h^2 M}{2}.$$

Referring to Lemma 5.5 and letting $s = hL$, $t = h^2 M/2$, and $a_j = |y_j - w_j|$, for each $j = 0, 1, \dots, N$, we see that

$$|y_{i+1} - w_{i+1}| \leq e^{(i+1)hL} \left(|y_0 - w_0| + \frac{h^2 M}{2hL} \right) - \frac{h^2 M}{2hL}$$

Because $|y_0 - w_0| = 0$ and $(i+1)h = t_{i+1} - t_0 = t_{i+1} - a$, this implies that

$$|y_{i+1} - w_{i+1}| \leq \frac{hM}{2L} (e^{(t_{i+1}-a)L} - 1)$$

for each $i = 0, 1, \dots, N-1$,

The weakness of Theorem 5.6 lies in the requirement that a bound be known for the second derivative of the solution. Although this condition often prohibits us from obtaining a realistic error bound, it should be noted that if both $\partial f/\partial t$ and $\partial f/\partial y$ exist, the chain rule for partial differentiation implies that

$$\begin{aligned} y''(t) &= \frac{dy'}{dt}(t) = \frac{df}{dt}(t, y(t)) \\ &= \frac{\partial f}{\partial t}(t, y(t)) + \frac{\partial f}{\partial y}(t, y(t)) \cdot f(t, y(t)). \end{aligned}$$

So, it is at times possible to obtain an error bound for $y''(t)$ without explicitly knowing $y(t)$.

Example 2. The solution to the initial-value problem

$$y' = y - t^2 + 1, \quad 0 \leq t \leq 2, \quad y(0) = 0.5,$$

was approximated in Example 1 using Euler's method with $h = 0.2$. Use the inequality in Theorem 5.6 to find a bound for the approximation errors and compare these to the actual errors.

Solution. Because $f(t, y) = y - t^2 + 1$, we have $\partial f(t, y)/\partial y = 1$ for all y , so $L = 1$. For this problem, the exact solution is $y(t) = (t + 1)^2 - 0.5e^t$, so $y''(t) = 2 - 0.5e^t$ and

$$|y''(t)| \leq 0.5e^2 - 2, \text{ for all } t \in [0, 2]$$

[Using the inequality in the error bound for Euler's method with $h = 0.2$, $L = 1$, and $M = 0.5e^2 - 2$ gives

$$|y_i - w_i| \leq (0.5e^2 - 2)(e^{ti} - 1)$$

Hence,

$$|y(0.2) - w_1| \leq 0.1 (0.5e^2 - 2)(e^{0.2} - 1) = 0.03752,$$

$$|y(0.4) - w_2| \leq 0.1 (0.5e^2 - 2)(e^{0.4} - 1) = 0.08334,$$

and so on. Table 5.2 lists the actual error found in Example 1, together with this error bound. Note that even though the true bound for the second derivative of the solution was used, the error bound is considerably larger than the actual error, especially for increasing values of t .

Table 5.2

t_i	0.2	0.4	0.6	0.8	1.0	1.2	1.4	1.6	1.8	2.0
Actual Error	0.02930	0.06209	0.09854	0.13875	0.18268	0.23013	0.28063	0.33336	0.38702	0.43969
Error Bound	0.03752	0.08334	0.13931	0.20767	0.29117	0.39315	0.51771	0.66985	0.85568	1.08264

Theorem 5.7

Let $y(t)$ denote the unique solution to the initial-value problem

$$\begin{aligned} y' &= f(t, y), & a \leq y \leq b, \\ y(a) &= \alpha, \end{aligned} \quad (12)$$

and $u_0, u_1, \dots, u_N$ be the approximations. If $|\delta_i| < \delta$ for each $i = 0, 1, \dots, N$ and the hypotheses of Theorem 5.6 hold for Eq. (12), then

$$\begin{aligned} |y(t_i) - u_i| &\leq \frac{1}{L} \left(\frac{hM}{2} + \frac{\delta}{h} \right) [e^{L(t_i-a)} - 1] \\ &\quad + |\delta_0| e^{L(t_i-a)} \end{aligned} \quad (13)$$

for each $i = 0, 1, \dots, N - 1$,

Example 3. Use Euler's method with $h = 0.1$ to solve the IVP $\frac{dy}{dt} = t^2 + y^2, 0 \leq t \leq 0.5, y(0) = 0$.

Solution. We have $f(t, y) = t^2 + y^2$, $h = 0.1, w_0 = 0, w_{i+1} = w_i + hf(t_i, w_i)$, where $t_i = 0.1i$

Thus $w_{i+1} = w_i + 0.1[t_i^2 + w_i^2]$ and will get the following table.

i	t_i	w_i
0	0	0
1	0.1	0
2	0.2	0.001
3	0.3	0.005
4	0.4	0.014
5	0.5	0.0300196

Example 4. Using Euler Method solve the equation $y' = 1 + 2ty$ with $y(0) = 0, h = 0.02$ for $x = 0.1$

Solution. Here we have $f(t, y) = 1 + 2ty, h = 0.02, t_0 = 0, w_0 = 0, t_i = 0.02i$,

$$w_{i+1} = w_i + hf(t_i, w_i) = w_i + h[1 + 2t_i w_i]$$

i	t_i	w_i
0	0	0
1	0.02	0.02
2	0.04	0.04

3	0.06	0.06
4	0.08	0.08
5	1	0.1

Example 5. Do by Euler's Method in five steps to find the value of y for $t = 1$ approximately for the IVP $y' = t + y$, $y(0) = 0$. Compare the result with the exact value.

Solution. With $N = 5$, we have $h = 0.2$ since $t_0 = 0$ and $t_N = 1$ and $h = \frac{t_N - t_0}{N} = \frac{1 - 0}{5} = 0.2$

Also, $w_0 = 0$ and $w_{i+1} = w_i + 0.2 (t_i + w_i)$

The exact solution to $y' = t + y$, $y(0) = 0$ is $y(t) = e^t - t - 1$.

Thus, we obtain the following table of values.

i	t_i	w_i	$y_i = y(t_i)$	$ y_i - w_i $
0	0	0	0	0
1	0.2	0	0.021	0.021
2	0.4	0.040	0.92	0.052
3	0.6	0.128	0.222	0.094
4	0.8	0.274	0.426	0.152
5	1	0.489	0.718	0.229

Thus, the approximate value of $y(1) = 0.489$ by Euler's method while the exact value is 0.718

Questions 5.2

1. Use Euler's method to approximate the solutions for each of the following initial-value problems.

a. $y' = te^{3t} - 2y$, $0 \leq t \leq 1$, $y(0) = 0$, with $h = 0.5$

b. $y' = 1 + (t - y)^2$, $2 \leq t \leq 3$, $y(2) = 1$, with $h = 0.5$

c. $y' = 1 + y/t$, $1 \leq t \leq 2$, $y(1) = 2$, with $h = 0.25$

d. $y' = \cos 2t + \sin 3t$, $0 \leq t \leq 1$, $y(0) = 1$, with $h = 0.25$

2. The actual solutions to the initial-value problems in Exercise 1 are given here. Compare the actual error at each step to the error bound.

a. $y(t) = \frac{1}{5}te^{3t} - \frac{1}{25}e^{3t} + \frac{1}{25}e^{-2t}$

b. $y(t) = t + \frac{1}{1-t}$

c. $y(t) = t \ln t + 2t$

d. $y(t) = \frac{1}{2}\sin 2t - \frac{1}{3}\cos 3t + \frac{4}{3}$

3. Use Euler's method to approximate the solutions for each of the following initial-value problems.

a. $y' = \frac{y}{t} - \left(\frac{y}{t}\right)^2$, $1 \leq t \leq 2$, $y(1) = 1$, with $h = 0.1$

b. $y' = 1 + \frac{y}{t} + \left(\frac{y}{t}\right)^2, \quad 1 \leq t \leq 3, \quad y(1) = 0, \quad \text{with } h = 0.1$

c. $y' = -(y + 1)(y + 3), \quad 0 \leq t \leq 2, \quad y(0) = -2, \quad \text{with } h = 0.2$

d. $y' = -5y + 5t^2 + 2t, \quad 0 \leq t \leq 1, \quad y(0) = \frac{1}{3}, \quad \text{with } h = 0.1$

4. The actual solutions to the initial-value problems in Exercise 3 are given here. Compute the actual error in the approximations of Exercise 3

a. $y(t) = \frac{1}{1 + \ln t}$

b. $y(t) = t \tan(\ln t)$

c. $y(t) = -3 + \frac{2}{1 + e^{-2t}}$

d. $y(t) = t^2 + \frac{1}{3}e^{-5t}$

5. Given the initial-value problem

$$y' = \frac{2}{t}y + t^2e^t, \quad 1 \leq t \leq 2, \quad y(1) = 0,$$

with exact solution $y(t) = t^2 (e^t - e)$:

a. Use Euler's method with $h = 0.1$ to approximate the solution and compare it with the actual values of y

b. Use the answers generated in part (a) and linear interpolation to approximate the following values of y and compare them to the actual values.

(i) $y(1.04)$
 $y(1.97)$

(ii) $y(1.55)$

(iii)

c. Compute the value of h necessary for $|y(t_i) - w_i| \leq 0.1$, using Eq. (11)

Answers

1. Euler's method gives the approximations in the following table.

a	i	l_i	w_i	$y(l_i)$	b.	i	l_i	w_i	$y(l_i)$
	1	0.500	0.0000000	0.2836165		1	2.500	2.0000000	1.8333333
	2	1.000	1.1204223	3.2190993		2	3.000	2.6250000	2.5000000
c	i	l_i	w_i	$y(l_i)$	d	i	l_i	w_i	$y(l_i)$
	1	1.250	2.7500000	2.7789294		1	0.250	1.2500000	1.3291498
	2	1.500	3.5500000	3.6081977		2	0.500	1.6398053	1.7304898
	3	1.750	4.3916667	4.4793276		3	0.750	2.0242547	2.0414720
	4	2.000	5.2690476	5.3862944		4	1.000	2.2364573	2.1179795

2.

a

t	Actual Error	Error Bound
0.5	0.2836165	11.3938
1.0	2.0986771	42.3654

b

.

t	Actual Error	Error Bound
2.5	0.166667	0.429570
3.0	0.125000	1.59726

c

.

t	Actual Error	Error Bound
1.25	0.0289294	0.0355032
1.50	0.0581977	0.0810902
1.75	0.0876610	0.139625
2.00	0.117247	0.214785

t	Actual Error
0.25	0.0791498
0.50	0.0906844
0.75	0.0172174
1.00	0.118478

3. Euler's method gives the approximations in the following tables.

a	i	li	wi	y(li)	b	i	li	wi	y(li)
	2	1.200	1.0082645	1.0149523	.	2	$\frac{1.40}{0}$	0.4388889	0.4896817
	4	1.400	1.0385147	1.0475339		4	$\frac{1.80}{0}$	1.0520380	1.1994386
	6	1.600	1.0784611	1.0884327		6	$\frac{2.20}{0}$	1.8842608	2.2135018
	8	1.800	1.1232621	1.1336536		8	$\frac{2.60}{0}$	3.0028372	3.6784753
	10	2.000	1.1706516	1.1812322		10	$\frac{3.00}{0}$	4.5142774	5.8741000

c	i	li	wi	y(li)	d	i	li	wi	y(li)
	2	1.400	-1.6080000	$\frac{-}{1.6200510}$		2	0.2	0.4083333	0.1626265
	4	0.800	-1.3017370	$\frac{-}{1.3359632}$		4	0.4	0.1620833	0.2051118
	6	1.200	-1.1274909	$\frac{-}{1.1663454}$		6	0.6	0.3455208	0.3765957
	8	1.600	-1.0491191	$\frac{-}{1.0783314}$		8	0.8	0.6213802	0.6461052

10	2.000	-1.0181518	-	1	1.0	0.9803451	1.002246
			1.0359724	0			0

4. The actual errors for the approximations in Exercise 3 are in the following tables

a.

t	Actual Error
1.2	0.0066879
1.5	0.0095942
1.7	0.010229
2.0	0.0105806

b.

t	Actual Error
1.4	0.0507928
2.0	0.2240306
2.4	0.4742818
3.0	1.3598226

c.

t	Actual Error
0.4	0.0120510
1.0	0.0391546
1.4	0.0349030
2.0	0.0178206

d.

t	Actual Error
0.2	0.0542931
0.5	0.0363200
0.7	0.0273054
1.0	0.0219009

5. Euler's method gives the approximations in the following table

a.

t	t ₁	w ₁	y(t ₁)
1	1.1	0.271828	0.345920
5	1.5	3.18744	3.96767
6	1.6	4.62080	5.70296
9	1.9	11.7480	14.3231
10	2.0	15.3982	18.6831

b.

t	Approximation	y(t)	Error
1.04	0.108731	0.119986	0.01126
1.55	3.90412	4.78864	0.8845
1.97	14.3031	17.2793	2.976

c. $h < 0.00064$

5.3 Higher-Order Taylor Methods.

Since the object of a numerical techniques is to determine accurate approximations with minimal effort, we need a means for comparing the efficiency of various approximation methods. The first device we consider is called the **local truncation error** of the method.

Consider the initial value problem

$$y' = f(t, y), \quad a \leq t \leq b, \quad y(a) = \alpha$$

Definition

The difference method

$$w_0 = \alpha$$

$$w_{i+1} = w_i + h\phi(t_i, w_i), \text{ for each } i = 0, 1, \dots, N-1,$$

has **local truncation error**

$$\begin{aligned}\tau_{i+1}(h) &= \frac{y_{i+1} - (y_i + h\phi(t_i, y_i))}{h} \\ &= \frac{y_{i+1} - y_i}{h} - \phi(t_i, y_i),\end{aligned}\quad (14)$$

for each $i = 0, 1, \dots, N-1$, where y_i and y_{i+1} denote the solution of the differential equation at t_i and t_{i+1} , respectively.

For example, Euler's method has local truncation error at the i th step

$$\tau_{i+1}(h) = \frac{y_{i+1} - y_i}{h} - f(t_i, y_i), \text{ for each } i = 0, 1, \dots, N-1.$$

This error is a local error because it measures the accuracy of the method at a specific step, assuming that the method was exact at the previous step. As such, it depends on the differential equation, the step size, and the particular step in the approximation.

By considering Eq. (7) in the previous section, we see that Euler's method has $\tau_{i+1}(h) = \frac{h}{2}y''(\xi_i)$, for some ξ_i in (t_i, t_{i+1}) .

When $y''(t)$ is known to be bounded by a constant M on $[a, b]$, this implies

$$\left| \tau_{i+1}(h) = \frac{h}{2} M \right|.$$

Since Euler's method was derived by using Taylor's Theorem with $n = 1$ to approximate the solution of the differential equation, our first attempt to find methods for improving the convergence properties of difference methods is to extend this technique of derivation to larger values of n .

Suppose the solution $y(t)$ to the initial-value problem

$$y' = f(t, y), \quad a \leq t \leq b, \quad y(a) = \alpha$$

has $(n + 1)$ continuous derivatives. If we expand the solution, $y(t)$, in terms of its n th Taylor polynomial about t_i and evaluate at t_{i+1} , we obtain

$$\begin{aligned} y(t_{i+1}) = & y(t_i) + hy'(t_i) + \frac{h^2}{2} y''(t_i) + \cdots + \frac{h^n}{n!} y^{(n)}(t_i) \\ & + \frac{h^{n+1}}{(n+1)!} y^{(n+1)}(\xi_i), \end{aligned} \quad (15)$$

for some ξ_i in (t_i, t_{i+1}) .

Successive differentiation of the solution, $y(t)$, gives

$$\begin{aligned} y'(t) &= f(t, y(t)), \quad y''(t) = f'(t, y(t)), \text{ and, generally,} \\ y^{(k)}(t) &= f^{(k-1)}(t, y(t)) \end{aligned}$$

Substituting these results into Eq. (15) gives

$$y(t_{i+1}) = y(t_i) + hf(t_i, y(t_i)) + \frac{h^2}{2} f'(t_i, (t_i)) + \cdots$$

$$+ \frac{h^n}{n!} f^{(n-1)}(t_i, y(t_i)) \\ + \frac{h^{n+1}}{(n+1)!} f^{(n)}(\xi_i, y(\xi_i)). \quad (16)$$

The difference-equation method corresponding to Eq. (16) is obtained by deleting the remainder term involving ξ_i

Taylor method of order n

$$w_0 = \alpha$$

$$w_{i+1} = w_i + hT^{(n)}(t_i, w_i), \text{ for each } i = 0, 1, \dots, N-1, \quad (17)$$

where

$$T^{(n)}(t_i, w_i) = f(t_i, w_i) + \frac{h}{2} f'(t_i, w_i) + \dots \\ + \frac{h^{n-1}}{n!} f^{(n-1)}(t_i, w_i).$$

Euler's method is Taylor's method of order one.

Example 1 Apply Taylor's method of orders (a) two and (b) four with $N = 10$ to the initial-value problem

$$y' = y - t^2 + 1, \quad 0 \leq t \leq 2, \quad y(0) = 0.5$$

Solution. (a) For the method of order two, we need the first derivative of $f(t, y(t)) = y(t) - t^2 + 1$ with respect to the variable t . Because $y' = y - t^2 + 1$, we have

$$f'(t, y(t)) = \frac{d}{dt}(y - t^2 + 1) = y' - 2t = y - t^2 + 1 - 2t$$

so

$$\begin{aligned}
 T^{(2)}(t_i, w_i) &= f(t_i, w_i) + \frac{h}{2}f'(t_i, w_i) \\
 &= w_i - t_i^2 + 1 + \frac{h}{2}(w_i - t_i^2 + 1 - 2t) \\
 &= \left(1 + \frac{h}{2}\right)(w_i - t_i^2 + 1) - ht_i
 \end{aligned}$$

Because $N = 10$, we have $h = 0.2$, and $t_i = 0.2i$ for each $i = 1, 2, \dots, 10$. Thus, the second-order method becomes

$$w_0 = 0.5,$$

$$\begin{aligned}
 w_{i+1} &= w_i + h \left[\left(1 + \frac{h}{2}\right)(w_i - t_i^2 + 1) - ht_i \right] \\
 &= w_i + 0.2 \left[\left(1 + \frac{0.2}{2}\right)(w_i - 0.04i^2 + 1) - 0.04i \right] \\
 &= 1.22w_i - 0.0088i^2 - 0.008i + 0.22.
 \end{aligned}$$

The first two steps give the approximations

$$\begin{aligned}
 y(0.2) &\approx w_1 = 1.22(0.5) - 0.0088(0)^2 - 0.008(0) + 0.22 \\
 &= 0.83
 \end{aligned}$$

and

$$\begin{aligned}
 y(0.4) &\approx w_2 = 1.22(0.83) - 0.0088(0.2)^2 - 0.008(0.2) \\
 &\quad + 0.22 = 1.2158.
 \end{aligned}$$

All the approximations and their errors are shown in Table 5.3.

(b) For Taylor's method of order 4, we need the first three derivatives of $f(t, y(t))$ with respect to t . Again using $y' = y - t^2 + 1$, we have

$$f'(t, y(t)) = y - t^2 + 1 - 2t,$$

$$f''(t, y(t)) = \frac{d}{dt}(y - t^2 + 1 - 2t) = y' - 2t - 2$$

$$= y - t^2 + 1 - 2t - 2 = y - t^2 - 2t - 1.$$

and

$$\begin{aligned} f'''(t, y(t)) &= \frac{d}{dt}(y - t^2 - 2t - 1) = y' - 2t - 2 \\ &= y - t^2 - 2t - 1, \end{aligned}$$

So

$$\begin{aligned} T^{(4)}(t_i, w_i) &= f(t_i, w_i) + \frac{h}{2}f'(t_i, w_i) + \frac{h^2}{6}f''(t_i, w_i) \\ &\quad + \frac{h^3}{24}f'''(t_i, w_i) \\ &= w_i - t_i^2 + 1 + \frac{h}{2}(w_i - t_i^2 + 1 - 2t_i) \\ &\quad + \frac{h^2}{6}(w_i - t_i^2 - 2t_i - 1) \\ &\quad + \frac{h^3}{24}(w_i - t_i^2 - 2t_i - 1) \end{aligned}$$

$$= \left(1 + \frac{h}{2} + \frac{h^2}{6} + \frac{h^3}{24}\right)(w_i - t_i^2) - \left(1 + \frac{h}{3} + \frac{h^2}{12}\right)(ht_i) + 1 + \frac{h}{2} - \frac{h^2}{6} - \frac{h^3}{24}$$

Hence, Taylor's method of order four is

$$w_0 = 0.5,$$

$$w_{i+1} = w_i + h \left[\left(1 + \frac{h}{2} + \frac{h^2}{6} + \frac{h^3}{24}\right)(w_i - t_i^2) - \left(1 + \frac{h}{3} + \frac{h^2}{12}\right)ht_i + 1 + \frac{h}{2} - \frac{h^2}{6} - \frac{h^3}{24} \right],$$

for $i = 0, 1, \dots, N - 1$

Because $N = 10$ and $h = 0.2$, the method becomes

$$w_{i+1} = w_i + 0.2 \left[\left(1 + \frac{0.2}{2} + \frac{0.04}{6} + \frac{0.008}{24}\right)(w_i - 0.04i^2) - \left(1 + \frac{0.2}{3} + \frac{0.04}{12}\right)(0.04i) + 1 + \frac{0.2}{2} - \frac{0.04}{6} - \frac{0.008}{24} \right]$$

$$= 1.2214w_i - 0.008856i^2 - 0.00856i + 0.2186,$$

for each $i = 0, 1, \dots, 9$. The first two steps give the approximations

$$y(0.2) \approx w_1 = 1.2214(0.5) - 0.008856(0)^2 - 0.00856(0) + 0.2186 = 0.8293$$

and $y(0.4) \approx w_2 = 1.2214(0.8293) - 0.008856(0.2)^2 - 0.00856(0.2) + 0.2186 = 1.214091$

Table 5.3		
	Taylor Order 2	Error
t_1	w_1	$[y(t_1) - w_1]$
0.0	0.500000	0
0.2	0.830000	0.000701
0.4	1.215800	0.001712
0.6	1.652076	0.003135
0.8	2.132333	0.005103
1.0	2.648646	0.007787
1.2	3.191348	0.011407
1.4	3.748645	0.016245
1.6	4.306146	0.022663
1.8	4.846299	0.031122
2.0	5.347684	0.042212

Table 5.4		
	Taylor Order 4	Error
t_1	w_1	$[y(t_1) - w_1]$
0.0	0.500000	0
0.2	0.829300	0.000001
0.4	1.214091	0.000003
0.6	1.648947	0.000006
0.8	2.127240	0.000010
1.0	2.640874	0.000015
1.2	3.179964	0.000023
1.4	3.732432	0.000032
1.6	4.283529	0.000045
1.8	4.815238	0.000062
2.0	5.305555	0.000083

Theorem 5.12

If Taylor's method of order n is used to approximate the solution to

$$y'(t) = f(t, y(t)), \quad a \leq t \leq b, \quad y(a) = 0.5$$

with step size h and if $y \in C^{n+1}[a, b]$, then the local truncation error is $O(h^n)$.

Proof. Note that Eq. (16) can be rewritten

$$\begin{aligned} y_{i+1} - y_i - hf(t_i, y_i) - \frac{h^2}{2}f'(t_i, y_i) - \dots - \frac{h^n}{n!}f^{(n-1)}(t_i, y_i) \\ = \frac{h^{n+1}}{(n+1)!}f^{(n)}(\xi_i, y(\xi_i)) \end{aligned}$$

for some ξ_i in (t_i, t_{i+1}) . So, the local truncation error is

$$\tau_{i+1}(h) = \frac{y_{i+1} - y_i}{h} - T^{(n)}(t_i, y_i) = \frac{h^n}{(n+1)!}f^{(n)}(\xi_i, y(\xi_i)),$$

for each $i = 0, 1, \dots, N-1$. Since $y \in C^{n+1}[a, b]$ we have $y^{(n+1)}(t) = f^{(n)}(t, y(t))$ is bounded on $[a, b]$ and $\tau_i(h) = O(h^n)$, for each $i = 1, 2, \dots, N$.

Questions 5.3

1. Use Taylor's method of order two to approximate the solutions for each of the following initial-value problems.

a. $y' = te^{3t} - 2y, \quad 0 \leq t \leq 1, \quad y(0) = 0, \quad \text{with } h = 0.5$

b. $y' = 1 + (t - y)^2, \quad 2 \leq t \leq 3, \quad y(2) = 1, \quad \text{with } h = 0.5$

c. $y' = 1 + y/t, \quad 1 \leq t \leq 2, \quad y(1) = 2, \quad \text{with } h = 0.25$

d. $y' = \cos 2t - \sin 3t, \quad 0 \leq t \leq 1, \quad y(0) = 1, \quad \text{with } h = 0.25$

2. Repeat Exercise 1 using Taylor's method of order four.

3. Use Taylor's method of order two to approximate the solution for each of the following initial-value problems.

a. $y' = \frac{y}{t} - \left(\frac{y}{t^2}\right), \quad 0 \leq t \leq 1.2, \quad y(1) = 1, \quad \text{with } h = 0.1$

b. $y' = \sin t + e^{-t}, \quad 0 \leq t \leq 1, \quad y(0) = 0, \quad \text{with } h = 0.5$

c. $y' = (y^2 + y)/t, \quad 1 \leq t \leq 3, \quad y(1) = -2, \quad \text{with } h = 0.5$

d. $y' = -ty + 4ty^{-1}, \quad 0 \leq t \leq 1, \quad y(0) = 1, \quad \text{with } h = 0.5$

4. Given the initial-value problem

$$y' = \frac{2}{t}y + t^2e^t, \quad 1 \leq t \leq 2, \quad y(1) = 0$$

with exact solution $y(t) = t^2(e^t - e)$

a. Use Taylor's method of order two with $h = 0.1$ to approximate the solution and compare it with the actual values of y .

5. Use the Taylor method of order two with $h = 0.1$ to approximate the solution to

$$y' = 1 + t \sin(ty), \quad 0 \leq t \leq 2, \quad y(0) = 0$$

Answers

1.

a.	t_i	w_i	$y(t_1)$
	0.50	0.12500000	0.28361652
	1.00	2.02323897	3.21909932

b.	t_i	w_i	$y(t_1)$
	2.50	1.75000000	1.83333333
	3.00	2.42578125	2.50000000

2.

c.	t_i	w_i	$y(t_1)$
	1.25	2.78125000	2.77892944
	1.50	3.61250000	3.60819766
	1.75	4.48541667	4.47932763
	2.00	5.39404762	5.38629436

d.	t_i	w_i	$y(t_1)$
	0.25	1.343751000	1.32914981
	0.50	1.77218707	1.73048976
	0.75	2.11067606	2.04147203
	1.00	2.20164395	2.11797955

3.

a.	Order 2			
	i	t_i	w_i	$y(t_1)$
	1	1.1	1.214999	1.215886
	2	1.2	1.465250	1.467570

b.	Order 2			
	i	t_i	w_i	$y(t_1)$
	1	0.5	0.5000000	0.5158868
	2	1.0	1.076858	1.091818

c.	Order 2				d.	Order 2			
	i	t_i	w_i	$y(t_i)$		i	t_i	w_i	$y(t_i)$
	1	1.5	2.000000	1.500000		1	0.25	1.093750	1.087088
	2	2.0	1.777776	1.333333		2	0.50	1.312319	1.289805
	3	2.5	1.585732	1.250000		3	0.75	1.538468	1.513490
	4	3.0	1.458882	1.200000		4	1.0	1.720480	1.701870

5. Taylor's method of order two gives the results in the following table.

i	t_i	w_i	$y(t_i)$
1	1.1	0.3397852	0.3459199
5	1.5	3.910985	3.967666
6	1.6	5.643081	5.720962
9	1.9	14.15268	14.32308
10	2.0	18.46999	18.68310

5. Taylor's method of order two gives the following;

t_i	w_i	$y(t_i)$
5	0.5	0.5146389
10	1.0	1.249305

15	1.5	2.152599
20	2.0	2.095185

5.4 Runge-Kutta method

The Taylor methods outlined in the previous section have the desirable property of high order local truncation error but the disadvantage of requiring the computation and evaluation of the derivatives of $f(t, y)$.

Runge-Kutta methods have the high-order local truncation error of the Taylor methods but eliminate the need to compute and evaluate the derivatives of $f(t, y)$

The difference-equation method resulting from replacing $T^{(2)}(t, y)$ in Taylor's method of order two by $f(t + (h/2), y + (h/2)f(t, y))$ is a specific Runge-Kutta method known as the **Midpoint method**.

Midpoint Method

$$w_0 = \alpha$$

$$w_{i+1} = w_i + hf\left(t_i + \frac{h}{2}, w_i + \frac{h}{2}f(t_i, w_i)\right), \text{ for } i = 0, 1, \dots, N-1$$

Only three parameters are present in $a_1(f + \alpha_1, y + \beta_1)$, and all are needed in the match of $T^{(2)}$. So, a more complicated form is required to satisfy the conditions for any of the higher-order Taylor methods.

The most appropriate four-parameter form for approximating

$$T^{(3)}(t, y) = f(t, y) + \frac{h}{2}f'^{(t,y)} + \frac{h^2}{6}f''(t, y)$$

is

$$\begin{aligned} & a_1 f(t, y) \\ & + a_2 f(t + \alpha_2 f, y) \\ & + \delta_2 f(t, y), \end{aligned} \quad (18)$$

and even with this, there is insufficient flexibility to match the term

$$\frac{h^2}{6} \left[\frac{\partial f}{\partial y}(t, y) \right]^2 f(t, y),$$

resulting from the expansion of $(h^2/6)f''(t, y)$. Consequently, the best that can be obtained from using (18) are methods with $O(h^2)$ local truncation error.

The fact that (18) has four parameters, however, gives a flexibility in their choice, so a number of $O(h^2)$ methods can be derived. One of the most important is the Modified Euler method, which corresponds to choosing $a_1 = a_2 = \frac{1}{2}$ and $\alpha_2 = \delta_2 = h$. It has the following difference-equation form.

Modified Euler Method

$$w_0 = \alpha$$

$$\begin{aligned} w_{i+1} &= w_i + hf[f(t_i, w_i) + f(t_{i+1}, w_i + hf(t_i, w_i))], \text{ for } i \\ &= 0, 1, \dots, N-1 \end{aligned}$$

Example 2

Use the Midpoint method and the Modified Euler method with $N = 10, h = 0.2, t_i = 0.2i$ and $w_0 = 0.5$ to approximate the solution to our usual example,

$$y' = y - t^2 + 1, \quad 0 \leq t \leq 2, \quad y(0) = 0.5$$

Solution The difference equations produced from the various formulas are

Midpoint method: $w_{i+1} = 1.22w_i - 0.0088i^2 - 0.008i + 0.218$

and

Modified Euler method: $w_{i+1} = 1.22w_i - 0.0088i^2 - 0.008i + 0.216$,

for each $i = 0, 1, \dots, 9$. The first two steps of these methods give

Midpoint method: $w_1 = 1.22(0.5) - 0.0088(0)^2 - 0.008(0) + 0.218 = 0.828$

and

Modified Euler method: $w_1 = 1.22(0.5) - 0.0088(0)^2 - 0.008(0) + 0.216 = 0.826$

and

Midpoint method: $w_2 = 1.22(0.828) - 0.0088(0.2)^2 - 0.008(0.2) + 0.218$

$$= 1.21136$$

and

Modified Euler method: $w_2 = 1.22(0.826) - 0.0088(0.2)^2 - 0.008(0.2) + 0.216$

$$= 1.20692.$$

Table 5.6 lists all the results of the calculations. For this problem, the Midpoint method is superior to the Modified Euler method.

Table 5.6

t_i	$y(t_i)$	Midpoint Method	Error	Modified Euler Method	Error
0.0	0.500000 0	0.500000 0	0	0.5000000	0
0.2	0.829298 6	0.828000 0	0.001298 6	0.8260000	0.003298 6
0.4	1.214087 7	1.211360 0	0.002727 7	1.2069200	0.007167 7
0.6	1.648940 6	1.644659 2	0.004281 4	1.6372424	0.011698 2
0.8	2.127229 5	2.121284 2	0.005945 3	2.1102357	0.016993 8
1.0	2.640859 1	2.633166 8	0.007692 3	2.6176876	0.023171 5
1.2	3.179941 5	3.170463 4	0.009478 1	3.1495789	0.030362 7

1. 4	3.732400 0	3.721165 4	0.011234 6	3.6936866 2	0.038713 8
1. 6	4.283483 8	4.270621 8	0.012862 0	4.2350972	0.048386 6
1. 8	4.815176 3	4.800958 6	0.014217 7	4.7556185	0.059557 7
2. 0	5.305472 0	5.290369 5	0.015102 5	5.2330546	0.072417 3

Runge-Kutta Order Four

$$w_0 = \alpha$$

$$k_1 = hf(t_i, w_i)$$

$$k_2 = hf\left(t_i + \frac{h}{2}, w_i + \frac{1}{2}k_1\right)$$

$$k_3 = hf\left(t_i + \frac{h}{2}, w_i + \frac{1}{2}k_2\right)$$

$$k_4 = hf(t_{i+1}, w_i + k_3)$$

$$w_{i+1} = w_i + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

for each $i = 0, 1, \dots, N - 1$.

This method has local truncation error $O(h^4)$, provided that the solution $y(t)$ has five continuous derivatives. We introduce the notation k_1, k_2, k_3, k_4 into the method to eliminate the need for successive nesting in the second variable of $f(t, y)$.

Example 3

Use the Runge-Kutta method of order four with $h = 0.2, N = 10$, and $t_i = 0.2i$ to obtain approximations to the solution of the initial-value problem

$$y' = y - t^2 + 1, \quad 0 \leq t \leq 2, \quad y(0) = 0.5$$

Solution. The approximation to $y(0.2)$ is obtained by

$$w_0 = 0.5$$

$$k_1 = 0.2f(0, 0.5) = 0.2(1.5) = 0.3$$

$$k_2 = 0.2f(0.1, 0.65) = 0.328$$

$$k_3 = 0.2f(0.1, 0.664) = 0.3308$$

$$k_4 = 0.2f(0.2, 0.8308) = 0.35816$$

$$\begin{aligned} w_1 &= 0.5 + \frac{1}{6}(0.3 + 2(0.328) + 2(0.3308) + 0.35816) \\ &= 0.8292933 \end{aligned}$$

The remaining results and their errors are listed in Table 5.8.

Table 5.8

t_1	Exact $y_1 = y(t_1)$	Runge-Kutta Order /Four w_1	Error $[y_1 - w_1]$
0.0	0.5000000	0.5000000	0

0.2	0.8292986	0.8292933	0.0000053
0.4	1.2140877	1.2140762	0.0000114
0.6	1.6489406	1.6489220	0.0000186
0.8	2.1272295	2.1272027	0.0000269
1.0	2.6408591	2.6408227	0.0000364
1.2	3.1799415	3.1798942	0.0000474
1.4	3.7324000	3.7323401	0.0000599
1.6	4.2834838	4.2834095	0.0000743
1.8	4.8151763	4.8150857	0.0000906
2.0	5.3054720	5.3053630	0.0001089

Example 4. Use the modified Euler's method, to approximate the solution of $y' = t^2 + y$, $y(0) = 1$ at $t = 0.1$ with $h = 0.05$

Solution. We have $f(t, y) = t^2 + y$, $t_0 = 0$, $w_0 = 1$.

$$t_i = 0.05i, \quad w_{i+1} = w_i + \frac{h}{2} [f(t_i, w_i) + f(t_{i+1}, w_i + hf(t_i, w_i))]$$

First step $w_0 + hf(t_0, w_0) = 1 + 0.05(1) = 1.05$

Then $w_1 = 1 + \frac{0.05}{2} [f(0, 1) + f(0.05, 1.05)]$

$$= 1 + 0.025[1 + 0.05^2 + 1.05]$$

$$= 1.0513$$

Second step $w_1 + hf(t_1, w_1)$

$$= 1.0513 + 0.05[0.05^2 + 1.0513]$$

$$= 1.1040$$

$$\text{Then } w_2 = 1.0513 + \frac{0.05}{2} [f(0.05, 1.0513) + f(0.1, 1.1040)]$$

$$= 1.0513 + 0.025[(0.05^2 + 1.0513) + (0.1^2 + 1.1040)]$$

$$= 1.1055$$

Thus $y(0.1) \approx 1.1055$

Example 5. Use the modified Euler's method to approximate the solution of $y' = t + \sqrt{y}$, $y(0) = 1$ at $t = 0.2$ with $h = 0.2$.

Solution. We have $f(t, y) = t + \sqrt{y}$,

$$t_0 = 0, \quad w_0 = 1, \quad h = 0.2, \quad t_1 = 0.2$$

$$w_1 = w_0 + \frac{h}{2} [f(t_0, w_0) + f(t_1, w_0 + hf(t_0, w_0))]$$

So first consider $w_0 + (t_0, w_0)$

$$= 1 + 0.2(0 + 1) = 1.2$$

$$\text{Then } w_1 = 1 + \frac{0.2}{2} [f(0, 1) + f(0.2, 1.2)]$$

$$= 1 + 0.1[1 + 0.2 + \sqrt{1.2}]$$

$$= 1.2295.$$

Thus $y(0.2) \approx 1.2295$

Example 6. Use the Midpoint method with $h = 0.1$ to approximate $y(0.2)$ where $y' = t^2 + y^2, y(0) = 0$.

Solution. $f(t, y) = t^2 + y^2$, $h = 0.1$, $t_0 = 0$, $t_i = 0.1i$

$$w_0 = 0, \quad w_{i+1} = w_i + hf\left[t_i + \frac{h}{2}, w_i + \frac{h}{2}f(t_i, w_i)\right] \quad i = 0, 1.$$

Step 1 Consider $w_0 + \frac{h}{2}f(t_0, w_0)$

$$= 0 + \frac{0.1}{2}[t_0^2 + w_0^2] = 0$$

$$\text{Then } w_1 = w_0 + 0.1 \times f\left[0 + \frac{0.1}{2}, 0\right]$$

$$= 0 + 0.1f(0.05, 0)$$

$$= 0.1[0.05^2] = 0.00025$$

Step 2 Consider $w_1 + \frac{h}{2}f(t_1, w_1)$

$$= 0.00025 + \frac{0.1}{2}[0.1^2 + 0.00025^2]$$

$$= 0.00075$$

$$\text{Then } w_2 = w_1 + 0.1f\left[t_1 + \frac{5}{2}, w_1 + \frac{h}{2}f(t_1, w_1)\right]$$

$$= 0.00025 + 0.1 \times f(0.1 + 0.05, 0.00075)$$

$$= 0.00025 + 0.1[0.15^2 + 0.00075^2]$$

$$= 0.0025$$

Thus $y(0.2) \approx 0.0025$

Example 7. Use the Runge-Kutta method of order four with $h = 0.1$ to obtain approximation to the solution of $y' = t^2 + y^2$, $y(0) = 0$ at $t = 0.2$

Solution. We obtain $f(t, y) = t^2 + y^2$, $t_0 = 0$, $w_0 = 0$,

$$h = 0.1, \quad t_i = 0.1i \quad [i.e., t_1 = 0.1, \quad t_2 = 0.2]$$

$$w_{i+1} = w_i + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4) \text{ where}$$

$$k_1 = hf(t_i, w_i), \quad k_2 = hf\left(t_i + \frac{h}{2}, w_i + \frac{1}{2}k_1\right),$$

$$k_3 = hf\left(t_i + \frac{h}{2}, w_i + \frac{1}{2}k_2\right), \quad k_4 = hf(t_{i+1}, w_i + k_3)$$

First step. The approximation to $y(0.1)$. [we have $t_0 = 0, w_0 = 0$]

$$k_1 = 0.1f(0,0) = 0.1[0 + 0] = 0$$

$$k_2 = 0.1f\left(0 + \frac{0.1}{2}, 0 + \frac{0}{2}\right) = 0.1[0.05^2 + 0] = 0.00025$$

$$\begin{aligned} k_3 &= 0.1f\left(0 + \frac{0.1}{2}, 0 + \frac{0.00025}{2}\right) \\ &= 0.1[0.05^2 + 0.000125^2] = 0.00025 \end{aligned}$$

$$\begin{aligned} k_4 &= 0.1f(0.1, 0 + 0.00025) = 0.1[0.1^2 + 0.00025^2] \\ &= 0.001 \end{aligned}$$

$$\begin{aligned} \therefore w_1 &= 0 + \frac{1}{6}[0 + 2 \times 0.00025 + 2 \times 0.00025 + 0.001] \\ &= 0.00033 \end{aligned}$$

Second step. The approximation to $y(0.2)$

We have $t_1 = 0.1$, $w_1 = 0.00033$

$$k_1 = 0.1f(t_1, w_1) = 0.1[0.1^2 + 0.00033^2] = 0.001$$

$$\begin{aligned} k_2 &= 0.1f\left(t_1 + 0.05, w_1 + \frac{0.0001}{2}\right) = 0.1[0.15^2 + 0.00083^2] \\ &= 0.00225 \end{aligned}$$

$$\begin{aligned} k_3 &= 0.1f\left(t_1 + 0.05, w_1 + \frac{0.00225}{2}\right) \\ &= 0.1[0.15^2 + 0.001455^2] = 0.00025 \end{aligned}$$

$$k_4 = 0.1f(t_2, w_1 + 0.00025) = 0.1[0.2^2 + 0.0058^2] = 0.004$$

$$\begin{aligned} \therefore w_2 &= 0.00033 + \frac{1}{6}[0.001 + 2 \times 0.00225 + 2 \times 0.00025 \\ &\quad + 0.004] = 0.002663 \end{aligned}$$

Thus $y(0.2) \approx 0.002663$

Example 8. Use Runge-Kutta method of order four with $h = 0.2$ to approximate $y' = 1 + y^2$, $0 \leq t \leq 0.6, y(0) = 0$

Solution. Here $f(t, y) = 1 + y^2$, $h = 0.2$, $t_0 = 0$, $w_0 = 0$, $t_i = 0.2i$ for $i = 1, 2, 3$

$$w_{i+1} = w_i + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4).$$

First step The approximation to $y(0.2)$

$$k_1 + 0.2(1 + 0) = 0.2$$

$$k_2 = 0.2 \left(1 + \left(w_0 + \frac{k_1}{2} \right)^2 \right) = 0.2[1 + 0.1^2] = 0.202$$

$$k_3 = 0.2 \left(1 + \left(w_0 + \frac{k_2}{2} \right)^2 \right) = 0.2[1 + 0.101^2] = 0.20204$$

$$k_4 = 0.2(1 + (w_0 + k_3)^2) = 0.2[1 + 0.20204^2] = 0.20816$$

$$\begin{aligned} \therefore w_1 &= 0 + \frac{1}{6}[0.2 + 2 \times 0.202 + 2 \times 0.20204 + 0.20816] \\ &= 0.2027 \end{aligned}$$

Second step The approximation to $y(0.4)$

$$k_1 = 0.2[1 + w_1^2] = 0.2[1 + 0.2027^2] = 0.2082$$

$$k_2 = 0.2 \left[1 + \left(w_1 + \frac{k_1}{2} \right)^2 \right] = 0.2[1 + 0.3068^2] = 0.2188$$

$$k_3 = 0.2 \left[1 + \left(w_1 + \frac{k_2}{2} \right)^2 \right] = 0.2[1 + 0.3121^2] = 0.2195$$

$$k_4 = 0.2[1 + (w_1 + k_3)^2] = 0.2[1 + 0.4222^2] = 0.2356$$

$$\therefore w_2 = 0.2027 + \frac{1}{6} [0.2082 + 2 \times 0.2188 + 2 \times 0.2195 + 0.2356] = 0.4228$$

Third step The approximation to $y(0.6)$

$$k_1 = 0.2[1 + w_2^2] = 0.2[1 + 0.4228^2] = 0.2358$$

$$k_2 = 0.2[1 + \left(w_2 + \frac{k_1}{2}\right)^2] = 0.2[1 + (0.5407)^2] = 0.2585$$

$$k_3 = 0.2[1 + \left(w_2 + \frac{k_2}{2}\right)^2] = 0.2[1 + (0.5521)^2] = 0.2609$$

$$k_4 = 0.2[1 + (w_2 + k_3)^2] = 0.2[1 + 0.6837^2] = 0.2935$$

$$\therefore w_3 = 0.4228 + \frac{1}{6} [0.2358 + 2 \times 0.2585 + 2 \times 0.2609 + 0.2935] = 0.6842$$

Thus $y(0.6) \approx 0.6842$

Example 9. Find the approximate value of y at $t = 1$ for the IVP $y' = t + y$ by Runge-Kutta Method of order four in five steps. Compare the result with the exact value.

Solution. Given $f(t, y) = t + y$, $t_0 = 0, w_0 = 0$,

$$t_N = 1 \quad \text{So } h = \frac{t_N - t_0}{N} = \frac{1 - 0}{5} = 0.2$$

After calculation, we will obtain the following values.

i	t_i	w_i	y_i	Error $ y_i - w_i $
0	0	0	0	0
1	0.2	0.021400	0.21403	0.000003
2	0.4	0.091818	0.091825	0.000007
3	0.6	0.222107	0.222119	0.000011
4	0.8	0.425521	0.425541	0.000020
5	1	0.718251	0.718282	0.000031

Questions 5.4

1. Use the Modified Euler method to approximate the solutions to each of the following initial-value problems and compare the results to the actual values.

a. $y' = te^{3t} - 2y$, $0 \leq t \leq 1$, $y(0) = 0$, with $h = 0.5$;
actual solution $y(t) = \frac{1}{5}te^{3t} - \frac{1}{25}e^{3t} + \frac{1}{25}e^{-2t}$

b. $y' = 1 + (t - y)^2$, $2 \leq t \leq 3$, $y(2) = 1$, with $h = 0.5$;
actual solution $y(t) = 1 + \frac{1}{1-t}$

c. $y' = 1 + y/t$, $1 \leq t \leq 2$, $y(1) = 2$, with $h = 0.25$;
actual solution $y(t) = t \ln t + 2t$.

d. $y' = \cos 2t - \sin 3t$, $0 \leq t \leq 1$, $y(0) = 1$, with $h = 0.25$;
actual solution $y(t) = \frac{1}{2}\sin 2t - \frac{1}{3}\cos 3t + \frac{4}{3}$

2. Use the Modified Euler method to approximate the solutions to each of the following initial-value problems and compare the results to the actual values.

a. $y' = \frac{y}{t} - \left(\frac{y}{t}\right)^2$, $1 \leq t \leq 1.2$, $y(1) = 1$, with $h = 0.1$;
actual solution $y(t) = \frac{2t+1}{t^2+1}$

b. $y' = 1 + \frac{y}{t} + \left(\frac{y}{t}\right)^2$, $1 \leq t \leq 3$, $y(1) = 0$, with $h = 0.2$;
actual solution $y(t) = t \tan(\ln t)$

c. $y' = -(y+1)(y+3)$, $0 \leq t \leq 2$, $y(0) = -2$, with $h = 0.2$; actual solution $y(t) = -3 + 2(1 + e^{-2t})^{-1}$

d. $y' = -5y + 5t^2 + 2t$, $0 \leq t \leq 1$, $y(0) = \frac{1}{3}$, with $h = 0.1$; actual solution $y(t) = t^2 + \frac{1}{3}e^{-5t}$.

3. Repeat Exercise 1 using the Midpoint method.

4. Repeat Exercise 2 using the Midpoint method.

5. Repeat Exercise 1 using the Runge-Kutta method of order four.

6. Repeat Exercise 2 using the Runge-Kutta method of order four.

Answers

1. 1.

a.	t	Modified Euler	y(t)
	0.5	0.5602111	0.2836165
	1.0	5.3014898	3.2190993

b.	t	Modified Euler	y(t)
	2.5	1.8125000	1.8333333
	3.0	2.4815531	2.5000000

c .	t	Modified Euler	y(t)	d .	t	Modified Euler	y(t)
	1.25	2.7750000	2.7789294		0.25	1.3199027	1.3291498
	1.50	3.6008333	3.6081977		0.50	1.7070300	1.7304898
	1.75	4.4688294	4.4793276		0.75	2.0053560	2.0414720
	2.00	5.3728586	5.38622944		1.00	2.0770789	2.1179795

2.

a .	t	Modified Euler	y(t)
	1.2	1.0147137	1.0149523
	1.5	1.0669093	1.0672624
	1.7	1.1102751	1.1106551
	2.0	1.1808345	1.1812322

b .	t	Modified Euler	y(t)
	1.4	0.4850495	0.4896817
	2.0	1.6384229	1.6612818
	2.4	2.8250651	2.8765514
	3.0	5.7075690	5.8741000

c .	t	Modified Euler	y(t)
	0.4	1.6229206	-1.6200510
	1.0	-1.2442903	-1.2384058
	1.4	-1.1200763	-1.1146484

d .	t	Modified Euler	y(t)
	0.2	0.1742708	0.1626265
	0.5	0.2878200	0.2773617
	0.7	0.5088359	0.5000658

	2.0	-1.0391938	-1.0359724
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	1.0	1.0096377	1.0022460
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3.

a .	t	Midpoint	y(t)
	0.5	0.2646250	0.2836165
	1.0	3.1300023	3.2190993

b .	t	Midpoint	y(t)
	2.5	1.7812500	1.8333333
	3.0	2.4550638	2.5000000

c.	t	Midpoint	y(t)
	1.25	2.7777778	2.7789294
	1.50	3.6060606	3.6081977
	1.75	4.4763015	4.4793276
	2.00	5.3824398	5.3862914

d.	t	Midpoint	y(t)
	0.25	1.3337962	1.3291498
	0.50	1.7422854	1.7304898
	0.75	2.0596374	2.0414720
	1.00	2.1385560	2.1179795

4.

a .	Midpoint		
	t	w ₁	y(t ₁)
	1.2	1.0153257	1.0149523
	1.5	1.0677427	1.0672624

b .	Midpoint		
	t	w ₁	y(t ₁)
	1.4	0.4861770	0.4896817
	2.0	1.6438889	1.6612818

	1.7	1.1111478	1.1106551
	2.0	1.1817275	1.1812322

	2.4	2.8364357	2.8765514
	3.0	5.7386475	5.8741000

c.	Midpoint		
	t	w ₁	y(t ₁)
	0.4	-1.6192966	-1.6200510
	1.0	-1.2402470	-1.2384058
	1.4	-1.1175165	-1.1146484
	2.0	-1.0382227	-1.0359724

d.	Midpoint		
	t	w ₁	y(t ₁)
	0.2	0.1722396	0.1626265
	0.5	0.2848046	0.2773617
	0.7	0.5056268	0.5000658
	1.0	1.0063347	1.0022460

5.

a	Rung-Kutta		
.	t	w ₁	y(t ₁)
	0.5	0.2969975	0.2836165
	1.0	3.3143118	3.2190993

b	Rung Kutta		
.	t	w ₁	y(t ₁)
	2.5	1.8333234	1.8333333
	3.0	2.4999712	2.5000000

c.	Rung-Kutta		
	t	w_1	$y(t_1)$
	1.25	2.7789095	2.7789294
	1.50	3.6081647	3.6081977
	1.75	4.4792846	4.4793276
	2.00	5.3862426	5.3862944

d.	Rung Kutta		
	t	w_1	$y(t_1)$
	0.25	1.3291650	1.3291498
	0.50	1.7305336	1.7304898
	0.75	2.0415436	2.0414720
	1.00	2.1180636	2.1179795

6.

a.	Rung-Kutta		
	t	w_1	$y(t_1)$
	1.2	1.0149520	1.0149523
	1.5	1.0672620	1.0672624
	1.7	1.1106547	1.1106551
	2.0	1.1812319	1.1812322

b.	Rung Kutta		
	t	w_1	$y(t_1)$
	1.4	0.4896842	0.4896817
	2.0	1.6612651	1.6612818
	2.4	2.8764941	2.8765514
	3.0	5.8738386	5.8741000

c.	Rung-Kutta		
	t	w_1	$y(t_1)$
	0.4	-	-
		1.6200576	1.6200510

d.	Rung Kutta		
	t	w_1	$y(t_1)$
	0.2	0.1627655	0.1626265

	1.0	- 1.2384307	1.2384056			0.5	0.2774767	0.2773617
	1.4	- 1.1146769	- 1.1146184			0.7	0.5001579	0.5000658
	2.0	- 1.0359922	- 1.0359724			1.0	1.0023207	1.0022160

5.5 Multistep methods.

The methods discussed to this point in the chapter are called **one-step methods** because the approximation for the mesh point t_{i+1} involves information from only one of the previous mesh points, t_i . Although these methods might use function-evaluation information at points between t_i and t_{i+1} they do not retain that information for direct use in future approximations. All the information used by these methods is obtained within the subinterval over which the solution is being approximated.

The approximate solution is available at each of the mesh points $t_0, t_1, \dots, t_i$ before the approximation at t_{i+1} is obtained, and because the error $|w_j - y(t_j)|$ tends to increase with j , so it seems reasonable to develop methods that use these more accurate previous data when approximating the solution at t_{i+1} .

Methods using the approximation at more than one previous mesh point to determine the approximation at the next point are called **multistep methods**. The precise definition of these methods follows, together with the definition of the two types of multistep methods

Definition

An m -step multistep method for solving the initial-value problem

$$y' = f(t, y), \quad a \leq t \leq b, \quad y(a) = \alpha, \quad (19)$$

has a difference equation for finding the approximation w_{i+1} at the mesh point t_{i+1} represented by the following equation, where m is an integer greater than 1:

$$\begin{aligned} w_{i+1} = & a_{m-1}w_i + a_{m-2}w_{i-1} + \cdots + a_0w_{i+1-m} \\ & + h[b_m f(t_{i+1}, w_{i+1}) + b_{m-1}f(t_i, w_i) + \cdots \\ & + b_0 f(t_{i+1-m}, w_{i+1-m})] \end{aligned} \quad (20)$$

For $i = m - 1, m, \dots, N - 1$, where $h = (b - a)/N$, the $a_0, a_1, \dots, a_{m-1}$ and $b_0, b_1, \dots, b_m$ are constants, and the starting values

$$w_0 = \alpha, \quad w_1 = \alpha_1, \quad w_2 = \alpha_2, \quad \dots, \quad w_{m-1} = \alpha_{m-1}$$

are specified.

When $b_m = 0$, the method is called **explicit**, or **open**, because Eq. (20) then gives w_{i+1} explicitly in terms of previously determined values. When $b_m \neq 0$, the method is called **implicit**, or **closed**, because w_{i+1} occurs on both sides of Eq. (20), so w_{i+1} is specified only implicitly.

For example, the equations

$$w_0 = \alpha, \quad w_1 = \alpha_1, \quad w_2 = \alpha_2, \quad w_3 = \alpha_3,$$

$$w_{i+1} = w_i + \frac{h}{24} [55f(t_i, w_i) - 59f(t_{i-1}, w_{i-1}) + 37f(t_{i-2}, w_{i-2}) - 9f(t_{i-3}, w_{i-3})] \quad (21)$$

for each $i = 3, 4, \dots, N - 1$, define an explicit four-step method known as the **fourth-order Adams-Bashforth technique**. The equations

$$w_0 = \alpha, \quad w_1 = \alpha_1, \quad w_2 = \alpha_2,$$

$$w_{i+1} = w_i + \frac{h}{24} [9f(t_{i+1}, w_{i+1}) + 19f(t_i, w_i) - 5f(t_{i-1}, w_{i-1}) + f(t_{i-2}, w_{i-2})] \quad (22)$$

for each $i = 2, 3, \dots, N - 1$, define an implicit three-step method known as the **fourth-order Adams-Moulton technique**.

The starting values in either Eq. (21) or Eq. (22) must be specified, generally by assuming $w_0 = \alpha$ and generating the remaining values by either a Runge-Kutta or the Taylor method. We will see that the implicit methods are generally more accurate than the explicit methods, but to apply an implicit method such as (21) directly, we must solve the implicit equation for w_{i+1} . This is not always possible, and even when it can be done, the solution for w_{i+1} may not be unique.

Example 1.

In Example 3 of Section 5.4 (see Table 5.8), we used the Runge-Kutta method of order four with $h = 0.2$ to approximate the solutions to the initial value problem

$$y' = y - t^2 + 1, \quad 0 \leq t \leq 2, \quad y(0) = 0.5$$

The first four approximations were found to be $y(0) = w_0 = 0.5$, $y(0.2) \approx w_1 = 0.8292933$, $y(0.4) \approx w_2 = 1.2140762$, and $y(0.6) \approx w_3 = 1.6489220$. Use these as starting values for the fourth-order Adams-Bashforth method to compute new approximations for $y(0.8)$ and $y(1.0)$ and compare these new approximations to those produced by the Runge-Kutta method of order four.

Solution.

For the fourth-order Adams-Bashforth method, we have

$$\begin{aligned} y(0.8) \approx w_4 &= w_3 \\ &+ \frac{0.2}{24} (55f(0.6, w_3) - 59f(0.4, w_2) \\ &+ 37f(0.2, w_1) - 9f(0, w_0)) \end{aligned}$$

$$\begin{aligned} &= 1.6489220 + \frac{0.2}{24} (55f(0.6, 1.6489220) \\ &\quad - 59f(0.4, 1.2140762) \end{aligned}$$

$$+ 37f(0.2, 0.8292933) - 9f(0, 0.5))$$

$$\begin{aligned} &= 1.6489220 + 0.0083333(55(2.2889220) \\ &\quad - 59(2.0540762) \end{aligned}$$

$$+ 37(1.7892933) - 9(1.5))$$

$$= 2.1272892$$

and

$$y(1.0) \approx w_5$$

$$\begin{aligned}
 &= w_4 + \frac{0.2}{24} (55f(0.8, w_4) - 59f(0.6, w_3) + 37f(0.4, w_2) \\
 &\quad - 9f(0.2, w_1)) \\
 &= 2.1272892 + \frac{0.2}{24} (55f(0.8, 2.1272892) \\
 &\quad - 59f(0.6, 1.6489220) \\
 &\quad + 37f(0.4, 1.2140762) - 9f(0.2, 0.8292933)) \\
 &= 2.1272892 + 0.0083333(55(2.4872892) \\
 &\quad - 59(2.2889220) \\
 &\quad + 37(2.0540762) - 9(1.7892933)) \\
 &= 2.6410533
 \end{aligned}$$

The errors for these approximations at $t = 0.8$ and $t = 1.0$ are, respectively

$$|2.1272295 - 2.1272892| = 5.97 \times 10^{-5} \quad \text{and} \quad |2.6410533 - 2.6408591| = 1.94 \times 10^{-4}$$

The corresponding Runge-Kutta approximations had errors

$$|2.1272027 - 2.1272892| = 2.69 \times 10^{-5} \quad \text{and} \quad |2.6408227 - 2.6408591| = 3.64 \times 10^{-5}$$

Adams-Bashforth Explicit Methods

Some of the explicit multistep methods, together with their required starting values are as follows.

Adams-Bashforth Two-Step Explicit Method

$$w_0 = \alpha, \quad w_1 = \alpha_1$$

$$w_{i+1} = w_i + \frac{h}{2} [3f(t_i, w_i) - f(t_{i-1}, w_{i-1})]. \quad (23)$$

where $i = 1, 2, \dots, N - 1$. The local truncation error is $\tau_{i+1}(h) = \frac{5}{12} y'''(\mu_i) h^2$, for some $\mu_i \in (t_{i-1}, t_{i+1})$

Adams-Bashforth Three-Step Explicit Method

$$w_0 = \alpha, \quad w_1 = \alpha_1, \quad w_2 = \alpha_2$$

$$w_{i+1} = w_i + \frac{h}{2} [23f(t_i, w_i) - 16f(t_{i-1}, w_{i-1}) + 5f(t_{i-2}, w_{i-2})]. \quad (24)$$

where $i = 2, 3, \dots, N - 1$. The local truncation error is $\tau_{i+1}(h) = \frac{3}{8} y^4(\mu_i) h^3$, for some $\mu_i \in (t_{i-2}, t_{i+1})$

Adams-Bashforth Four-Step Explicit Method

$$w_0 = \alpha, \quad w_1 = \alpha_1, \quad w_2 = \alpha_2, \quad w_3 = \alpha_3$$

$$\begin{aligned} w_{i+1} &= w_i \\ &+ \frac{h}{24} [55f(t_i, w_i) - 59f(t_{i-1}, w_{i-1}) + 37f(t_{i-2}, w_{i-2}) \\ &- 9f(t_{i-3}, w_{i-3})]. \end{aligned} \quad (25)$$

where $i = 3, 4, \dots, N - 1$. The local truncation error is $\tau_{i+1}(h) = \frac{251}{720} y^{(5)}(\mu_i) h^4$, for some $\mu_i \in (t_{i-3}, t_{i+1})$

Adams-Moulton Implicit Methods

Implicit methods are derived by using $(t_{i+1}, f(t_{i+1}, y(t_{i+1})))$ as an additional interpolation node in the approximation of the integral

$$\int_{t_i}^{t_{i+1}} f(t, y(t)) dt$$

Some of the more common implicit methods are as follows.

Adams-Moulton Two-Step Implicit Method

$$w_0 = \alpha, \quad w_1 = \alpha_1,$$

$$w_{i+1} = w_i + \frac{h}{12} [5f(t_{i+1}, w_{i+1}) + 8f(t_i, w_i) - f(t_{i-1}, w_{i-1})]. \quad (26)$$

where $i = 1, 2, \dots, N - 1$. The local truncation error is

$$\tau_{i+1}(h) = -\frac{1}{24} y^{(4)}(\mu_i) h^3, \text{ for some } \mu_i \in (t_{i-1}, t_{i+1})$$

Adams-Moulton Three-Step Implicit Method

$$w_0 = \alpha, \quad w_1 = \alpha_1, \quad w_2 = \alpha_2$$

$$w_{i+1} = w_i + \frac{h}{24} [9f(t_{i+1}, w_{i+1}) + 19f(t_i, w_i) - 5f(t_{i-1}, w_{i-1}) + f(t_{i-2}, w_{i-2})] \quad (27)$$

where $i = 2, 3, \dots, N - 1$. The local truncation error is

$$\tau_{i+1}(h) = -\frac{19}{720}y^{(5)}(\mu_i)h^4, \text{ for some } \mu_i \in (t_{i-2}, t_{i+1})$$

Adams-Moulton Four-Step Implicit Method

$$w_0 = \alpha, \quad w_1 = \alpha_1, \quad w_2 = \alpha_2, \quad w_3 = \alpha_3$$

$$w_{i+1} = w_i + \frac{h}{720} [251f(t_{i+1}, w_{i+1}) + 646f(t_i, w_i) - 264f(t_{i-1}, w_{i-1}) + 106f(t_{i-2}, w_{i-2}) - 19f(t_{i-3}, w_{i-3})]. \quad (28)$$

where $i = 3, 4, \dots, N - 1$. The local truncation error is

$$\tau_{i+1}(h) = -\frac{3}{160}y^{(6)}(\mu_i)h^5, \text{ for some } \mu_i \in (t_{i-3}, t_{i+1})$$

Example 4.

Consider the initial-value problem

$$y' = y - t^2 + 1, \quad 0 \leq t \leq 2, \quad y(0) = 0.5$$

Use the exact values given from $y(t) = (t + 1)^2 - 0.5e^t$ as starting values and $h = 0.2$ to compare the approximations from (a) by the explicit Adams-Bashforth four-step method and (b) the implicit Adams-Moulton three-step method.

Solution. (a) The Adams-Bashforth method has the difference equation

$$w_{i+1} = w_i + \frac{h}{24} [55f(t_i, w_i) - 59f(t_{i-1}, w_{i-1}) + 37f(t_{i-2}, w_{i-2}) - 9f(t_{i-3}, w_{i-3})],$$

for $i = 3, 4, \dots, 9$. When simplified using $f(t, y) = y - t^2 + 1$, $h = 0.2$, and $t_i = 0.2i$, it becomes

$$w_{i+1} = \frac{1}{24} [35w_i - 11.8w_{i-1} + 7.4w_{i-2} - 1.8w_{i-3} - 0.192i^2 - 0.192i + 4.736].$$

(b) The Adams-Moulton method has the difference equation

$$w_{i+1} = w_i + \frac{h}{24} [9f(t_{i-1}, w_{i+1}) + 19f(t_i, w_i) - 5f(t_{i-1}, w_{i-1}) + f(t_{i-2}, w_{i-2})]$$

for $i = 2, 3, \dots, 9$ This reduces to

$$w_{i+1} = \frac{1}{24} [1.8w_{i+1} + 27.8w_i - w_{i-1} + 0.2w_{i-2} - 0.192i^2 - 0.192i + 4.736].$$

To use this method explicitly, we need to solve the equation explicitly solve for w_{i+1} .

This gives

$$w_{i+1} = \frac{1}{22.2} [27.8w_i - w_{i-1} + 0.2w_{i-2} - 0.192i^2 - 0.192i + 4.736]$$

for $i = 2, 3, \dots, 9$

The results in Table 5.13 were obtained using the exact values from $y(t) = (t + 1)^2 - 0.5e^t$ for $\alpha, \alpha_1, \alpha_2$ and α_3 in the explicit Adams-Bashforth case and for α, α_1 and α_2 in the implicit Adams-Moulton case. Note that the implicit Adams-Moulton method gives consistently better results.

Table 5.13

t_1	Exact	Adams – Bashfort h w_1	Error	Adams- Moulton w_1	Error
0.0	0.500000 0				
0.2	0.8292986				
0.4	1.2140877				
0.6	1.6489406			1.648934 1	0.000006 5
0.8	2.1272295	2.127312 4	0.000082 8	2.127213 6	0.000016 0
1.0	2.6408591	2.641081 0	0.000221 9	2.640829 8	0.000029 3
1.2	3.1799415	3.180348 0	0.000406 5	3.179893 7	0.000047 8
1.4	3.7324000	3.733060 1	0.000660 1	3.732327 0	0.000073 1
1.6	4.2834838	4.284493 1	0.001009 3	4.283376 7	0.000107 1

1.8	4.8151763	4.816657 5	0.001481 2	4.815023 6	0.000152 7
2.0	5.3054720	5.307583 8	0.002111 9	5.305258 7	0.000213 2

Predictor-Corrector Methods

In Example 4, the implicit Adams-Moulton method gave better results than the explicit Adams-Bashforth method of the same order. Although this is generally the case, the implicit methods have the inherent weakness of first having to convert the method algebraically to an explicit representation for w_{i+1} . This procedure is not always possible, as can be seen by considering the elementary initial-value problem

$$y' = e^y, \quad 0 \leq t \leq 0.25, \quad y(0) = 1$$

Because $f(t, y) = e^y$, the three-step Adams-Moulton method has

$$w_{i+1} = w_i + \frac{h}{24} [9e^{w_{i+1}} + 19e^{w_i} - 5e^{w_{i-1}} + e^{w_{i-2}}]$$

as its difference equation, and this equation cannot be algebraically solved for w_{i+1} .

We could use Newton's method or the secant method to approximate w_{i+1} , but this complicates the procedure considerably. In practice, implicit multistep methods are not used as described above. Rather, they are used to improve approximations obtained by explicit methods. The combination

of an explicit method to predict and an implicit method to improve the prediction is called a **predictor-corrector method**.

Consider the following fourth-order method for solving an initial-value problem. The first step is to calculate the starting values w_0, w_1, w_2 , and w_3 for the four-step explicit Adams-Bashforth method. To do this, we use a fourth-order one-step method, the Runge-Kutta method of order four. The next step is to calculate an approximation, w_{4p} , to $y(t_4)$ using the explicit Adams-Bashforth method as predictor:

$$w_{4p} = w_3 + \frac{h}{24} [55f(t_3, w_3) - 59f(t_2, w_2) + 37f(t_1, w_1) - 9f(t_0, w_0)]$$

This approximation is improved by inserting w_{4p} in the right side of the three-step implicit Adams-Moulton method and using that method as a corrector. This gives

$$w_4 = w_3 + \frac{h}{24} [9f(t_4, w_{4p}) + 19f(t_3, w_3) - 5f(t_2, w_2) + f(t_1, w_1)]$$

The only new function evaluation required in this procedure is $f(t_4, w_{4p})$ in the corrector equation; all the other values of f have been calculated for earlier approximations.

The value w_4 is then used as the approximation to $y(t_4)$, and the technique of using the Adams-Bashforth method as a predictor and the Adams-Moulton method as a corrector is repeated to find w_{5p} and w_5 , the initial and final approximations to $y(t_5)$. This process is continued until we obtain an approximation w_N to $y(t_N) = y(b)$.

Example 5.

Apply the Adams fourth-order predictor-corrector method with $h = 0.2$ and starting values from the Runge-Kutta fourth-order method to the initial-value problem

$$y' = y - t^2 + 1, \quad 0 \leq t \leq 2, \quad y(0) = 0.5$$

Solution.

This is a continuation and modification of the problem considered in Example 1

at the beginning of the section. In that example, we found that the starting approximations from Runge-Kutta are

$$y(0) = w_0 = 0.5, \quad y(0.2) \approx w_1 = 0.8292933, \quad y(0.4) \approx w_2 = 1.2140762, \text{ and}$$

$$y(0.6) \approx w_3 = 1.6489220,$$

and the fourth-order Adams-Bashforth method gave

$$\begin{aligned} y(0.8) &\approx w_{4p} = w_3 \\ &\quad + \frac{0.2}{24} (55f(0.6, w_3) - 59f(0.4, w_2) \\ &\quad + 37f(0.2, w_1) - 9f(0, w_0)) \\ &= 1.6489220 + \frac{0.2}{24} (55f(0.6, 1.6489220) \\ &\quad - 59f(0.4, 1.2140762) \\ &\quad + 37f(0.2, 0.8292933) - 9f(0, 0.5)) \end{aligned}$$

$$\begin{aligned}
 &= 1.6489220 + 0.0083333(55(2.2889220) \\
 &\quad - 59(2.0540762) \\
 &+ 37(1.7892933) - 9(1.5)) \\
 &= 2.1272892.
 \end{aligned}$$

We will now use w_{4p} as the predictor of the approximation to $y(0.8)$ and determine the corrected value w_4 , from the implicit Adams-Moulton method. This gives

$$\begin{aligned}
 y(0.8) &\approx w_4 = w_3 \\
 &\quad + \frac{0.2}{24} \left(9f(0.8, w_{4p}) + 19f(0.6, w_3) \right. \\
 &\quad \left. - 5f(0.4, w_2) + f(0.2, w_1) \right) \\
 &= 1.6489220 + \frac{0.2}{24} (9f(0.8, 2.1272892) \\
 &\quad + 19f(0.6, 1.6489220) \\
 &\quad - 5f(0.4, 1.2140762) + f(0.2, 0.8292933)) \\
 &= 1.6489220 + 0.0083333(9(2.4872892) \\
 &\quad + 19(2.2889220) - 5(2.0540762) \\
 &\quad + (1.7892933)) \\
 &= 2.1272056.
 \end{aligned}$$

Now we use this approximation to determine the predictor, w_{5p} , for $y(1.0)$ as $y(1.0) \approx w_5 = w_4 + \frac{0.2}{24} \left(55f(0.8, w_{4p}) - 59f(0.6, w_3) + 37f(0.4, w_2) - 9f(0.2, w_1) \right)$

$$\begin{aligned}
 &= 2.1272056 + \frac{0.2}{24} (55f(0.8, 2.1272056) \\
 &\quad - 59f(0.6, 1.6489220) \\
 &\quad + 37f(0.4, 1.2140762) - 9f(0.2, 0.8292933)) \\
 &= 2.1272056 + 0.0083333(55(2.4872056) \\
 &\quad - 59(2.2889220) \\
 &\quad + 37(2.0540762) - 9(1.7892933)) \\
 &= 2.6409314
 \end{aligned}$$

and correct this with

$$\begin{aligned}
 y(1.0) &\approx w_5 = w_4 \\
 &\quad + \frac{0.2}{24} (9f(1.0, w_{5p}) + 19f(0.8, w_4) \\
 &\quad - 5f(0.6, w_3) + f(0.4, w_2)) \\
 &= 2.1272056 + \frac{0.2}{24} (9f(1.0, 2.6409314) \\
 &\quad + 19f(0.8, 2.1272892) \\
 &\quad - 5f(0.6, 1.6489220) + f(0.4, 1.2140762)) \\
 &= 2.1272056 + 0.0083333(9(2.6409314) + 19(2.4872056) \\
 &\quad - 5(2.2889220) \\
 &\quad + (2.0540762)) \\
 &= 2.6408286.
 \end{aligned}$$

In Example 1, we found that using the explicit Adams-Bashforth method alone produced results that were inferior to those of

Runge-Kutta. However, these approximations to $y(0.8)$ and $y(1,0)$ are accurate, respectively, to within

$$|2.1272295 - 2.1272056| = 2.39 \times 10^{-5} \text{ and } |2.6408286 - 2.64085911| = 3.05 \times 10^{-5},$$

compared to those of Runge-Kutta, which were accurate, respectively, to within

$$|2.1272027 - 2.1272892| = 2.69 \times 10^{-5} \text{ and } |2.6408227 - 2.64085911| = 3.64 \times 10^{-5},$$

The remaining predictor-corrector approximations are shown in Table 5.14.

Table 5.14

t_1	$y_1 = y(t_1)$	w_1	Error [$y_1 - w_1$]
0.0	0.5000000	0.5000000	0
0.2	0.8292986	0.8292933	0.0000053
0.4	1.2140877	1.2140762	0.0000114
0.6	1.6489406	1.6489220	0.0000186
0.8	2.1272295	2.1272027	0.0000239
1.0	2.6408591	2.6408227	0.0000305
1.2	3.1799415	3.1798942	0.0000474

1.4	3.7324000	3.7323505	0.0000495
1.6	4.2834838	4.2834208	0.0000630
1.8	4.8151763	4.8150964	0.0000799
2.0	5.3054720	5.3053707	0.0001013

Questions 5.5

1. Use all the Adams-Bashforth methods to approximate the solutions to the following initial-value problems. In each case, use exact starting values and compare the results to the actual values.

a. $y' = te^{3t} - 2y$, $0 \leq t \leq 1$, $y(0) = 0$, with $h = 0.2$;
actual solution $y(t) = \frac{1}{5}te^{3t} - \frac{1}{25}e^{3t} + \frac{1}{25}e^{-2t}$

b. $y' = 1 + (t - y)^2$, $2 \leq t \leq 3$, $y(2) = 1$, with $h = 0.2$;
actual solution $y(t) = t + \frac{1}{1-t}$

c. $y' = 1 + y/t$, $1 \leq t \leq 2$, $y(1) = 2$, with $h = 0.2$; actual solution $y(t) = t \ln t + 2t$.

d. $y' = \cos 2t + \sin 3t$, $0 \leq t \leq 1$, $y(0) = 1$, with $h = 0.2$; actual solution $y(t) = \frac{1}{2}\sin 2t - \frac{1}{3}\cos 3t + \frac{4}{3}$

2. Use each of the Adams-Bashforth methods to approximate the solutions to the following initial-value problems. In each case, use starting values obtained from the Runge-Kutta method of order four. Compare the results to the actual values.

a. $y' = \frac{y}{t} - \left(\frac{y}{t}\right)^2, \quad 1 \leq t \leq 2, \quad y(1) = 1, \quad \text{with } h = 0.1;$
actual solution $y(t) = \frac{t}{1 + \ln t}$

b. $y' = 1 + \frac{y}{t} + \left(\frac{y}{t}\right)^2, \quad 1 \leq t \leq 3, \quad y(1) = 0, \quad \text{with } h = 0.2;$
actual solution $y(t) = t \tan(\ln t)$

c. $y' = -(y + 1)(y + 3), \quad 0 \leq t \leq 2, \quad y(0) = -2, \quad \text{with } h = 0.1;$ actual solution $y(t) = -3 + 2 / (1 + e^{-2t})$

d. $y' = -5y + 5t^2 + 2t, \quad 0 \leq t \leq 1, \quad y(0) = \frac{1}{3}, \quad \text{with } h = 0.1;$ actual solution $y(t) = t^2 + \frac{1}{3}e^{-5t}.$

3. Use all the Adams-Moulton methods to approximate the solutions to the Exercises 1(a), 1(c), and 1(d). In each case, use exact starting values and explicitly solve for w_{i+1} . Compare the results to the actual values.

Answers

1. The Adams-Bashforth methods give the results in the following tables,

a.						
	t	2-step	3-step	4-step	5-step	y(t)
	0.2	0.0268128	0.0268128	0.0268128	0.0268128	0.0268128
	0.4	0.1200522	0.1507778	0.1507778	0.1507778	0.1507778

	0.6	0.4153551	0.4613866	0.4960196	0.4960196	0.4960196
	0.8	1.1462844	1.2512447	1.2961260	1.3308570	1.3308570
	1.0	2.8241683	3.0360680	3.1461400	3.1854002	3.2190993

b.						
	t	2-step	3-step	4-step	5-step	y(t)
	2.2	1.3666667	1.3666667	1.3666667	1.3666667	1.3666667
	2.4	1.6750000	1.6857143	1.6857143	1.6857143	1.6857143
	2.6	1.9632431	1.9794407	1.9750000	1.9750000	1.9750000
	2.8	2.2323184	2.2488759	2.2423065	2.2444444	2.2444444
	3.0	2.4884512	2.5051340	2.4980306	2.5011406	2.5000000

c.						
	t	2-step	3-step	4-step	5-step	y(t)
	1.2	2.6187859	2.6187859	2.6187859	2.6187859	2.6187859
	1.4	3.2734823	3.2710611	3.2710611	3.2710611	3.2710611
	1.6	3.9567107	3.9514231	3.9520058	3.9520058	3.9520058
	1.8	4.6647738	4.6569191	4.6582078	4.6580160	4.6580160

	2.0	5.3949416	5.3848058	5.3866452	5.3862177	5.3862944
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d.						
	t	2-step	3-step	4-step	5-step	y(t)
	0.2	1.2529306	1.2529306	1.2529306	1.2529306	1.2529306
	0.4	1.5986417	1.5712255	1.5712255	1.5712255	1.5712255
	0.6	1.9386951	1.8827238	1.8750869	1.8750869	1.8750869
	0.8	2.1766821	2.0844122	2.0698063	2.0789180	2.0789180
	1.0	2.2369407	2.1115540	2.0998117	2.1180642	2.1179795

2.The Adams-Bashforth methods give the results in the following tables,

a.

t	2-step	3-step	4-step	5-step	$y(t)$
1.2	1.0161982	1.0149520	1.0149520	1.0149520	1.0149523
1.4	1.0497665	1.0468730	1.0477278	1.0475336	1.0475339
1.6	1.0910204	1.0875837	1.0887567	1.0883045	1.0884327
1.8	1.1363845	1.1327465	1.1340093	1.1334967	1.1336536
2.0	1.1840272	1.1803057	1.1815967	1.1810689	1.1812322

b.

t	2-step	3-step	4-step	5-step	$y(t)$
1.4	0.4867550	0.4896842	0.4896842	0.4896842	0.4896817
1.8	1.1856931	1.1982110	1.1990422	1.1994320	1.1994386
2.2	2.1753785	2.2079987	2.2117448	2.2134792	2.2135018
2.6	3.5849181	3.6617484	3.6733266	3.6777236	3.6784753
3.0	5.6491203	5.8268008	5.8589944	5.8706101	5.8741000

c.

t	2-step	3-step	4-step	5-step	$y(t)$
0.5	-1.5357010	-1.5381988	-1.5379372	-1.5378676	-1.5378828
1.0	-1.2374093	-1.2389605	-1.2383734	-1.2383693	-1.2384058
1.5	-1.0952910	-1.0950952	-1.0947925	-1.0948481	-1.0948517
2.0	-1.0366643	-1.0359996	-1.0359497	-1.0359760	-1.0359724

d.

t	2-step	3-step	4-step	5-step	$y(t)$
0.2	0.1739041	0.1627655	0.1627655	0.1627655	0.1626265
0.4	0.2144877	0.2026399	0.2066057	0.2052405	0.2051118
0.6	0.3822803	0.3747011	0.3787680	0.3765206	0.3765957
0.8	0.6491272	0.6452640	0.6487176	0.6471458	0.6461052
1.0	1.0037415	1.0020894	1.0064121	1.0073348	1.0022460

3. The Adams-Moulton methods give the results in the following tables,

a.					
	t	2-step	3-step	4-step	$y(t)$
	0.2	0.0268128	0.0268128	0.0268128	0.0268128
	0.4	0.1533627	0.1507778	0.1507778	0.1507778
	0.6	0.5030068	0.4979042	0.4960196	0.4960196
	0.8	1.3463142	1.3357923	1.3322919	1.3308570

	1.0	3.2512866	3.2298092	3.2227484	3.2190993
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c.					
	t	2-step	3-step	4-step	y(t)
	1.2	2.6187859	2.6187859	2.6187859	2.6187859
	1.4	3.2711394	3.2710611	3.2710611	3.2710611
	1.6	3.9521454	3.9519886	3.9520058	3.9520058
	1.8	4.6582061	4.6579866	4.6580211	4.6580160
	2.0	5.3865293	5.3862558	5.3863027	5.3862944

d.					
	t	2-step	3-step	4-step	y(t)
	0.2	1.2529306	1.2529306	1.2529306	1.2529306
	0.4	1.5700866	1.5712255	1.5712255	1.5712255
	0.6	1.8738414	1.8757546	1.8750869	1.8750869
	0.8	2.0787117	2.0803067	2.0789471	2.0789180
	1.0	2.1196912	2.1199024	2.1178679	2.1179795